PLOS NEGLECTED TROPICAL DISEASES

RESEARCH ARTICLE

Tick borne relapsing fever - a systematic review and analysis of the literature

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Abstract

Tick borne relapsing fever (TBRF) is a zoonosis caused by various *Borrelia* species transmitted to humans by both soft-bodied and (more recently recognized) hard-bodied ticks. In recent years, molecular diagnostic techniques have allowed to extend our knowledge on the global epidemiological picture of this neglected disease. Nevertheless, due to the patchy occurrence of the disease and the lack of large clinical studies, the knowledge on several clinical aspects of the disease remains limited. In order to shed light on some of these aspects, we have systematically reviewed the literature on TBRF and summarized the existing data on epidemiology and clinical aspects of the disease. Publications were identified by using a predefined search strategy on electronic databases and a subsequent review of the reference lists of the obtained publications. All publications reporting patients with a confirmed diagnosis of TBRF published in English, French, Italian, German, and Hungarian were included. Maps showing the epidemiogeographic mosaic of the different TBRF *Borrelia* species were compiled and data on clinical aspects of TBRF were analysed.

The epidemiogeographic mosaic of TBRF is complex and still continues to evolve. Ticks harbouring TBRF *Borrelia* have been reported worldwide, with the exception of Antarctica and Australia. Although only molecular diagnostic methods allow for species identification, microscopy remains the diagnostic gold standard in most clinical settings. The most suggestive symptom in TBRF is the eponymous relapsing fever (present in 100% of the cases). Thrombocytopenia is the most suggestive laboratory finding in TBRF. Neurological complications are frequent in TBRF. Treatment is with beta-lactams, tetracyclines or macrolids. The risk of Jarisch-Herxheimer reaction (JHR) appears to be lower in TBRF (19.3%) compared to louse-borne relapsing fever (LBRF) (55.8%). The overall case fatality rate of TBRF (6.5%) and LBRF (4–10.2%) appears to not differ. Unlike LBRF, where perinatal fatalities are primarily attributable to abortion, TBRF-related perinatal fatalities appear to primarily affect newborns.

OPEN ACCESS

Citation: Jakab Á, Kahlig P, Kuenzli E, Neumayr A (2022) Tick borne relapsing fever - a systematic review and analysis of the literature. PLoS Negl Trop Dis 16(2): e0010212. https://doi.org/10.1371/journal.pntd.0010212

Editor: Jon Blevins, UAMS, UNITED STATES

Received: November 26, 2021

Accepted: January 27, 2022

Published: February 16, 2022

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Data Availability Statement: All relevant data are within the manuscript and its Supporting Information files.

Funding: The author(s) received no specific funding for this work.

Competing interests: The authors have declared that no competing interests exist.

Author summary

Tick-borne relapsing fever (TBRF) is a bacterial disease characterized by eponymous recurrent fever episodes. The disease is common on all continents except Australia and Antarctica and is caused by several species of *Borrelia* bacteria. The *Borrelia* bacteria causing relapsing fever circulate naturally between ticks and various animal hosts (usually small rodents). Humans become infected when they are accidentally bitten by an infected tick.

Although the disease has been known since 1904, many aspects of the disease have never been investigated in larger studies and are therefore still not conclusively understood. To shed light on some of these aspects, we reviewed the published literature on TBRF and analysed all reported data on the geographic distribution of the different TBRF-causing *Borrelia* spp. as well as on the clinical presentations of the disease, its complications, its diagnosis and treatment and its outcome, and compiled them in this review.

Introduction

Two febrile illnesses related to human pathogenic spirochetes belonging to the genera *Borrelia* present as "relapsing fevers", louse-borne relapsing fever (LBRF) and tick-borne relapsing fever (TBRF). While LBRF is an anthroponotic disease exclusively caused by *Borrelia recurrentis*, TBRF is a zoonotic disease caused by various *Borrelia* species. In different regions of the world different *Borrelia* spp. have been identified to be endemic. They differ in their natural enzootic cycles (involving different tick species and their hosts, mostly small rodents) and they are capable of infecting humans as accidental dead-end hosts [1]. An exemption is *B*. *duttonii*, for which humans may be the reservoir [2]. Since LBRF and TBRF present clinically identical and LBRF- and TBRF-*Borrelia* are microscopically indistinguishable, differentiation between the two diseases was historically limited to the epidemiological circumstances (LBRF: outbreaks, epidemics, occurrence in vulnerable populations exposed to body lice; TBRF: sporadic cases in persons exposed to ticks). Finally, the advent of molecular diagnostic techniques not only enabled to distinguish TBRF from LBRF, but also significantly changed the understanding of the diversity and epidemiology of TBRF. To summarize the current knowledge on epidemiology and clinical relevant aspects of TBRF we reviewed and analysed the existing literature, analogously to our recently published review on LBRF [3,4].

Epidemiology

Historically, in most regions of the world TBRF has always been overshadowed by LBRF which was more prominent and epidemiologically relevant because of its epidemic occurrence. With the decline of lice-infested populations in most regions of the world, LBRF became a rare disease, while TBRF received increasing attention, especially in recent years.

TBRF has been recognized in Africa since 1904, owing to the researches of Ross, Dutton and others [5]. In the early 1920s, TBRF was also recognized as an endemic disease in the United States of America (USA), although a tick vector was not recognized until 1930 [6]. In the following years, case reports of TBRF showed the extend of endemic areas in the USA and various tick species were identified as vectors [7]. Fig 1 shows a map with the assumed global distribution of TBRF and LBRF published in 1971 [8].

Today, TBRF is reported from all continents except Australia and Antarctica [9] and constitutes an important public health problem in some parts of the world. In Western Africa, TBRF

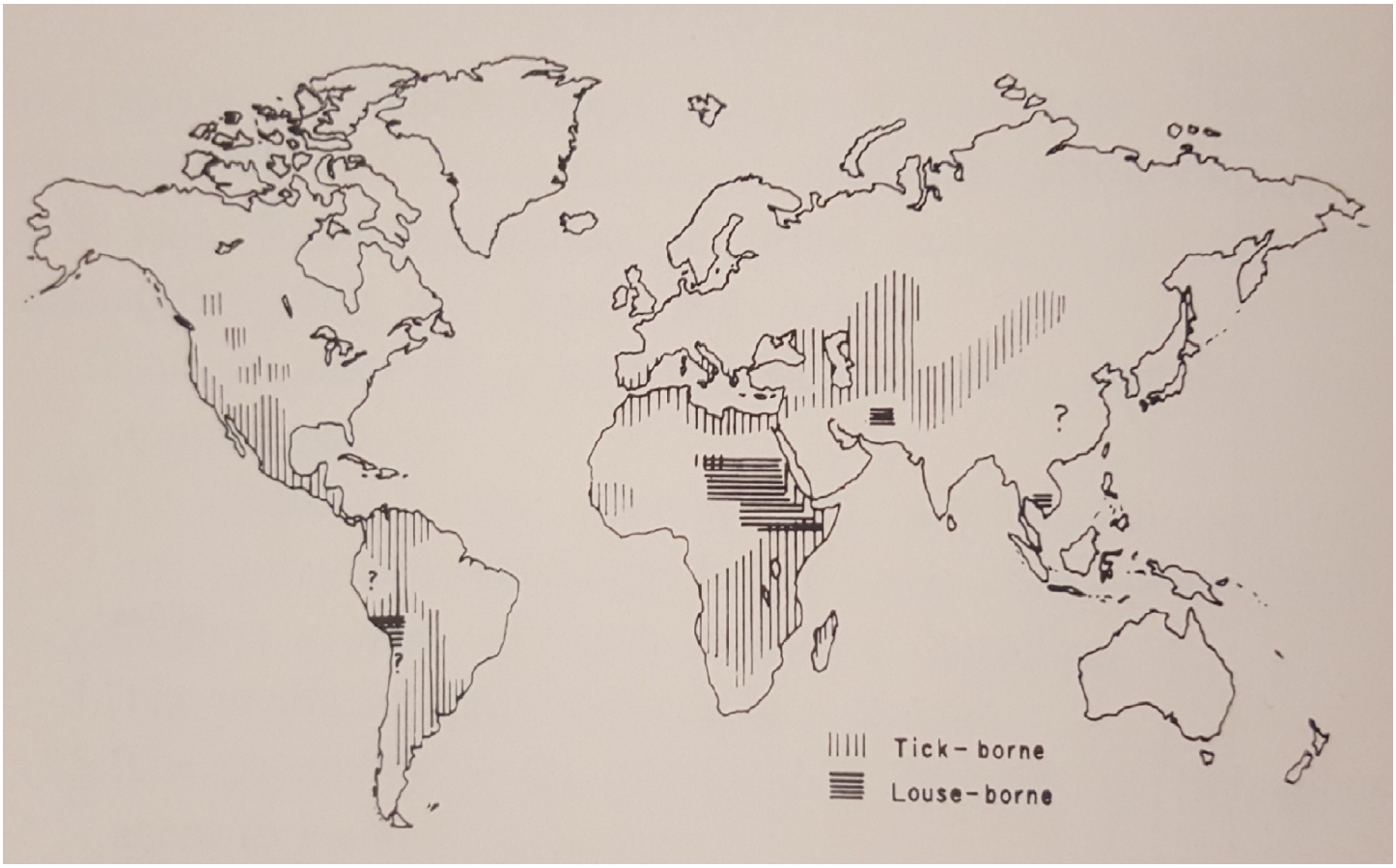


Fig 1. Assumed global distribution of TBRF and LBRF, 1950–1969 (Felsenfeld O. *Borrelia*; Strains, Vectors, Human and Animal Borreliosis. St. Louis: Warren H. Green; 1971[8]).

https://doi.org/10.1371/journal.pntd.0010212.g001

accounts for about 13% of febrile illnesses [10] and in endemic regions of East Africa, TBRF is one of the diseases with the highest lethality among children [11].

Tick vectors

Historically, TBRF was considered to be exclusively transmitted by soft ticks (*Ornithodoros* spp.) [8]. In 2011, this paradigm changed when *Borrelia miyamotoi*, a *Borrelia* species discovered in Japan in 1995 [12], was reported to cause TBRF transmitted by hard ixodid ticks in Russia [4], a finding later confirmed in Europe, Japan and the USA [13–15]. Nevertheless, since most TBRF *Borrelia* are transmitted by soft ticks, several distinct and epidemiological relevant differences between soft and hard ticks deserve to be highlighted. Soft ticks differ from hard ticks not only by the eponymous lack of a hard shell around the mouthparts, but also by the fact that they do not wait on leaves or blades of grass for their prey to walk by. Instead, they live in close proximity to their small mammal hosts (e.g. mice, rats, squirrels, rabbits) and rarely leave the confines of their hosts' nest or burrow. Humans may be targeted by these night active ticks when sleeping close to their habitats. Because soft ticks feed rapidly (15–90 minutes) and then return to the place from which they came, their attack is rarely noticed. Persistent infection of the ticks' salivary glands [16] allows quick transmission of TBRF *Borrelia* during the short feeding period, possibly after only 30 seconds of attachment [17]. Soft tick

females lay clutches of eggs after each blood meal. This reproductive pattern is strikingly different from that of hard ticks, where adult females reproduce only once in their lifetime [18]. The life cycle stages of soft ticks include egg, larva, several successive nymphs and the adult. After hatching, all stages are obligate blood feeders and capable of transmitting *Borrelia* [18]. Once infected, ticks remain infectious for the duration of their life. Since soft ticks may live for more than 10 years and survive up to 5 years without feeding [19], they can outlive their rodent hosts and infect several cohorts of rodents over the course of their lifespan [20].

Clinical picture

The incubation period of TBRF is 4–18 days. Thereafter, up to 12 recurrent febrile episodes occur. These fever episodes last 2–7 days and are separated by afebrile periods of up to 10 days [18,21,22]. A broad range of accompanying unspecific symptoms (e.g. headache, myalgia, chills, nausea, vomiting, arthralgia) as well as neurologic complications (e.g. meningitis, encephalitis, hemiplegia, facial palsy, radiculopathy, occasionally subarachnoid hemorrhage) may occur. They generally become more prominent after the second febrile episode [1,23,24]. The characteristic disease pattern of recurrent febrile episodes, interspersed with afebrile episodes, is attributable to the antigenic variation of different, sequentially expressed versions of the bacterium's outer-membrane lipoprotein (vmp), allowing the bacterium to temporarily evade the host's humoral immune response. Once the host's immune system mounts antibodies against a specific vmp variant, a new vmp variant is expressed by the *Borrelia*, camouflaging itself, until antibodies are also generated against the new vmp variant [25]. The clinical presentation of LBRF is very similar to TBRF and the pathophysiological mechanism of the recurrent fever episodes is identical. However, the number of recurrent fever episodes is overall lower in LBRF (mostly less than 2) compared to TBRF ($\geq$2), while the paroxysms last longer in LBRF (up to 10 days) compared to TBRF ($\leq$7 days). The frequently observed neurological complications in TBRF are rare in LBRF, and TBRF is usually milder and lethality reportedly lower compared to LBRF [26].

Diagnostics

Relapsing fever *Borrelia* cause massive, microscopically visible bacteremia during febrile episodes. Therefore, the microscopic examination of blood smears (Fig 2) has been the diagnostic method of choice since relapsing fever *Borrelia* were first microscopically detected in the blood of patients by Obermeier in 1873 [3]. The optimum time to obtain blood is during the presence of fever, as *Borrelia* are usually not detectable once the temperature is decreasing or back to normal. Thick and thin blood films are taken and stained with e.g. Giemsa, May-Grünwald Giemsa, Wright, Wright-Giemsa, Field's or Diff-Quick stain. Various techniques, including centrifugation of the blood samples before microscopy [27], quantitative buffy coat (QBC) preparation [28], dark-field microscopy and direct or indirect immunofluorescence [29] have been used to improve the sensitivity of microscopic detection.

No commercial serological assays have been developed to diagnose TBRF. This is due to cross-reactivity among *Borrelia* spp., including LBRF, Lyme disease and other spirochetes (e.g. *Treponema pallidum*) [30] as well as the fact that serology is not helpful to diagnose acute infection due to the time to seroconversion. Culture of *Borrelia* spp. is difficult and time-consuming and thus largely remains restricted to research institutions. Animal inoculation was considered as a putative adjunct diagnostic tool in the late 1940s [31] but, being cumbersome, was never routinely used for diagnostic reasons alone.

With the introduction of polymerase chain reaction (PCR) and sequencing techniques in the 1980s, highly sensitive and specific diagnostic tools became available. However, the availability

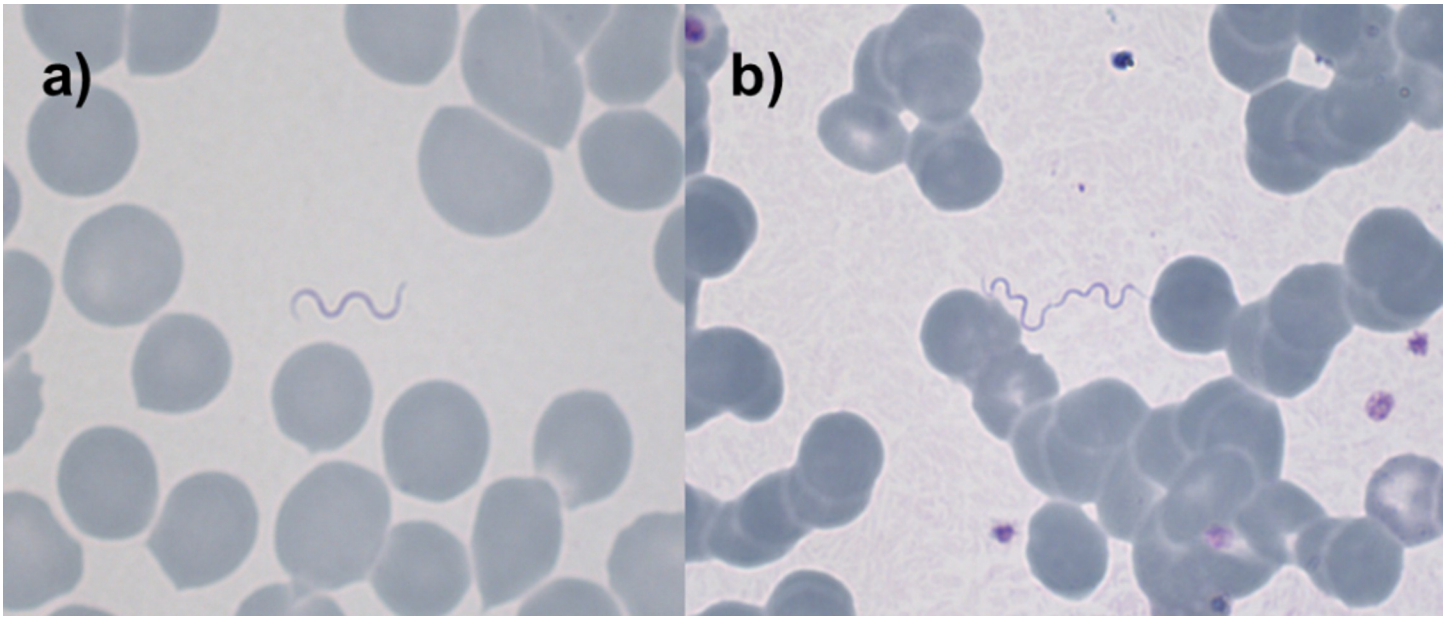


Fig 2. Microscopical detection of TBRF *Borrelia* in blood films. Microscopic images of Giemsa-stained thin blood films (original magnifications ×1'000) showing TBRF *Borrelia* in a patient suffering from TBRF fever due to *Borrelia persica* (courtesy of Dr. Veronika Muigg).

https://doi.org/10.1371/journal.pntd.0010212.g002

of these molecular diagnostic tools still remains largely restricted to research institutions and microscopy remains the diagnostic gold standard for TBRF even in affluent countries [32,33]. Table 1 summarizes the advantages and disadvantages of the different diagnostic methods.

Molecular epidemiology

Historically, TBRF *Borrelia* species are geographically grouped into Old World species (e.g. *Borrelia duttonii*, *B. persica*, *B. hispanica*, *B. crocidurae* a.o.) and New World species (e.g. *B.*

Table 1. Overview of laboratory methods applied in TBRF and their advantages, disadvantages and use.

Method	Advantage	Disadvantage	Use
PCR	Species specific; high sensitivity allows to differentiate TBRF- from LBRF-*Borrelia* and among TBRF-*Borrelia*	Currently no standardized protocol available; availability in resource-poor countries limited	Largely restricted to research institutions
Microscopy	Fast; widely available	Variable sensitivity (spirochete density, inter-observer variability, methodological differences); does not allow species differentiation	Diagnostic gold standard
Culture	Isolation and growth of *Borrelia* spp.	Time and resource demanding; overall challenging	Largely restricted to research institutions
Animal inoculation	Enhanced sensitivity in cases with negative microscopy; allows differentation between TBRF and LBRF*	Time and resource demanding	Historical research method; formerly also used to "transport" *Borrelia*
Serology	Allows retrospective evaluation of infection	Not useful as acute diagnostic method due to delayed seroconversion; cross-reactivity with other non-RF *Borrelia*	Restricted to epidemiological studies

LBRF, louse borne relapsing fever; PCR, polymerase chain reaction; TBRF, tick borne relapsing fever; RF: relapsing fever.

* Note: rodents are susceptible to TBRF *Borrelia* spp. but not susceptible to *B. recurrentis* infection.

(Table adapted from [3])

https://doi.org/10.1371/journal.pntd.0010212.t001

hermsii, *B. turicatae*, *B. parkeri* a.o.). For some *Borrelia* species (identified in vectors and animal hosts only) their humanpathogenic potential still remains to be determined (e.g. *B. cachapoal*, *B. osphepa* a.o. [34,35]).

With the advent of PCR and sequencing techniques, it became not only possible to differentiate TBRF from LBRF, but these techniques also allowed genetic characterization of the different TBRF *Borrelia* species. Currently, 12 different *Borrelia* spp. and an additional 4 proposed "*Candidatus*" spp. have been reported to cause TBRF. The *Candidatus* status is used for newly discovered species for which more than a mere nucleic acid sequence is available but for which characteristics required for description according to the *International Code of Nomenclature of Bacteria* are lacking [36,37]. In general, to confirm the novelty of a bacterial species, 16S rRNA gene sequencing is performed and the sequence is compared to archived reference sequences. A threshold of 98.7% of 16S rRNA gene sequence similarity with the phylogenetically closest species with standing in the nomenclature was suggested by Stackebrandt and Ebers to classify a new bacterial species [38]. With the increasing availability of molecular diagnostic techniques, the number of reported species is likely to continue expanding in the future.

Treatment, Jarisch-Herxheimer reaction (JHR) and outcome

In the first half of the 20th century, arsenicals and emetine bismuth iodide were the only available drugs for the treatment of relapsing fever [39]. After the discovery of penicillin, treatment shifted towards this antimicrobial agent in the second half of the century with alternative therapeutic agents becoming available over time (i.e. tetracyclines, macrolides). Today, the preferred antibiotics to treat TBRF are tetracyclines, β-lactams and macrolids [40] to which *Borrelia* are invariably susceptible [41].

Treatment may be complicated by Jarisch-Herxheimer reaction (JHR), which mostly occurs after administering the first dose of the antibiotic. JHR is characterized by intense chills and a rise in temperature about 1–2 hours after initiating antibiotic treatment and may be complicated by hypotension. JHR shares pathophysiological features of a classic endotoxin reaction mediated by proinflammatory cytokines (tumor necrosis factor α [TNF-α], interleukin 6 (IL-6), IL-8) [42]. JHR is not restricted to relapsing fever, but may also occur when treating other spirochete infections like syphilis, leptospirosis, and Lyme disease. In TBRF, JHR is reported to occur in up to 54.1% of cases [43]. Symptoms usually resolve within a few hours. Although JHR is rarely fatal, it is a clinically relevant complication which may require appropriate clinical therapeutic measures [4].

The lethality of untreated TBRF is reported to be 2–10% [44]. With antibiotic treatment the lethality is reported to be <2% [45]. Of note, TBRF is more serious in expatriates and visitors to an endemic area compared to indigenous people, who have usually been exposed to the pathogen previously [26].

TBRF infection during pregnancy is associated with an increased risk of death in pregnant women [46,47]. Infections during pregnancy are claimed to cause up to 10–15% of neonatal deaths worldwide and a perinatal lethality of up to 43.6% has been reported [1,23,44,48–50].

The aim of this study is to review and analyse the existing literature on TBRF and to summarize the epidemiological, clinical, diagnostic and treatment aspects of the disease, including its transmission through ticks, its vector reservoir and its clinical outcome.

Methods

We performed a systematic literature search of the databases Biosis Citation Index, Biosis Previews, CINAHL, Cochrane, Current Contents Connect, Data Citation Index, Derwent

Innovations Index, EMBASE Elsevier, EMBASE Ovid, Inspec, Medline, PMC, PubMed, SciELO Citation Index, Scopus, Web of Science, and Zoological Record on 04/Dec/2020, using the search term ("tick" OR "ticks" OR "tick borne" OR "*Ornithodoros*" OR "*Borrelia*" OR "*Borrelia miyamotoi*" OR "*Borrelia turicatae*" OR "*Borrelia hermsii*" OR "*Borrelia parkeri*" OR "*Borrelia persica*" OR "*Borrelia hispanica*" OR "*Borrelia crocidurae*" OR "*Borrelia duttonii*" OR "*Borrelia caucasica*" OR "*Borrelia microti*" OR "*Borrelia brasiliensis*" OR "*Borrelia mazzottii*" OR "*Borrelia venezuelensis*" OR "*Borrelia graingeri*" OR "*Borrelia latyschewii*" OR "*Borrelia dugesii*" OR "*Borrelia* infections" OR "*Borrelia*") AND ("relapsing fever" OR "recurrent fever" OR "relapsing fever disease") adapted to the search format of the different databases.

A detailed description of the literature search is available in S2 Text. After removing duplicates by EndNote (Version X9.2, Clarivate Analytics) and manually, the publications were pre-screened by title and abstract, removing those not concerning TBRF or not including the objectives of this study (epidemiology, transmission, vector, clinic, diagnostic, treatment, outcome). A full-text review of the remaining publications was then performed excluding those not meeting the inclusion criteria, according to the systematic review protocol (concerning TBRF and the objectives of the study, published in English, German, French, Italian or Hungarian), as shown in S1 Text. Publications that could neither be retrieved through the respective journals, nor by contacting libraries, or after contacting the authors, were classified as 'not retrievable' and excluded. During the full-text review, the reference lists of the articles were screened for additional relevant publications not identified previously («snowball-search» strategy). From the finally identified eligible studies, the following data were extracted: author, title, year of publication, type of study, study location, study period, location of acquisition/infection of *Borrelia*, *Borrelia* species, tick species, vector, percentage of ticks or vectors infected with *Borrelia*, hospital location for diagnosis, diagnostic method (microscopy, serology, molecular diagnostic, animal inoculation), grade of diagnostic certainty, number of patients, age of patient(s) (median and range), gender, symptoms, number of fever relapses, pregnancies, complications, used drug(s) and treatment regimen(s), number of treated or untreated patients, lethality of treated or untreated patients, frequency of JHR. To minimize bias, the same reviewer conducted a second full data extraction one month after the first extraction. Discrepancies and unclear cases were resolved by consulting a second reviewer. The probability of a correctly diagnosed TBRF was graded according to the diagnostic method used in the different studies, with PCR

Table 2. Diagnostic grading system to judge the certainty of the correct diagnosis of TBRF.

Diagnostic method	Grade of diagnostic certainty	Case classification	Comment
PCR	A	Confirmed diagnosis	Highest level of evidence, detection even at low level of spirochetemia
Microscopy	B	Microscopic diagnosis	High level of evidence, easy to carry out, examiner-dependent, likelihood of detection depends on level of spirochetemia
Culture	B	Microscopic diagnosis	High level of evidence, difficult to carry out, time demanding
Animal inoculation	B	Microscopic diagnosis	High level of evidence, difficult to carry out, time demanding
Serology	C	Indirect evidence	Intermediate level of evidence, not standardized, cross-reaction with other *Borrelia* (e.g. Lyme disease) possible

PCR, polymerase chain reaction.

https://doi.org/10.1371/journal.pntd.0010212.t002

having the highest (grade A) and serology the lowest (grade C) evidence for a correct diagnosis (Table 2).

The data extraction sheet is available in S1 Table.

To visualize the worldwide distribution of TBRF cases, the causative TBRF *Borrelia* spp. and the transmitting tick species, we used the free online geographic application *Mapchart* (www.mapchart.net).

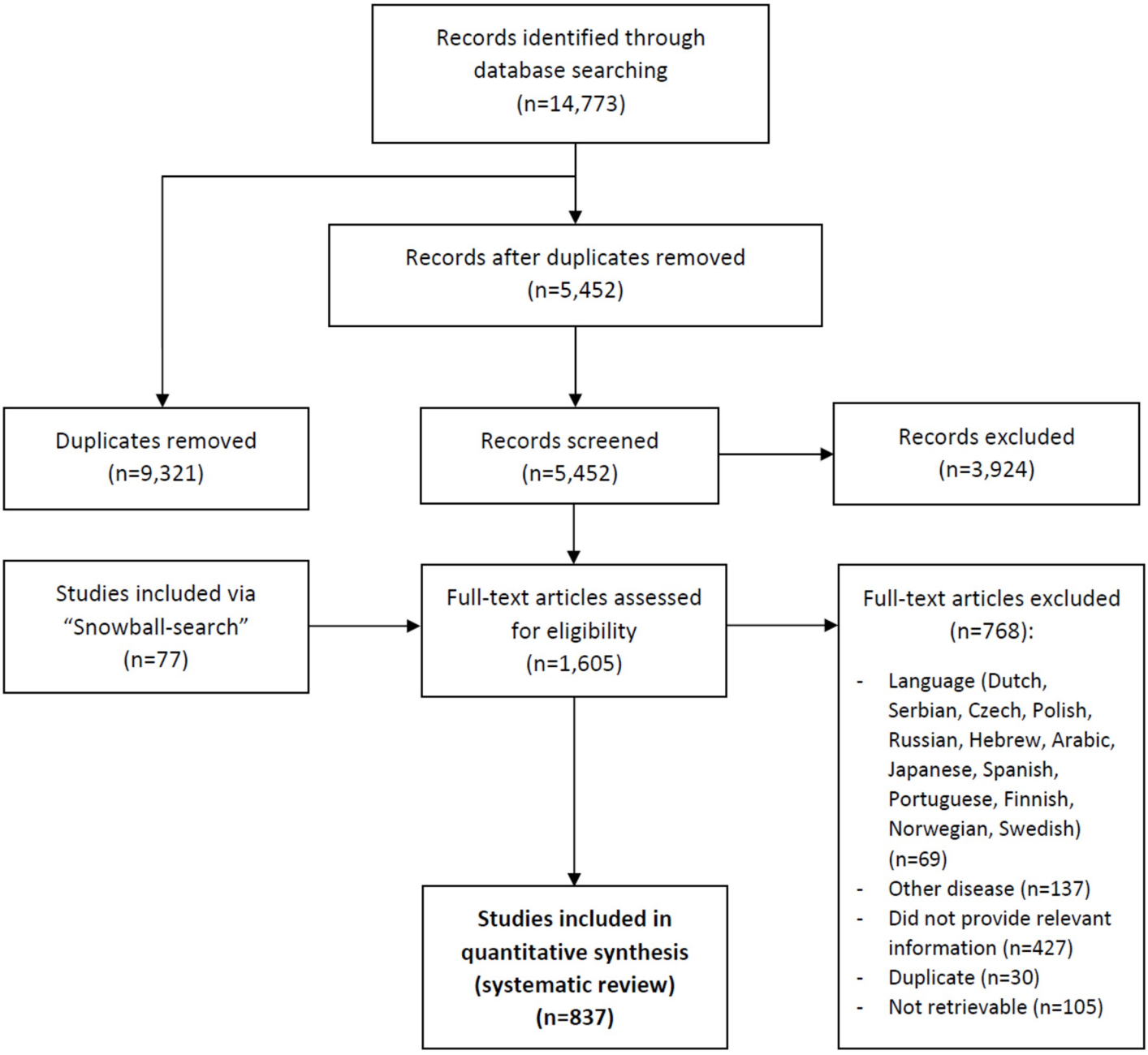


Fig 3. Flow diagram of search and selection of eligible publications.

https://doi.org/10.1371/journal.pntd.0010212.g003

Results

Our search identified 14,773 publications, of which 837 proved to be eligible for inclusion in the review (Fig 3). The reference list of the included and excluded publications and the PRISMA Checklist for systematic reviews are available in S3 Text and S1 PRISMA Checklist.

Fig 4 shows the number of TBRF case studies published from 1906 to 2020.

Geographic distribution of human TBRF cases and worldwide prevalence of TBRF-transmitting ticks and *Borrelia* species

385 of the 837 analysed studies reported the distribution of either ticks, *Borrelia* spp. or both. Some of the reported *Borrelia* are not yet acknowledged as official species and are currently considered "*Candidatus*" species. In North America, four *Borrelia* species (*B. miyamotoi*, *B. hermsii*, *B. turicatae*, *Candidatus* B. johnsonii) causing TBRF in humans are reported [15,43,51–104], in Central and South America four species (*B. obermeieri*, *B. neotropicalis*, *B. turicatae*, *B. parkeri*) [105–107], in Africa eight species (*B. crocidurae*, *B. hispanica*, *B. merionesi*, *B. parkeri*, *B. duttonii*, *Candidatus* B. algerica, *Candidatus* B. fainii, *Candidatus* B. kalaharica) [10,50,108–141], in Europe three species (*B. miyamotoi*, *B. hispanica*, *B. crocidurae*) [13,142–154] and in Asia three species (*B. miyamotoi*, *B. persica*, *B. microti*) [14,155–176]. No cases of TBRF or ticks known to transmit TBRF *Borrelia* are reported in Australia. The detailed list of *Borrelia* and ticks reported in the different continents and regions can be found in S1 Data. The worldwide distribution of reported TBRF cases by country and the causative *Borrelia* species are shown in Fig 5. The worldwide distribution of reported TBRF cases caused by unidentified *Borrelia* species is shown in Fig 6. The prevalence of *Borrelia* species causing TBRF in America, Africa, Europe, and Asia

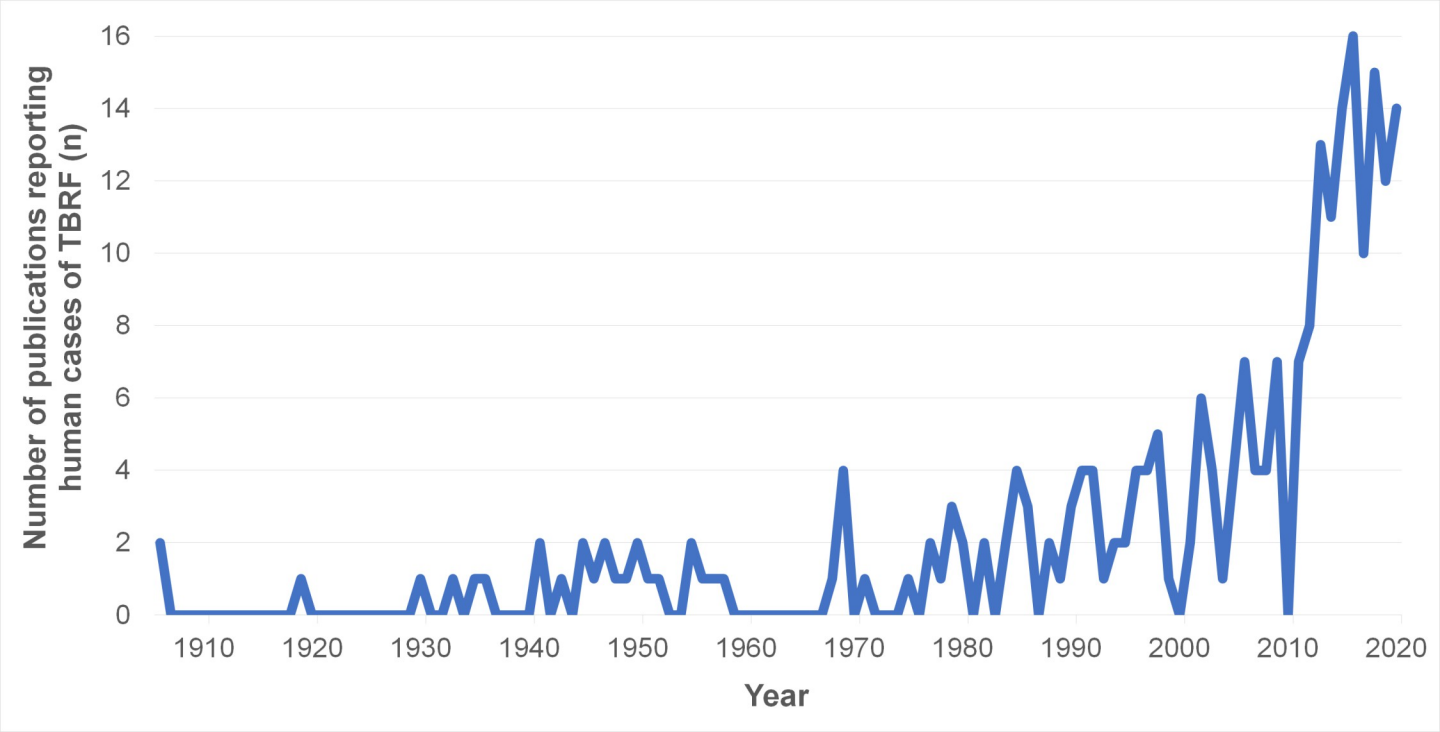


Fig 4. Number of TBRF case studies published from 1906 to 2020. TBRF, tick borne relapsing fever.

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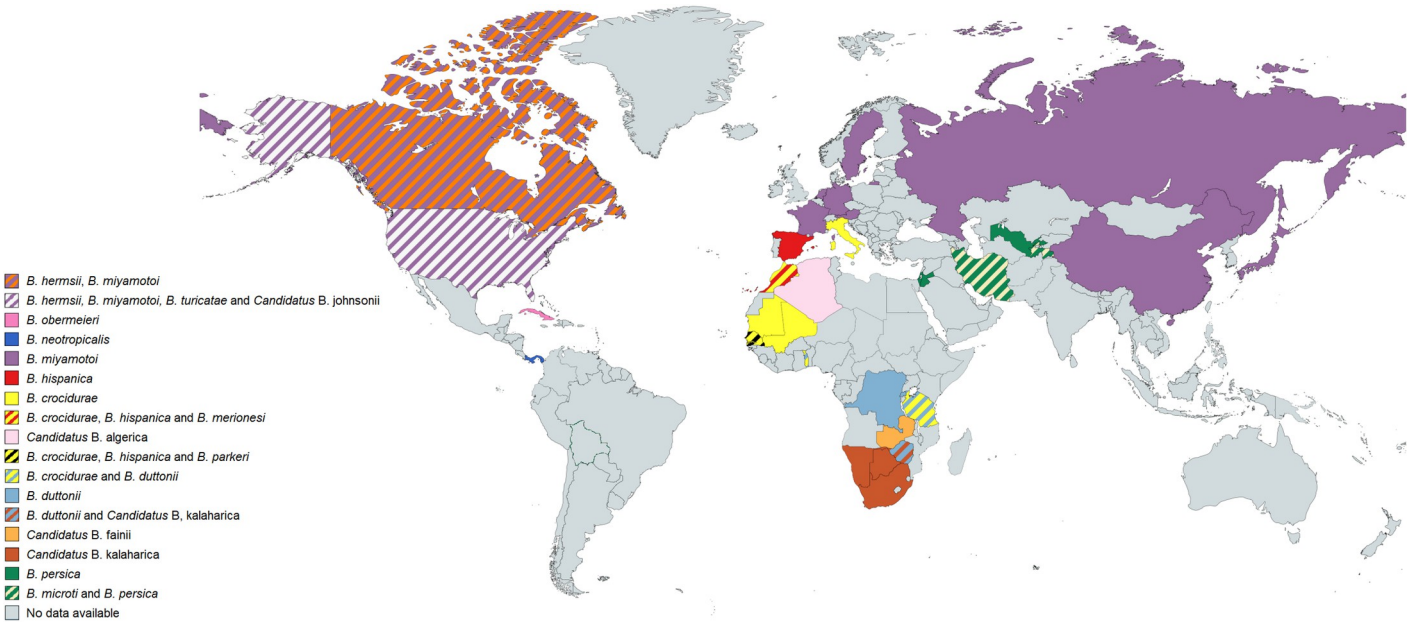


Fig 5. Reported TBRF cases by country and causative *Borrelia* species. B., *Borrelia*. Map created on www.mapchart.net.

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(based on detection in animal blood samples and/or ticks) is shown in Figs 7, 8, 9 and 10, respectively. The prevalence of competent vector ticks for TBRF *Borrelia* in America, Africa, Europe, and Asia can be found in S1 Fig.

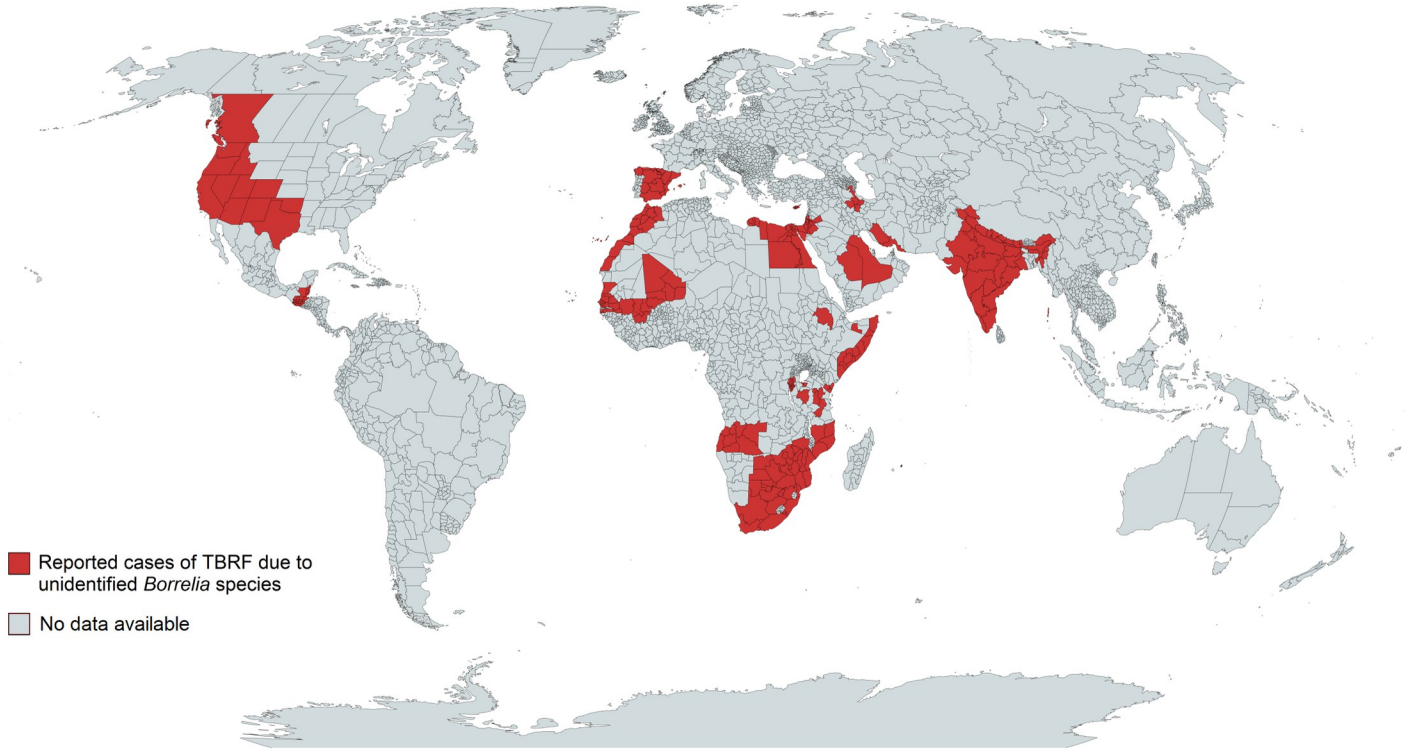


Fig 6. Reported TBRF cases caused by unidentified *Borrelia* species. TBRF, Tick borne relapsing fever. Map created on www.mapchart.net.

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Fig 7. Reported presence of TBRF *Borrelia* species in ticks and animal hosts in America. B., *Borrelia*. Map created on www.mapchart.net.

https://doi.org/10.1371/journal.pntd.0010212.g007

Known and putative TBRF spp. and their animal host(s) and transmitting tick species

124 studies reported data on *Borrelia* spp. and their associated animal hosts and transmitting ticks. Table 3 lists the known humanpathogenic TBRF *Borrelia* spp. as well as *Borrelia* spp. with yet unknown humanpathogenic potential, their known animal hosts and transmitting tick species.

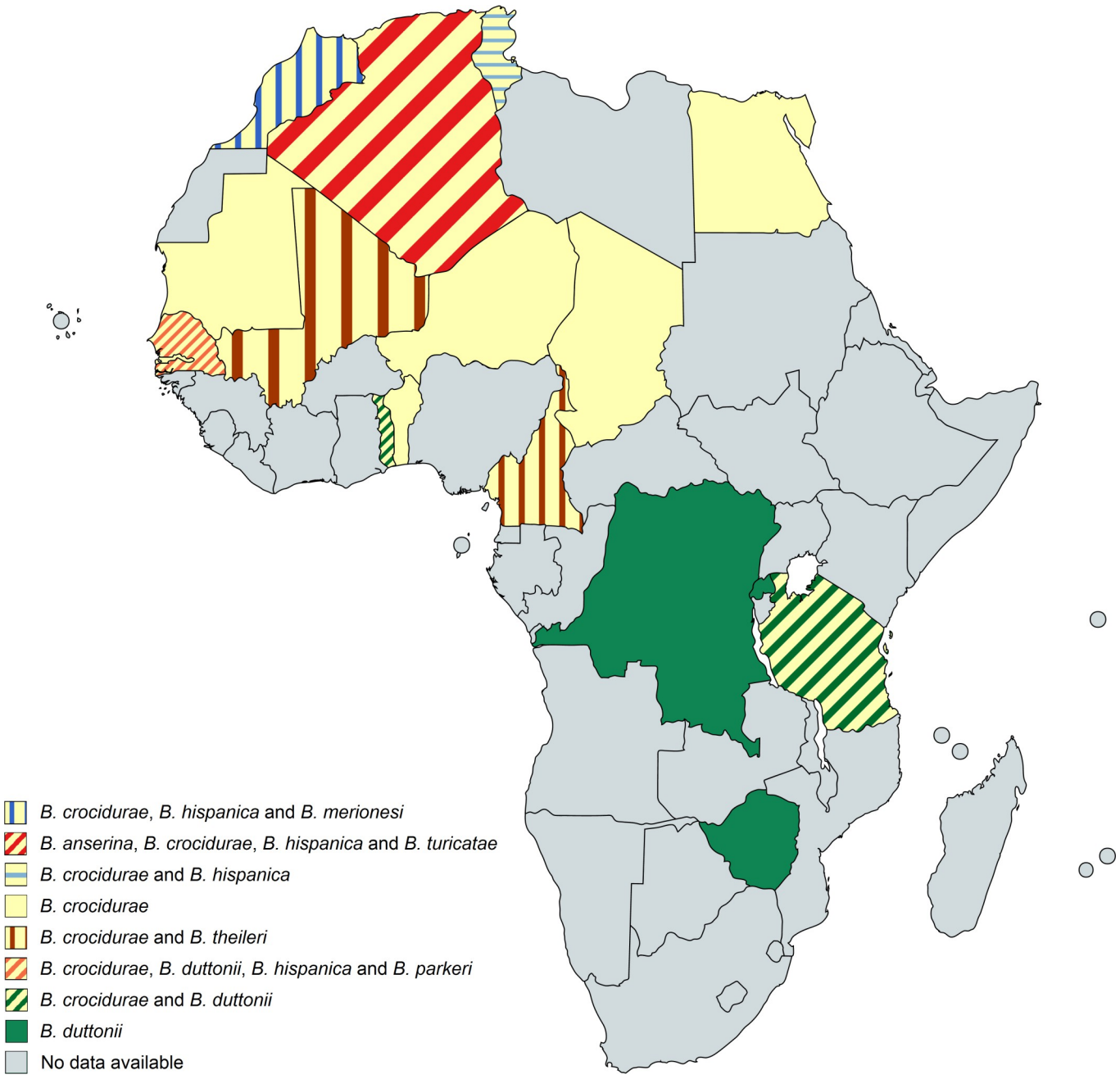


Fig 8. Reported presence of TBRF *Borrelia* species in ticks and animal hosts in Africa. B., *Borrelia*. Map created on www.mapchart.net.

https://doi.org/10.1371/journal.pntd.0010212.g008

TBRF case studies

228 (27.2%) of the 837 analysed publications reported a total of 9,372 human TBRF cases. For 5,755 cases, the patients' gender was reported: 3,164 (55.0%) were male, 2,591 (45.0%) were female. For 2,775 cases, the patient's age was reported: the median age of male patients was 32.7 years (range <1–90), the median age of female patients was 34.6 years (range <1–90). Table 4 lists the countries where the infections were acquired.

Information about travel-related TBRF cases are shown in Table 5.

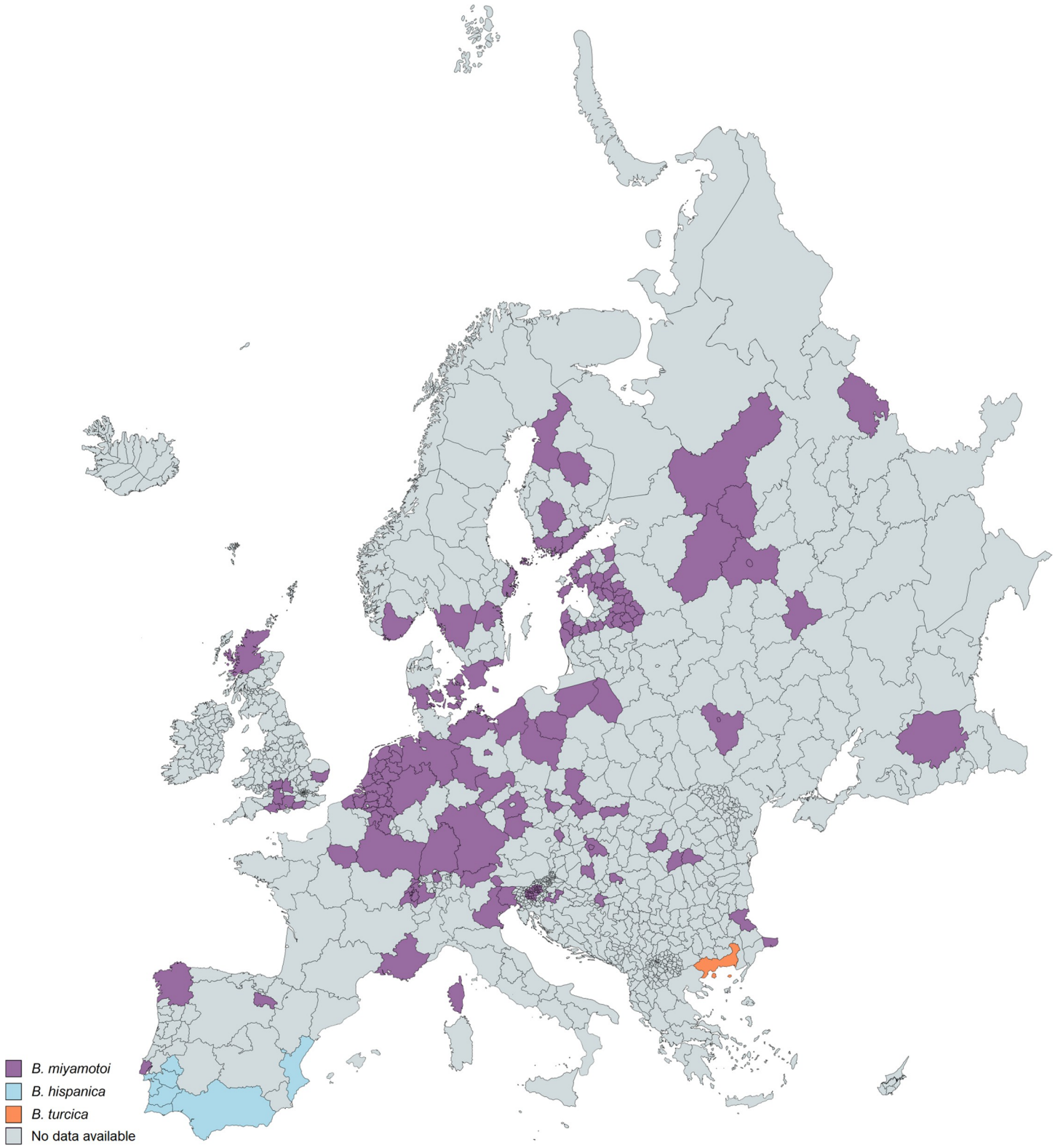


Fig 9. Reported presence of TBRF *Borrelia* species in ticks and animal hosts in Europe. B., *Borrelia*. Map created on www.mapchart.net.

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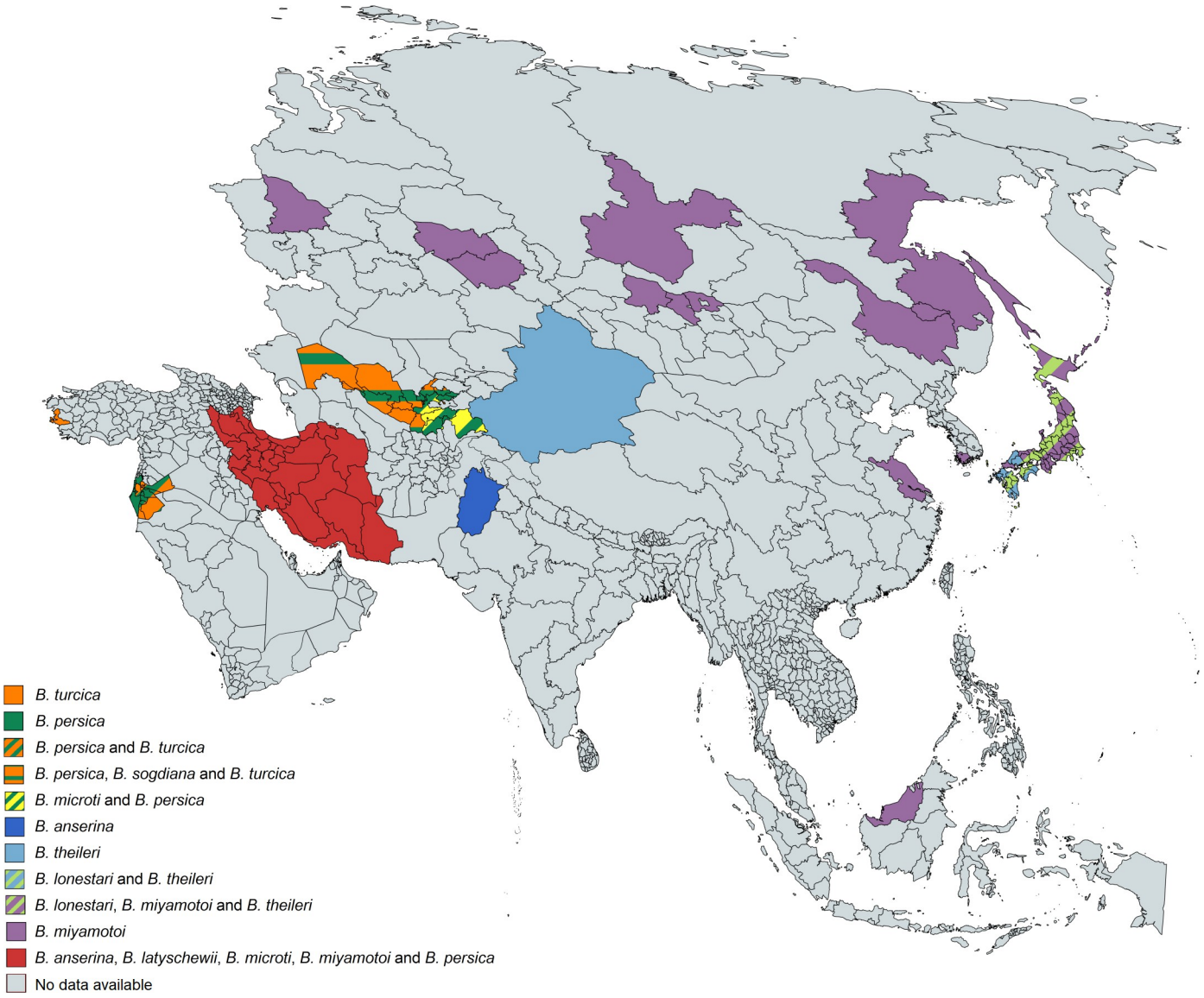


Fig 10. Reported presence of TBRF *Borrelia* species in ticks and animal hosts in Asia. B., *Borrelia*. Map created on www.mapchart.net.

https://doi.org/10.1371/journal.pntd.0010212.g010

Symptoms related to TBRF

A total of 152 publications reported specific symptoms related to TBRF. Fig 11 shows the relative frequency of these symptoms.

The number of relapsing fever episodes was reported in 67 publications (Fig 12).

Abnormal laboratory findings, were described in 65 studies (Fig 13).

Information on complications other than preterm delivery (61 cases), was available in 47 studies for 433 of the analysed 9,372 TBRF cases (Fig 14).

Diagnostic

Details on the diagnostic methods used to diagnose TBRF was available for 7,612 (81.2%) of the analysed 9,372 cases (Table 6).

Table 3. Known and putative TBRF *Borrelia* spp. and their animal host(s) and transmitting tick species.

***Borrelia* spp. causing TBRF (number of reported human cases with unequivocal species identification*)**	**Animal host(s)**	**Transmitting tick species**
***B. crocidurae* (425)**	Rodents, shrews	*O. erraticus*, *O. sonrai*
***B. duttonii* (141)**	Chicken, pigs	*O. moubata*, *O. porcinus*
***B. hermsii* (616)**	Chipmunks, deer, dogs, owls, rodents, squirrels	*O. hermsii*
***B. hispanica* (128)**	Cats, cattle, dogs, hedgehogs, pigs, rodents, sheep, warblers	*O. erraticus*, *O. marocanus*, *O. occidentalis*
B. merionesi	?	?
***B. microti* (1)**	Hedgehogs, rodents, toads	*O. erraticus*
***B. miyamotoi* (639)**	Birds, cats, cattle, deer, dogs, hedgehogs, ponies, rodents, sheep, squirrels, wild boar	*Am. americanum*, *D. reticulatus*, *D. variabilis*, *Ha. concinna*, *Ha. inermis*, *Ha. longicornis*, *Ha. punctata*, *I. dentatus*, *I. hexagonus*, *I. nipponensis*, *I. pacificus*, *I. pavlovskyi*, *I. persulcatus*, *I. ricinus*, *I. scapularis*
***B. neotropicalis* (106)**	?	?
***B. obermeieri* (1)**	?	?
B. parkeri	Horses	*O. parkeri*
***B. persica* (415)**	Camel, cat, cattle, dog, hyrax	*O. tholozani*
***B. turicatae* (4)**	Birds, coyotes, dogs, foxes, rats, tortoises	*C. capensis*, *C. kelleyi*, *O. turicata*
***Candidatus* B. algerica (1)**	?	?
***Candidatus* B. fainii (1)**	Rodents	?
***Candidatus* B. johnsonii (1)**	Bats	*C. kelleyi*
***Candidatus* B. kalaharica (2)**	?	*O. savigni*
***Borrelia* spp. with yet unknown human-pathogenic potential**	**Animal host(s)**	**Transmitting tick species**
B. anserina	Birds	*Ar. minatus*, *Ar. persicus*
B. baltazardii	?	?
B. brasiliensis	?	*O. brasiliensis*
B. caucasica	?	?
B. coriaceae	Deer	*O. coriaceus*
B. dugesii	Rats	?
B. graingeri	?	*O. graingeri*
B. latyschewii	Birds	*O. tartakovskyi*
B. lonestari	Birds, deer, dogs	*Am. americanum*, *C. capensis*
B. lonestari-like	Deer	*Ha.* spp.
B. mazzottii	Rats	*O. talaje*
B. osphepa	?	*O. spheniscus*
B. sogdiana	Rodents	*O. papillipes*
B. theileri	Bats, deer	*Rh. geigyi*
B. turcica	Birds, camels, cattle, tortoises	*Am. aureolatum*, *Am. longirostre*, *Hy. aegyptium*
B. venezuelensis	?	*O. rudis*
***Candidatus* B. mvumi**	?	*O. porcinus*
***Candidatus* B. texasensis**	Coyotes	*D. variabilis*
Unidentified *Borrelia* spp.	Bats, buffalos, cats, cattle, chipmunks, deer, dogs, lizards, penguins, rabbits, rodents, sheep, shrews, snakes, tortoises, turtles, wild boar	Multiple tick species

Am., *Amblyomma*; Ar., *Argas*; B., *Borrelia*; C., *Carios*; D., *Dermacentor*; Ha, *Haemaphysialis*; Hy., *Hyalomma*; I., *Ixodes*; O., *Ornithodoros*; Rh., *Rhipicephalus*; spp., species (plural);?, unknown.

*In total, we found 9,372 reported cases of TBRF in the literature. The table contains only the unequivocally attributable (PCR confirmed) number of cases caused by the respective *Borrelia* species.

https://doi.org/10.1371/journal.pntd.0010212.t003

Table 4. Number of publications on TBRF cases by country where the infections were most likely acquired (n = 240 studies).

Number of publications	Country where the TBRF cases reported in the publication acquired their infection (number of cases)
84	USA (1,341; 182*)
20	Senegal (229; 238*)
14	Iran (2,538), Israel (753)
13	Spain (267; 3*)
12	Tanzania (930)
7	Canada (55; 182*)
6	Morocco (131; 3*), Russia (317)
5	India (158; 1*)
4	Japan (5), Mali (3; 238*), South Africa (23; 3*), Tajikistan (2; 2*)
3	Botswana (3*), Cyprus (111), France (58), Mauritania (3; 238*), Namibia (1; 2*), Netherland (3), Rwanda (109), Uzbekistan (1; 2*), Jordan (237), Zimbabwe (14; 1*)
2	Egypt (1; 9*), Libya (4; 9*), Mexico (2), Saudi Arabia (3), Somalia (1,147)
1	Algeria (1), Angola (4), Austria (1), Belize (1*), Burundi (1), China (14), Cuba (1), Democratic Republic of the Congo (13), Ethiopia (262), Germany (1), Guatemala (1*), Italy (1), Kenya (49), Mozambique (1*), Nepal (1*), Palestine (4), Panama (106), Sweden (2), Togo (21), Zambia (1)

TBRF, tick borne relapsing fever; USA, United States of America.

* Number of additional cases which may have contracted TBRF in the respective country, but since the ill person visited additional countries within the presumed incubation period, the infection may have also been acquired elsewhere.

https://doi.org/10.1371/journal.pntd.0010212.t004

Treatment

Information on antimicrobial treatment was available for 1,274 (13.6%) of the analysed 9,372 TBRF cases. 1,238 patients received antimicrobial treatment, 36 patients received no antimicrobial treatment. Fig 15 shows the use of the different antimicrobial compounds/drugs, as reported in the studies, from 1930 until today. Detailed data on the used treatment regimens (frequency, dosage, length of treatment) can be found in S2 Table. Because of the large heterogeneity and the lack of precise data, a detailed analysis of the used treatment regimens was omitted.

JHR and outcome

Information on the occurrence of JHR was available for 1,189 (12.7%) of the 9,372 analysed TBRF cases. JHR occurred in 230 (19.3%) of antimicrobially treated patients. Data on antibiotic treatment and the occurrence/absence of JHR was reported in 65 studies (Table 7). JHR was fatal in 15 (6.5%) cases [46,133,188].

Information on the clinical outcome was available for 1,454 (15.5%) of the analysed 9,372 TBRF cases. 95 (6.5%) of the 1,454 cases were fatal [39,46,115,133–135,188–198]. 88 fatal cases were reported from Africa (Tanzania 72, Ethiopia 12, Democratic Republic of Congo 1, Egypt 1, Rwanda 1, Senegal 1), 5 from the USA, and 2 from Israel. Table 8 lists the outcome of TBRF in different patient groups.

Discussion

Publications on TBRF

Over the last 30 years, the number of published case studies on TBRF has increased significantly (Fig 4). The increasing number of publications over time might be attributable to the

Table 5. Case analysis on TBRF in travelers.

Year	No. cases	Infection acquired in	Imported to	*Borrelia* spp.	Complications	Ref.
1982	1	Namibia	South Africa	?	None reported	[177]
1985	1	Cyprus	England	?	None reported	[178]
1988	1	Israel	USA	?	JHR (n = 1)	[179]
1991	2	Senegal	Belgium	?	Meningoencephalitis (n = 1), JHR (n = 1)	[180]
1993	3	USA	Canada	*B. hermsii*	JHR (n = 1)	[67]
1995	1	Saudi Arabia	USA	?	None reported	[181]
1996	1	Nepal or India	Denmark	?	None reported	[182]
1999	2	Gambia or Senegal	Netherlands	*B. crocidurae*	Meningitis (n = 1)	[116]
1999	1	Senegal	Italy	?	None reported	[183]
2005	3	Spain or Morocco	France	*B. crocidurae*, *B. hispanica*	None reported	[140]
2006	1	Guatemala or Belize	Netherlands	?	None reported	[184]
2006	1	Senegal	Italy	*B. crocidurae*	None reported	[125]
2007	1	Mali	France	?	None reported	[185]
2008	4	Senegal	France	?	Meningoencephalitis (n = 1), JHR (n = 1)	[186]
2009	1	Senegal	France	*B. crocidurae*	None reported	[120]
2010	1	Senegal	Belgium	*B. crocidurae*	Meningoencephalitis (n = 1), JHR (n = 1)	[112]
2010	1	Uzbekistan	Japan	*B. persica*	None reported	[164]
2011	1	Uzbekistan or Tajikistan	France	*B. persica*	None reported	[175]
2009–2011	4	Senegal	France	*B. crocidurae*	Encephalitis (n = 2), meningitis (n = 3)	[119]
2015	1	Southern Africa	Germany	*Candidatus* B. kalaharica	None reported	[109]
2016	1	Southern Africa	Germany	*Candidatus* B. kalaharica	JHR (n = 1)	[110]
2017	1	Morocco	Belgium	*B. hispanica*	None reported	[137]
2017	1	USA	Japan	*B. miyamotoi*	None reported	[95]
2018	1	Senegal	France	*B. crocidurae*	None reported	[118]
2019	1	Botswana or South Africa	Netherlands	?	None reported	[187]
2019	1	Tajikistan	Switzerland	*B. persica*	JHR (n = 1)	[174]
2020	1	Jordan	USA	*B. persica*	None reported	[170]
2020	2	Mali	France	*B. crocidurae*	None reported	[117]
2020	1	Tajikistan	Italy	*B. microti*	None reported	[155]

B., *Borrelia*; JHR, Jarisch-Herxheimer reaction; Ref., reference; spp., species (plural); USA, United States of America; ?, unknown.

https://doi.org/10.1371/journal.pntd.0010212.t005

advent of molecular diagnostic tools as well as the increasing awareness and recognition of the disease, which previously was very likely underdiagnosed.

Epidemiology

With the advent of molecular diagnostic techniques and the resulting identification of multiple new TBRF causing *Borrelia* spp., as well as of previously unknown epidemic areas, the historic classification of TBRF *Borrelia* into Old World and New World TBRF *Borrelia* spp. has been replaced by a more complex picture. This is particularly evident when comparing the map outlining the TBRF endemic areas as assumed by Felsenfeld five decades ago (Fig 1) with maps compiled from confirmed data (Figs 5 and 6). Interestingly, some regions of the world, like e.g. Europe or Japan, emerged as endemic regions. Contrastingly, other regions like South America or central Africa were apparently considered more endemic in the past than they actually are (Figs 5 and 6). However, rating the relevance of TBRF in a particular region of the world demands not only to integrate the mere presence of TBRF *Borrelia* and the occurrence of human cases but also needs to take into account differences in available diagnostic capacities

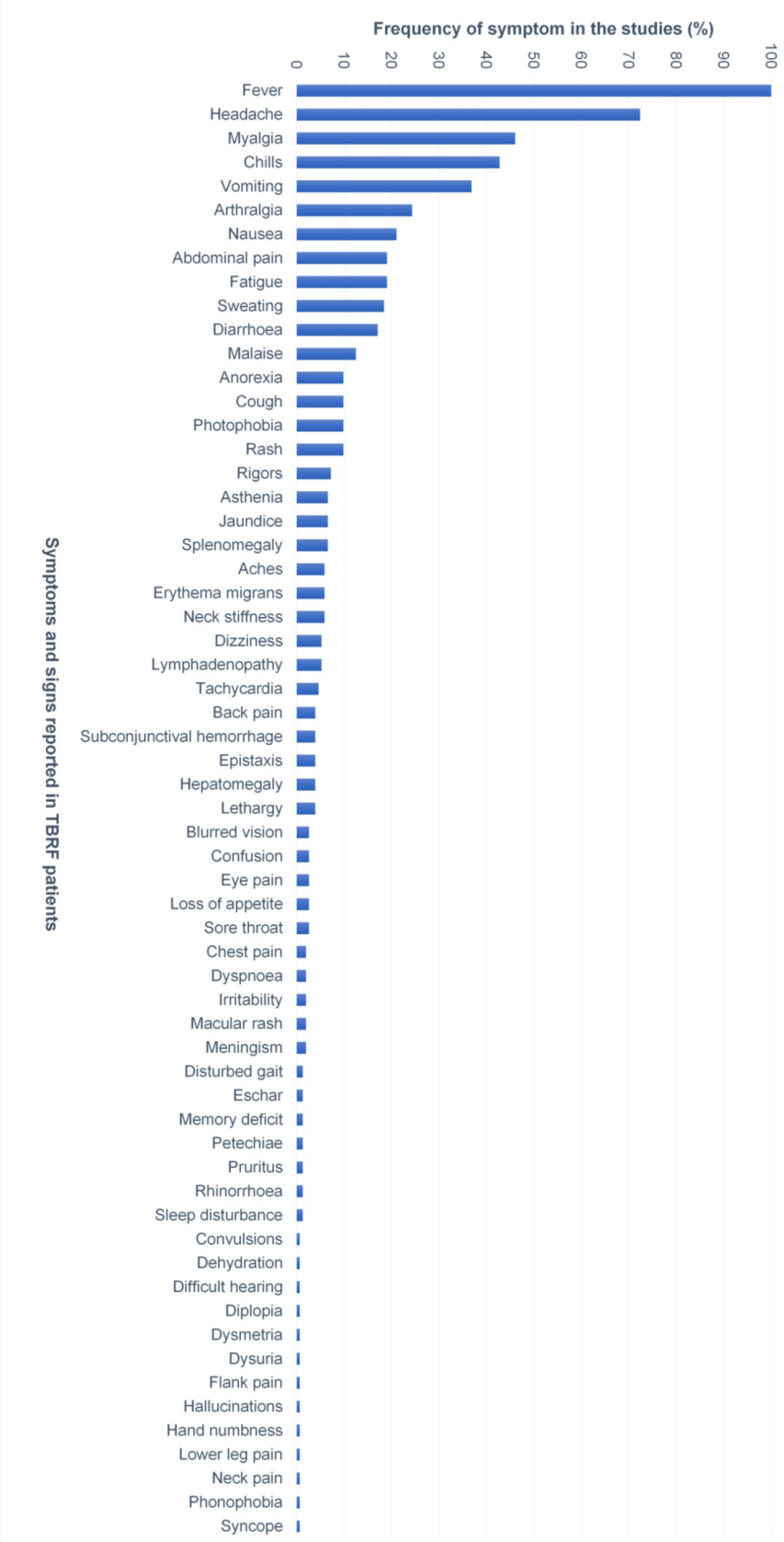

Frequency of symptom in the studies (%)
0
10
20
30
40
50
60
70
80
90
100
Symptoms and signs reported in TBRF patients
Fever
Headache
Myalgia
Chills
Vomiting
Arthralgia
Nausea
Abdominal pain
Fatigue
Sweating
Diarrhoea
Malaise
Anorexia
Cough
Photophobia
Rash
Rigors
Asthenia
Jaundice
Splenomegaly
Aches
Erythema migrans
Neck stiffness
Dizziness
Lymphadenopathy
Tachycardia
Back pain
Subconjunctival hemorrhage
Epistaxis
Hepatomegaly
Lethargy
Blurred vision
Confusion
Eye pain
Loss of appetite
Sore throat
Chest pain
Dyspnoea
Irritability
Macular rash
Meningism
Disturbed gait
Eschar
Memory deficit
Petechiae
Pruritus
Rhinorrhoea
Sleep disturbance
Convulsions
Dehydration
Difficult hearing
Diplopia
Dysmetria
Dysuria
Flank pain
Hallucinations
Hand numbness
Lower leg pain
Neck pain
Phonophobia
Syncope

Fig 11. Relative frequency of signs and symptoms (in %) related to TBRF (n = 152 studies). TBRF, tick borne relapsing fever.

https://doi.org/10.1371/journal.pntd.0010212.g011

as well as a reporting bias accruing from differences in academic publishing traditions (Table 4). It is to be expected that in the future additional endemic areas, additional *Candidatus*- as well as proven TBRF-*Borrelia* spp., additional animal host reservoirs, and additional transmitting tick species will emerge (Table 3) and further expand the granularity of our picture of TBRF.

The rather recent discovery of *B. miyamotoi*'s wide geographic distribution (Figs 5, 7, 9 and 10), as well the surprising finding that this *Borrelia* sp. is not only transmitted by various soft tick species [156,199–202] but also by hard ticks (*Ixodes* spp. [152,203]), which were previously considered non-vector-competent for TBRF *Borrelia*, serve as good example. The demand for an adequate ecological niche serving not only a specific animal host but also the transmitting tick species may be the main reason why TBRF is restricted to certain geographic areas. There are two ways for TBRF *Borrelia* spp. to spread geographically: either the infected host animals move into new areas with locally prevalent vector competent tick species or infected ticks, attached to overland migrating host animals or to migratory birds, translocate into new areas with suitable animal hosts and a suitable habitat allowing the ticks survival. Its potential to infect a broad range of host animals, and especially migratory birds, may be the reason why *B. miyamotoi* shows a wide geographic distribution all around the Northern hemisphere (Figs 5, 7, 9 and 10). Since migration by birds demands prolonged attachment of the vector tick species, the unusual vector competence of

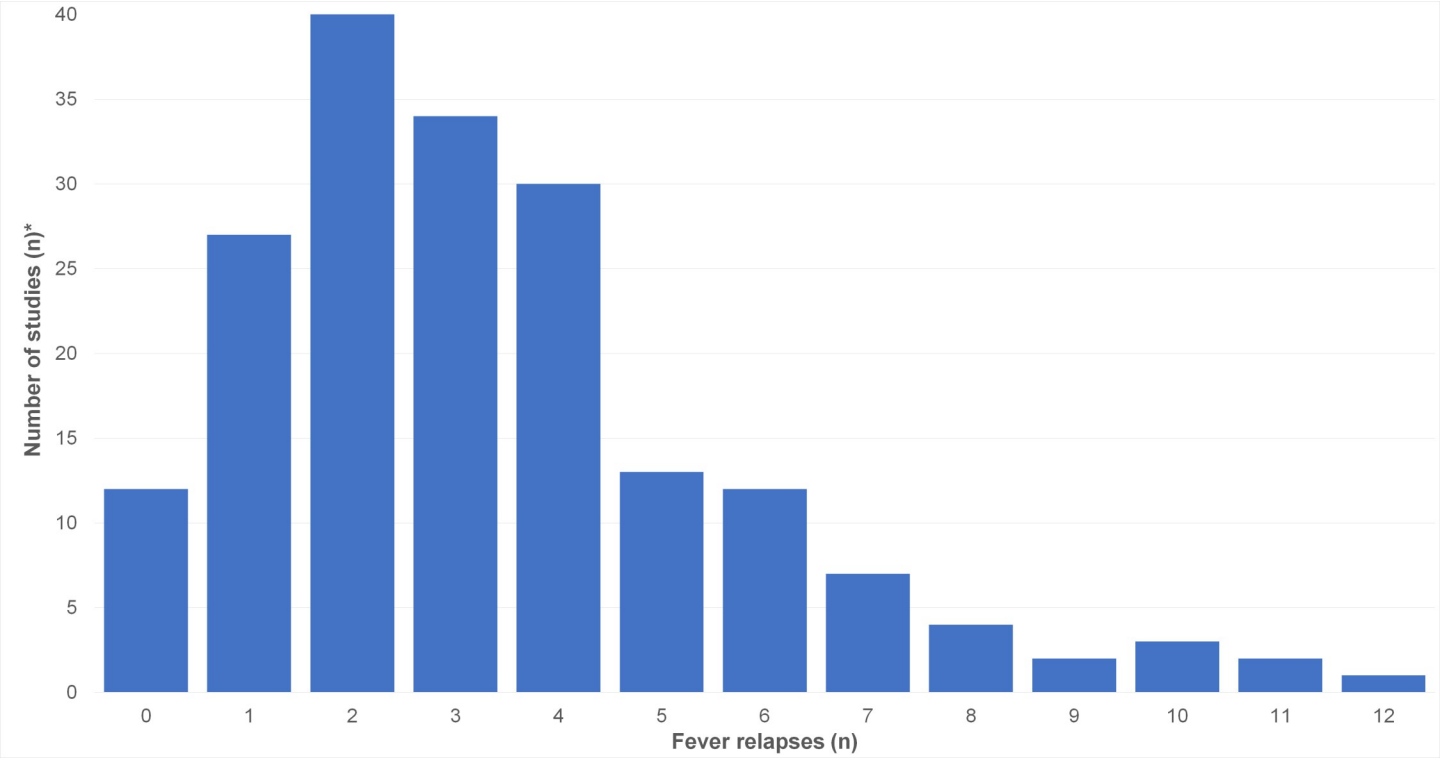


Fig 12. Number of relapsing fever episodes in studies on TBRF (n = 67 studies). * Note: Since the number of relapsing fever episodes within single studies was mostly reported as median, an evaluation per case was not possible.

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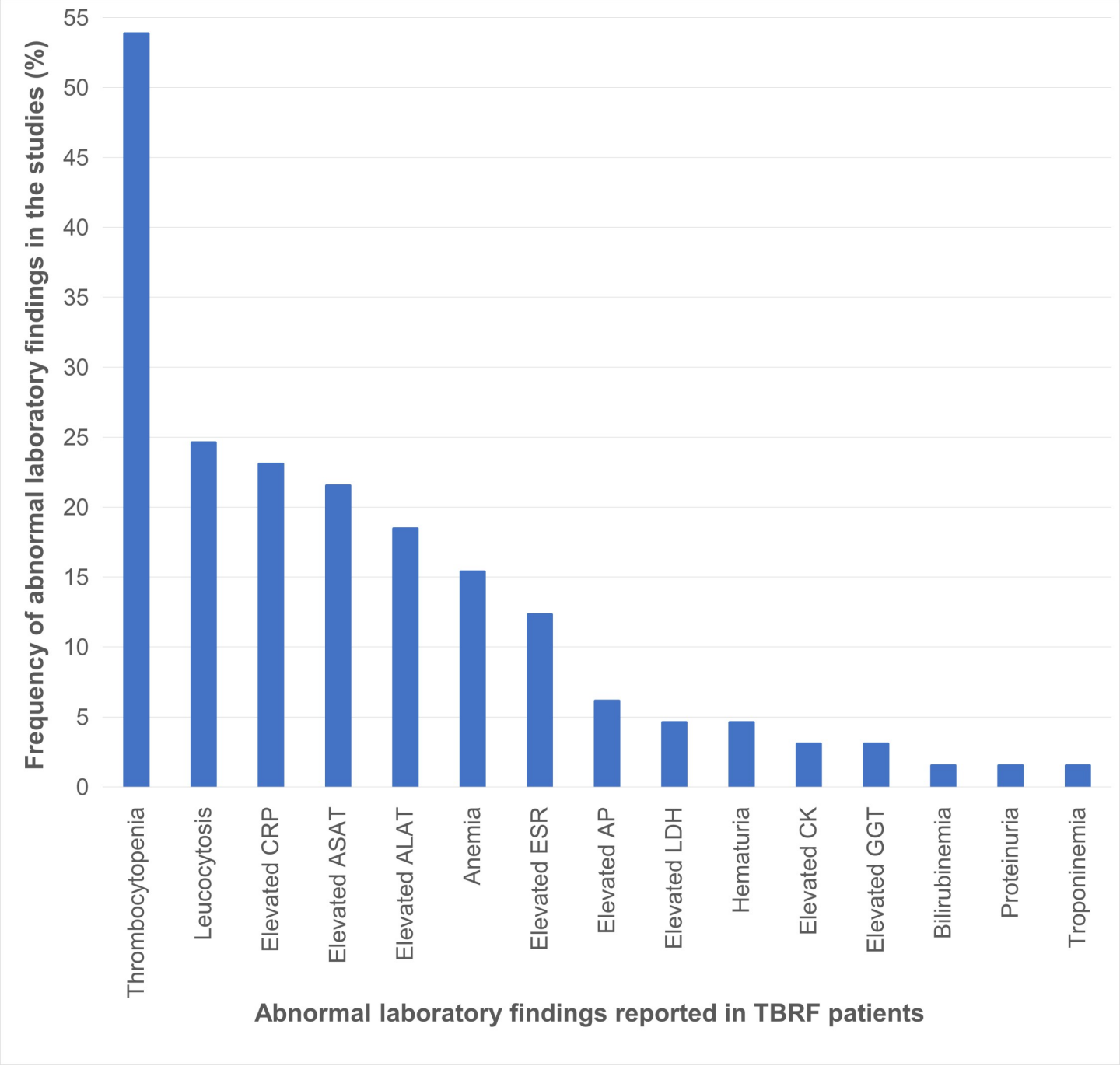


Fig 13. Abnormal laboratory findings related to TBRF (n = 65 studies). ALAT, alanine aminotransferase; AP, alkaline phosphatase; ASAT, aspartate transaminase; CK, creatine kinase; CRP, C-reactive protein; ESR, erythrocyte sedimentation rate; GGT, gamma-glutamyltransferase; LDH, lactate dehydrogenase; TBRF, tick borne relapsing fever.

https://doi.org/10.1371/journal.pntd.0010212.g013

hard ticks may also play a pivotal role in the extend geographic distribution of *B. miyamotoi*. Unlike the rather short attachment and feeding time of soft ticks, the prolonged attachment and feeding of hard ticks would be favourable for migration over large distances [1,204]. In contrast, TBRF *Borrelia* spp. with a narrow and spatially limited host range are rather

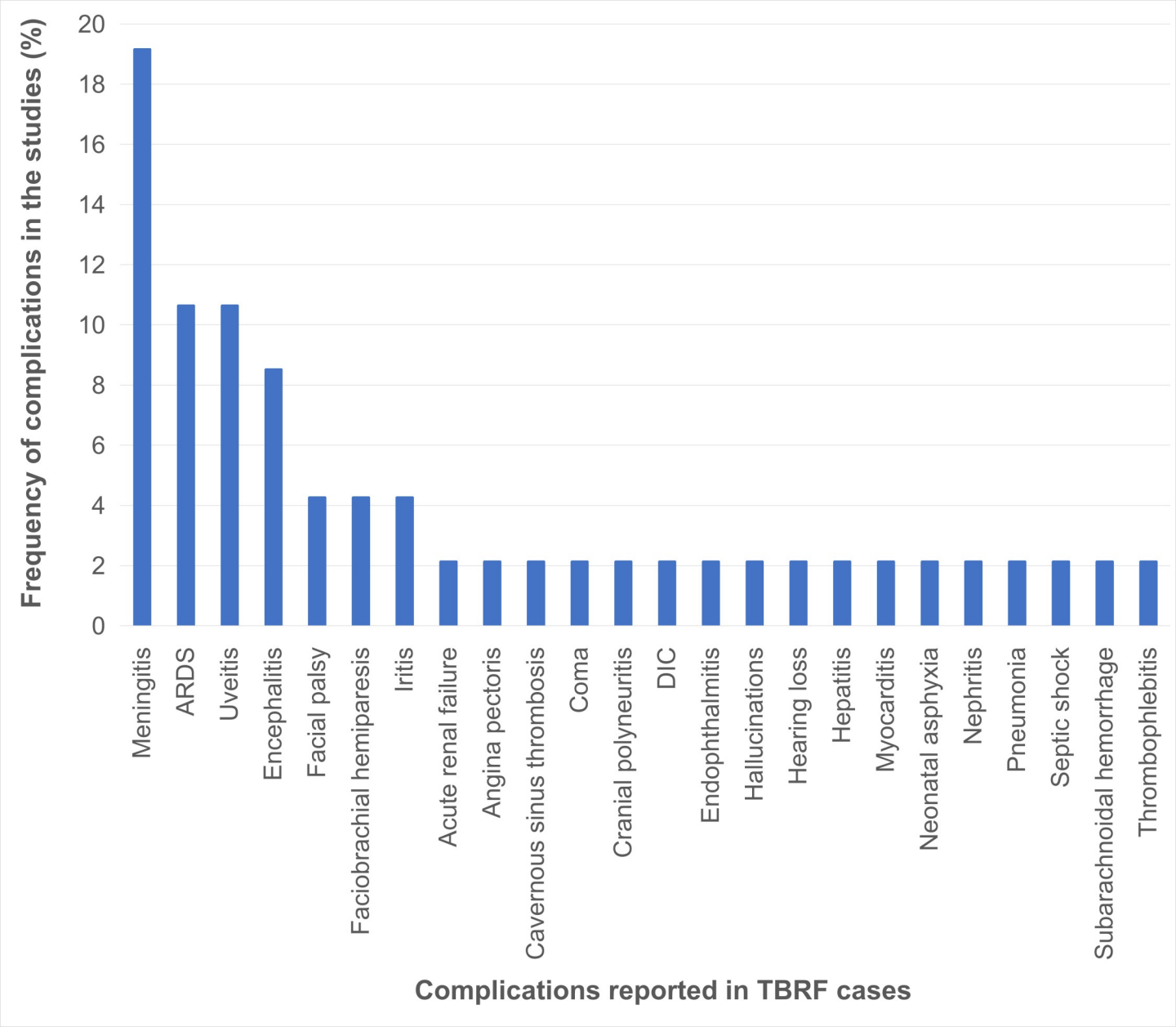


Fig 14. Complications of TBRF (n = 47 studies). ARDS, acute respiratory distress syndrome; DIC, disseminated intravascular coagulation; TBRF, tick borne relapsing fever.

https://doi.org/10.1371/journal.pntd.0010212.g014

unlikely to extend their geographic range, as for instance in the case of *B. coriaceae*. With deer being the only associated animal host, this *Borrelia* species is to date only reported in the Western part of the USA (California, Nevada, Oregon). Whether and to what extend climatic changes will influence the epidemiology and possible spread of TBRF in the future remains to be seen [205,206]. Reports of TBRF in international travelers are rare. To date, only 42 cases have been published (Table 5). Although underdiagnosing and underreporting is likely, existing surveillance data on infectious diseases in travelers confirm the apparently overall low exposure risk and the rare occurrence of TBRF in travelers [207].

Table 6. Diagnostic methods used to diagnose TBRF in 7,612 cases (n = 240 studies).

Diagnostic method	Grade of diagnostic certainty	Number of cases in which this diagnostic method was applied	Number of cases diagnosed only by this method	Number of cases diagnosed by a combination of diagnostic methods	Number of cases where this method was the method with the highest grade of diagnostic certainty
PCR	A	3,443	2,051	1,392	3,443
Microscopy	B	5,159	2,732	2,427	3,792
Culture	B	129	0	129	0
Animal inoculation	B	756	0	756	0
Serology	C	1,139	377	762	377

PCR, polymerase chain reaction.

Note: in 2,452 (32%) of the 7,612 cases a combination of diagnostic tests was used to establish the diagnosis. Thus, the number of tests exceeds the number of cases.

https://doi.org/10.1371/journal.pntd.0010212.t006

Signs, symptoms and complications

Presenting similar to a multitude of other febrile infectious diseases without specific signs and symptoms (Fig 11), TBRF may easily be missed or misdiagnosed. Furthermore, the under- or misdiagnosis of TBRF may also be caused by its mostly benign and, even without antimicrobial treatment, self-limiting course. The sensitivity of TBRF *Borrelia* to standard antibiotics widely available and used for empirically treatment may also contribute to underdiagnosing.

Even in endemic regions, cases presenting without the diagnosis-suggestive relapsing fever episodes or without complications leading to a thorough diagnostic work-up, are likely to be

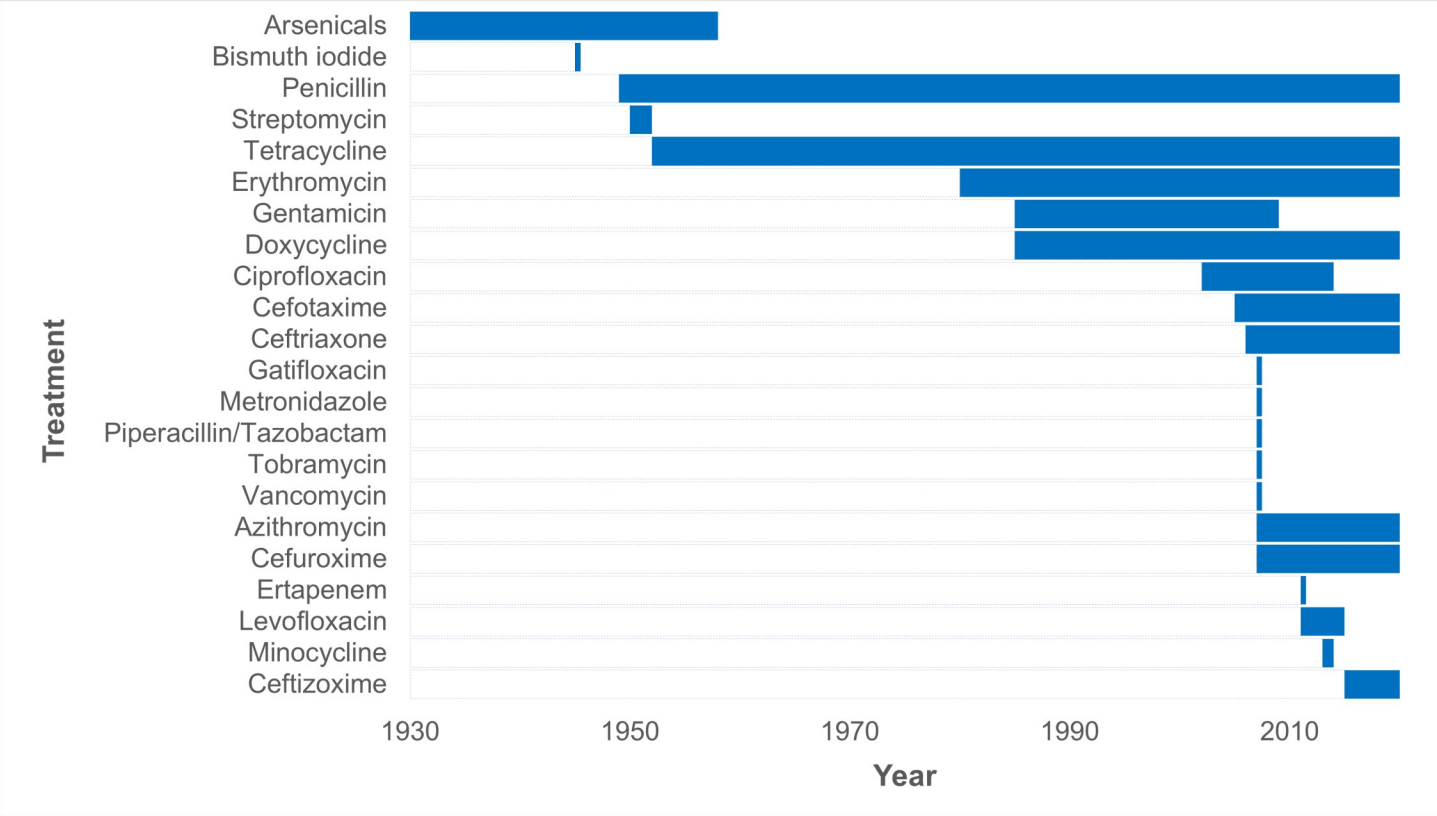


Fig 15. Use of different antimicrobial compounds/drugs to treat TBRF as reported from 1930 until today (n = 172 studies).

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Table 7. Treatment specific frequency of JHR in TBRF (n = 65 studies).

Antimicrobial treatment regimen	Number of patients with reported treatment regimen and reported occurrence/ absence of JHR (N)	Number of reported JHR (N)	Frequency of JHR (%)
Tetracyclines	116	28	24.1
Doxycycline	*83*	*18*	*21.7*
Tetracycline	*33*	*10*	*30.3*
β-lactams	26	4	15.4
Penicilline	*12*	*1*	*8.3*
Ceftriaxon	*12*	*2*	*16.7*
Cefuroxim	*2*	*1*	*50.0*
Erythromycin	13	4	30.8

JHR, Jarisch-Herxheimer-reaction.

https://doi.org/10.1371/journal.pntd.0010212.t007

missed. The most helpful symptom to differentiate TBRF from other febrile illnesses are the recurrent fever episodes, which are found in the majority of cases (Fig 12). When looking at Fig 12 it needs to be emphasized that the presented data do not reflect the natural course of TBRF as almost all reported and analysed cases received antimicrobial treatment. Thus, the number of fever relapses in untreated TBRF is very likely higher. In addition, it must be kept in mind that the number of fever relapses indirectly reflects the time elapsing before a patient seeks medical help or has access to medical care. This may differ widely across countries and health care systems. Likewise, it is very likely that in endemic areas physicians familiar with the disease will suspect, diagnose and treat TBRF much earlier compared to their colleagues in non-endemic areas.

To date, there is largely insufficient data to compare different TBRF *Borrelia* species with regard to possible differences in their clinical manifestations. We performed a subgroup analysis regarding the signs and symptoms, complications and number of fever relapses reported in TBRF caused by *B. hermsii* and *B. crocidurae* as for these the most data were available (S2 Fig). However, due to the low number of cases as well as the inhomogeneity of the available data, conclusions regarding possible differences in clinical manifestations of different TBRF *Borrelia* species remain difficult.

The only exception may be the reported rates of neurologic complications: in contrast to LBRF, neurological complications are common in TBRF [26], reported to occur in 10–40% of

Table 8. Case fatality analysis of TBRF (n = 17 studies).

Patient group	Number of fatal cases (%)	Number of documented fatal cases among cases with documented antimicrobial treatment (N = 629)	Number of documented fatal cases among cases with unknown antimicrobial treatment status (N = 816)	Number of documented fatal cases among documented untreated cases (N = 9)
All cases of TBRF not infected during pregnancy and not infected in the fetal or peripartal period (N = 992)	43 (4.3%)	38	5	0
Pregnant women (N = 231)	11 (4.8%)	0	11	0
Fetuses and neonates (N = 231)				0
Intrauterine death	11 (4.8%)	2	9	0
Postpartal death	30 (13.0%)	12	18	0
Total	95 (6.5%)	52	43	0

CFR, case fatality rate.

https://doi.org/10.1371/journal.pntd.0010212.t008

the cases [33,208]. Our analysis confirms the prominence of central nervous system (CNS) involvement among the reported complications of TBRF (Fig 14). A comprehensive review on neurologic and ophthalmologic involvement and complications in TBRF has been published by Cadavid and Barbour in 1998 [33]. Generalized neurologic symptoms like dizziness, apathy or delirium are considered attributable to spirochetemia and high fever rather than to direct invasion of the CNS by *Borrelia* and were reported in LBRF as well as TBRF. Neuropsychiatric abnormalities not solely attributable to high fevers have been reported for both TBRF and LBRF and cases of encephalitis or encephalopathy are occasionally observed in TBRF and LBRF [33]. Neurologic complications differ in their frequency and pathogenesis among the two diseases. While neurologic complications in LBRF are rare and primarily attributed to CNS hemorrhage and not to direct invasion of the CNS by the pathogen, neurologic complications in TBRF are frequently observed and attributed to direct invasion of the CNS invasion by *Borrelia* [33]. Neurologic manifestations are more likely to present during subsequent, rather than the initial febrile period [33]. A frequently reported complication in TBRF (and not reported in LBRF) is cranial neuritis, most often presenting in the form of facial palsy. Its frequency varies with different TBRF *Borrelia* species between 3% (7/230) of *B. hispanica*-related cases to 38% (8/21) of *B. turicatae*-related cases [33]. However, as mentioned above, data are limited, inhomogeneous, and come from times when molecular species identification was not possible. Nevertheless, data from animal studies also suggest that the different TBRF *Borrelia* species are neuroinvasive to varying degrees [33].

Reports of *B. miyamotoi* associated meningoencephalitis in immunocompromised patients suggests that the pathogens may behave like an opportunistic pathogen in this population [13,15,153]. Whether and to what extend *B. miyamotoi* may also cause neurological complications in non-immunocompromised patients is currently unknown.

Limited data suggest possible differences regarding the clinical presentation of soft tick-borne RF and hard tick-borne/*B. miyamotoi* RF ("Cases with the characteristic recurring febrile episodes interspersed with non-febrile intervals that typify classical RF have only been described sporadically [in *B. miyamotoi* RF]. . . . Furthermore, unlike RF spirochetes, epistaxis, abortion, jaundice and major organ failure have not appeared as features of *B. miyamotoi* infection" [209]), but the currently available data is limited and has yet to prove itself. Pooled clinical data, like presented in Figs 11–14, may thus not necessarily reflect the true picture on species level, but as mentioned above, the overall low number of reported cases and the inhomogeneity of the available data do not allow for robust subgroup analyses. Ocular involvement in TBRF includes iritis, cyclitis, choroiditis, and optic neuritis (Figs 11 and 14). When eye involvement is reported in TBRF, it is bilateral in one-third of the cases and almost always occurs after the third or fourth febrile episode. Involvement of the eyes during LBRF has not been reported [33].

The historical statement that the occurrence of vomiting in TBRF is exclusively related to meningitis [26] cannot be confirmed, as overall, gastrointestinal symptoms are quite common in TBRF (Fig 11).

Interestingly, Erythema migrans, a symptom highly specific for Lyme disease, has been reported in some cases of TBRF (Fig 11). Thus, it may be speculated that coinfections with other locally endemic tick-borne pathogens could lead to overlapping presentations making it difficult to attribute signs and symptoms to a specific pathogen. This speculation is strongly supported by the fact that all reported TBRF cases presenting Erythema migrans were reported from Russia, the Netherlands, and Japan and caused by *B. miyamotoi*, the only TBRF *Borrelia* transmitted by hard ticks and thus, by ticks plausibly capable of co-transmitting Lyme *Borrelia*. In analogy, the report of an eschar (Fig 11), a symptom primarily associated with rickettsial infections, suggests coinfection.

The frequency and relevance of such coinfections remains unclear. An outbreak of a febrile illness in West Texas was initially wrongly attributed to Lyme disease, based on a combination of facial palsy and a positive *B. burgdorferi* serology in some of the cases. However, in the end the disease was identified as TBRF due to *B. turicatae* and the serological results recognized as cross-reactivity [210].

Bleeding signs like petechiae, epistaxis, subconjunctival hemorrhage or hemorrhagic complications like subarachnoidal hemorrhage or disseminated intravascular coagulation (DIC, Fig 14) are only rarely reported in TBRF (Fig 11) when compared to LBRF, where subconjunctival hemorrhages and epistaxis are common (25%) and severe hemorrhage (hemoptysis, gastrointestinal bleeding, retinal hemorrhages) and DIC (leading to intracranial, massive gastrointestinal, pulmonary or peripartum hemorrhage) are feared complications [211].

Laboratory findings

Like the signs and symptoms, the laboratory findings are rather unspecific. Only thrombocytopenia is a feature present with a rather high frequency, possibly helping to support the tentative diagnosis (Fig 13). However, fever and thrombocytopenia occur in a variety of infections, including malaria, a broad range of common viral infections and notably also in many other tick-borne diseases (e.g., rickettsioses, ehrlichiosis, anaplasmosis, tularemia, Q fever, babesiosis, arboviral infections).

Diagnostic

To date, microscopy of thin and thick blood smears remains the most frequently reported diagnostic method for diagnosing TBRF (Table 6). Regarding the sensitivity of microscopy, it is important to remember that the positivity thresholds of thin and thick smear preparations are estimated at 10^5 and 10^4 spirochetes per mL of blood, respectively [28] and that the number of *Borrelia* in the peripheral blood is considered to be lower in TBRF compared to LBRF (an observation repeatedly quoted, but for which clear evidence is missing) [3]. To improve the sensitivity of microscopy using equipment that is easily available in small health centers, a method based on enrichment of bacteria by centrifugation followed by Giemsa staining was developed. This methods reduces the detection level to fewer than 10 spirochetes per mL of blood [27]. RF *Borrelia* are not infrequently detected in blood smears ordered because of the clinical suspicion of malaria. With the trend to progressively replace microscopy with rapid diagnostic tests (RDTs) to diagnose malaria, the incidental finding of RF *Borrelia* in malaria smears will become less, potentially further increasing the underdiagnosing of this disease.

PCR has grown in importance and is now the second most frequently reported diagnostic method. PCR is the most sensitive and specific diagnostic method available and the only diagnostic method to definitively differentiate between TBRF and LBRF (although, in most cases of microscopically detected RF *Borrelia* the epidemiological circumstances will allow to conclude whether TBRF or LBRF is the more likely diagnosis [3]) and to differentiate the different TBRF *Borrelia* species. However, because *Borrelia* are highly related at the molecular level (16S rRNA gene sequence variability $\leq$1%), the development of discrimination PCR assays is challenging and they will not always be able to provide species identification [128,182,212,213]. For instance, for *B. duttonii* and *B. recurrentis*, which are genetically and genomically very closely related, even PCR assays fail to provide species discrimination [214]. Nevertheless, several studies have been conducted using multiplex real-time PCR assays allowing the detection and speciation of several RF *Borrelia* (*B. crocidurae*, *B. duttonii/B. recurrentis*, *B. hispanica*) found in Africa [215]. Although the successful introduction of PCR as point-of-care routine

diagnostic in rural Senegal has been reported [114], the availability of PCR still remains largely restricted in resource poor settings.

Serology plays no relevant role in diagnosing TBRF for several reasons. Within endemic areas, seroprevalence is high, which may not necessarily reflect acute infection but previous infection (seroscars). Furthermore, the time to seroconversion is too long to influence treatment decisions in acutely ill patients. Additionally, as mentioned above, cross-reactivity of assays may confuse TBRF borrelioses and Lyme borreliosis [216]. The latter issue can be circumvented by using an assay detecting antibodies to the GlpQ protein, which is produced by RF *Borrelia* species, but not by Lyme *Borrelia* species [217].

Culture of TBRF *Borrelia* is restricted to very few laboratories in the world, has primarily been used in the context of research and has no role in routine diagnostic.

Treatment

Over time, many antibiotic compounds have been evaluated for the treatment of TBRF (Fig 15). However, as with LBRF [218], neither well-designed studies to determine the best treatment regimens nor comparative studies of the efficacy of different antimicrobial agents are available. Data evaluating putative differences between different TBRF *Borrelia* species regarding antimicrobial susceptibility or treatment response are scarce or non-existing.

Due to their successful use in patients with syphilis, the arsenic compounds arsphenamine (salvarsan; the first marketed antibiotic which cured a bacterial infection [219]) and its less toxic derivative neoarsphenamine (neosalvarsan) developed by Ehrlich and Hata [220] were the first antimicrobial compounds used to treat relapsing fever *Borrelia* infections.

In the 1940s, the considerably less toxic and more effective penicillin became available and replaced the arsenical compounds for the treatment of the spirochetal infections syphilis and RF. It quickly became apparent that unlike LBRF, for which a single administration of intramuscular procaine penicillin proved highly effective [218], TBRF required prolonged and sufficiently high-dose penicillin treatments to prevent relapse and achieve cure [221–224]. In this regard, and from the pathogen's neurotropic persistence demonstrated in animal models, an early analogy between TBRF and syphilis was drawn [225]. This analogy, as well as the marked differences in the treatment response of LBRF and TBRF, strengthened the suggestion that sufficient antibiotic target levels in the CNS are critical to successfully treat TBRF. Due to the lack of emerging resistance in spirochetal infections, penicillin remains an option for these infections up until today. However, the often restricted availability of procaine penicillin for intramuscular injection and the need to dose intravenously administered penicillin several times per day to achieve sufficient blood and tissue levels restricts the drug's use in clinical practice. Today, the use of β-lactams is mostly restricted to the treatment of TBRF patients with CNS involvement, similar to early CNS involvement in Lyme disease or neurosyphilis, and ceftriaxone (2g once daily for 10–14 days) is preferred over penicillin in these cases [226]. Of note, in vitro data suggesting resistance of *B. miyamotoi* to amoxicillin but susceptibility to ceftriaxone have been reported [227]. However, the validity and generalizability of these findings is called into question by the successful treatment of a case of *B. miyamotoi* TBRF with amoxicillin (and sultamicillin) [158].

In the 1950s, tetracycline was introduced and became the drug of choice for oral treatment of uncomplicated TBRF. Similar to β-lactams, a correlation between administered dose and length of treatment and relapse rate/treatment success exists [102,228]. In the absence of CNS involvement, oral or parenteral treatment with a tetracycline (tetracycline 500mg every 6 hours for 10 days [226] or doxycycline 100mg every 12 hours for 7–10 days [229]) is the recommended treatment for adults. While tetracycline remains contraindicated in children due

to the risk of irreversible dental staining, the administration of doxycycline is considered safe for up to 21 days regardless of age [229–231]. The recommended pediatric dose of doxycycline is 4.4mg/kg body weight/day divided in 2 doses (max. 200mg/day) [229]. Several studies have evaluated antibiotic postexposure prophylaxis/preemptive therapy with doxycycline to prevent TBRF following tick bites within endemic areas. Studies on preemptive therapy with a short course of doxycycline (day 1: 200mg/d, day 2–5: 100mg/d) were found to be highly effective in this regard [40,232,233]. A more recent study suggests, that even a single dose of doxycycline is sufficient and as effective [234].

For patients unable to take a β-lactam or a tetracycline, erythromycin (500mg or 12.5 mg/kg body weight every 6 hours for 7–10 days) is the most widely recommended alternative [226,229]. It is likely that the better tolerated azithromycin is equally effective, but dosing data are lacking [227].

Given the lack of comparative studies on antimicrobial treatment regimens of TBRF, well-designed studies evaluating different therapeutic regimens in the different TBRF species would be desirable.

JHR, outcome

The pathogenesis and frequency of JHR in spirochete infections has repeatedly been reviewed by several authors [42,235,236]. The reported frequency of JHR in spirochete infections varies widely (Lyme disease: 5–30%, syphilis: 8–75%, leptospirosis: 9–83%, LBRF: 0–100%, TBRF: 1–39%) [42]. However, due to the lack of a uniform definition and a standardised assessment of JHR, reliable data on the true incidence and possible differences in incidence of JHR in spirochete infections remain missing [4]. Thus, a critical appraisal of the incidence of JHR in TBRF is difficult. Nevertheless, compared to LBRF, where we found a JHR incidence rate of 55.8% [4], we found a considerably lower JHR incidence rate of 19.3% in TBRF. Considering the proposed underlying pathomechanisms, the lower incidence of JHR in TBRF may primarily be attributable to the overall lower number of *Borrelia* in the peripheral blood compared to LBRF [3,236]. Experimental animal data suggest that TBRF Borrelia species can differ in their degree of spirochaemia. This suggests that there may also be a species-specific risk of JHR. Unfortunately, the existing data are not sufficient to confirm or refute such an assumption.

The choice of antibiotics used for the treatment of spirochetal infections is considered to affect the incidence and severity of JHR, although studies in this regard provide conflicting views [237–239]. In their systematic review and meta-analysis comparing different antibiotic regimens in LBRF, Guerrier and Doherty found a benefit in favour of penicillin when comparing the rate of JHR (in 3/5 eligible studies) and concluded that treatment with a tetracycline appears to be associated with a higher rate of JHR [218]. Our analysis suggests that this may also be true in TBRF, with penicillin showing a considerably lower rate of reported JHR when compared to tetracyclines or erythromycin (Table 7). Overall, existing data suggest that in RF tetracycline treatment is associated with a higher rate of JHR but a lower relapse rate and penicillin treatment is associated with low rate of JHR but a higher relapse rate [211,218]. Overall, we found a TBRF-related CFR of 6.5%. This is in line with the generally reported TBRF-related CFR range of 2–10% [44]. The frequently encountered postulation that TBRF is less fatal than LBRF [18,240,241] may simply reflect the fact that the CFR of LBRF estimated in the literature has been too high [4]. This assumption is supported by our review on LBRF, were we found a CFR of 4% (treated)–10.2% (untreated) [4]. Therefore, we speculate that under similar medical conditions the overall CFR of TBRF and LBRF is not significantly different. Mortality in TBRF appears to

be primarily due to neurologic complications and ARDS, although reported data on attributable causes of death are largely lacking. It appears that JHR contributes little to the overall death rate, as only 6.5% of cases with JHR (from the 19.3% of antimicrobial-treated patients) die from it. In a clinical trial setting involving 184 patients with LBRF in Ethiopia, the CFR attributed to JHR was 3.3% [242]. Considering the probably above average quality of care in such clinical trial settings, CFR due to JHR in TBRF and LBRF may, overall, not be significantly different.

Regarding adverse pregnancy outcomes in TBRF, rates between 30% and 44% have been reported [192,243–246]. This is in analogy with LBRF, where adverse pregnancy outcomes, primarily in the form of abortions, are reported to occur in at least 70.9% of the cases [4]. Interestingly, our analysis suggests that the CFR for unborn children and for pregnant women does not differ from the CFR of other patients. Only the CFR of newborns appears to be considerably higher compared to non-neonatal cases (Table 8).

Table 9 comparatively summarizes the disease specific characteristics of TBRF and LBRF.

Our analysis has several limitations. First, data and results of studies and case series were often reported as overall numbers, medians or percentages and thus attributing data to individual cases was not possible. Second, the heterogeneity of the reviewed studies from very different geographic regions and clinical settings leads to the inherent problem of incomplete and not always compatible data, limiting the overall validity of the analysis. Third, the overall small number in subgroup analyses limits their validity and results may not reflect the true picture.

Key learning points

- TBRF is widespread worldwide, with transmission occurring by soft as well as hard ticks
- although only PCR-based methods allow for species identification, microscopy remains the diagnostic gold standard in most clinical settings

Table 9. Summary of characteristics of TBRF compared to LBRF.

	TBRF	LBRF [3,4]
Causative *Borrelia* spp.	Various *Borrelia* spp.	*B. recurrentis* (only)
Epidemiology	Occurrence of sporadic cases (affecting persons exposed to ticks)	Occurrence of outbreaks/epidemics (affecting vulnerable populations exposed to body lice)
Number of relapsing fever episodes	Mostly ≥2	Mostly <2
Duration of febrile episodes	Mostly ≤7 days	Up to 10 days
Treatment	Prolonged antibiotic treatment demanded (7–10 days; in the case of CNS involvement 10–14 days)	Single dose antibiotic treatment sufficient
Complications	Neurological complications are common (attributable to direct CNS invasion by *Borrelia*)	Neurologic complications are rare (attributable to hemorrhagic diathesis/bleeding complications rather than direct CNS invasion by *Borrelia*)
	Ocular involvement may occur	No ocular involvement reported
	Hemorrhagic diathesis/bleeding complications are rare	Subconjunctival hemorrhages and epistaxis are common.
Risk of JHR	19.3%	55.8%
Overall CFR	6.5%	4–10.2%
Perinatal fatalities	Primarily postpartal complications/affecting newborns	Primarly prepartal complications/affecting fetuses

TBRF, tick-borne relapsing fever; LBRF, louse-borne relapsing fever; CNS, central nervous system; JHR, Jarisch-Herxheimer reaction; CFR, case fatality rate.

https://doi.org/10.1371/journal.pntd.0010212.t009

- the risk of JHR is apparently lower in TBRF compared to LBRF
- the overall case fatality rate of TBRF and LBRF appears not to differ
- unlike LBRF, where perinatal fatalities are primarily attributable to abortion, TBRF-related perinatal fatalities appear to primarily affect newborns

Supporting information

S1 Text. Systematic review protocol. Established to conduct this systematic review.
(PDF)

S2 Text. List of databases with search terms used. Terms used for the study research in the different databases.
(PDF)

S3 Text. Reference list. Reference list of included and excluded publications.
(PDF)

S1 Table. Data extraction sheet. Used for screening and selecting eligible publications.
(PDF)

S2 Table. TBRF treatment details. Treatment details: antimicrobial treatment regimen, dosage and duration.
(PDF)

S1 PRISMA Checklist. PRISMA checklist. Twenty-seven-item checklist for systematic reviews. PRISMA, Preferred Reporting Items for Systematic Reviews and Meta-Analyses.
(PDF)

S1 Data. List of *Borrelia* spp. and ticks. List of *Borrelia* spp. and ticks associated to TBRF reported worldwide.
(XLSX)

S2 Data. Data master sheet. Excel sheet containing the underlying numerical data.
(XLSX)

S1 Fig. Supporting maps. Distribution of competent vector ticks for TBRF *Borrelia* spp.
(PDF)

S2 Fig. Subgroup analysis. Subgroup analysis of *B. crocidurae* and *B. hermsii*.
(PDF)

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PLOS NEGLECTED TROPICAL DISEASES

REVIEW

Louse-borne relapsing fever—A systematic review and analysis of the literature: Part 1—Epidemiology and diagnostic aspects

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Abstract

Louse-borne relapsing fever (LBRF) is a classical epidemic disease, which in the past was associated with war, famine, poverty, forced migration, and crowding under poor hygienic conditions around the world. The disease's causative pathogen, the spirochete bacterium *Borrelia recurrentis*, is confined to humans and transmitted by a single vector, the human body louse *Pediculus humanus*. Since the disease has had its heyday before the days of modern medicine, many of its aspects have never been formally studied and to date, remain incompletely understood. In order to shed light on some of these aspects, we have systematically reviewed the accessible literature on LBRF, since the recognition of its mode of transmission in 1907, and summarized the existing data on epidemiology and diagnostic aspects of the disease. Publications were identified by using a predefined search strategy on electronic databases and a subsequent review of the reference lists of the obtained publications. All publications reporting patients with a confirmed diagnosis of LBRF published in English, French, German, and Spanish since 1907 were included. Data extraction followed a predefined protocol and included a grading system to judge the certainty of the diagnosis of reported cases. Historically, Ethiopia is considered a stronghold of LBRF. The recognition of LBRF among East African migrants (originating from Somalia, Eritrea, and Ethiopia) arriving to Europe in the course of the recent migration flow from this region suggests that this epidemiological focus ostensibly persists. Currently, there is neither evidence to support or refute active transmission foci of LBRF elsewhere on the African continent, in Latin America, or in Asia. Microscopy remains the most commonly used method to diagnose LBRF. Data are lacking on sensitivity and specificity of most diagnostic methods.

Author summary

Louse-borne relapsing fever (LBRF) is an ancient epidemic disease, with descriptions dating back to Hippocrates' times. Linked to war, famine, poverty, forced migration, and crowding under poor hygienic conditions, the disease has accompanied humankind throughout history and, until 100 years ago, the disease was well recognized among

OPEN ACCESS

Citation: Kahlig P, Paris DH, Neumayr A (2021) Louse-borne relapsing fever—A systematic review and analysis of the literature: Part 1—Epidemiology and diagnostic aspects. PLoS Negl Trop Dis 15(3): e0008564. https://doi.org/10.1371/journal.pntd.0008564

Editor: Jon Blevins, UAMS, UNITED STATES

Published: March 11, 2021

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Funding: The authors received no specific funding for this work.

Competing interests: The authors have declared that no competing interests exist.

physicians in European and North American countries. Since then, the disease has fallen into oblivion, and reports of the disease are largely confined to Ethiopia. However, the disease has recently resurfaced when it was recognized as a cause of fever in East African migrants (originating from Somalia, Eritrea, and Ethiopia) arriving to Europe. Since the disease has had its heyday before the days of modern medicine, many aspects of the disease have never been formally studied and to date, remain incompletely understood. In order to shed light on some of these aspects, we have reviewed and analyzed the accessible literature on LBRF since its mode of transmission was recognized in 1907 and compiled this 2-part review focusing on epidemiology and diagnostic aspects (part 1) and mortality, Jarisch–Herxheimer reaction (JHR), and impact on pregnancy (part 2). We deliberately did not include an analysis on the antimicrobial treatment of LBRF, as Guerrier and Doherty have published a comprehensive meta-analysis on this aspect in 2011.

Introduction

Louse-borne relapsing fever (LBRF) is an ancient epidemic disease, with descriptions dating back to Hippocrates' times [1]. Linked to war, famine, poverty, forced migration, and crowding under poor hygienic conditions, the disease has accompanied mankind throughout history and was once even described as the "most epidemic among the epidemic diseases" [2]. The use of the name "relapsing fever" was first documented by Craigie and Henderson during the epidemic which occurred in Edinburgh from 1843 until 1848 [3,4], reviewed by Greig 100 years later [5]. Milestones in the disease's history were the discovery of the causative organism by Obermeier in Berlin in 1873 [6], the discovery of the organism in the vector by Mackie in India in 1907 [7], and the description of the mode of transmission by Sergent and Foley in Algeria in the same year [8,9]. In history, LBRF had a massive impact, especially following political crisis, socioeconomic disaster, and war [10,11].

Since the disease had its peak incidence and prevalence before the days of modern medicine, many aspects of the disease have never been formally studied and remain incompletely understood to date. In order to shed light on epidemiological and diagnostic aspects, we reviewed and analyzed the available published data on LBRF since its transmission was identified in 1907.

Epidemiology

With the reduction of the vector *Pediculus humanus*, due to improved living standards along with the introduction of the insecticide dichlorodiphenyltrichloroethane (DDT) in the 1940s, LBRF declined and finally disappeared from most regions of the world, as well as from most medical text books, over the past century [10,12].

In the last decades, reports of cases were almost exclusively limited to the Horn of Africa [10,11] and LBRF was increasingly considered a disappearing, neglected tropical disease (NTD) [12] until the disease recently resurfaced as non-malarial febrile illness in East African migrants arriving from Somalia, Eritrea, and Ethiopia to Europe [13–17]. Although several authors extensively reviewed the epidemiology of LBRF in Africa and Europe [18–26] and several studies and book chapters describe remaining endemic foci of LBRF in Africa, South America, and Asia [13,22,24,25,27–36], there is very little reliable data on the disease's true past and present epidemiology, especially in Latin America and Asia (Fig 1).

In order to shed light on the global epidemiology of LBRF over the past 100 years, we reviewed all available published reports on LBRF cases and summarized their number, their time of occurrence, and the grade of evidence for their correct diagnosis. Additionally, we

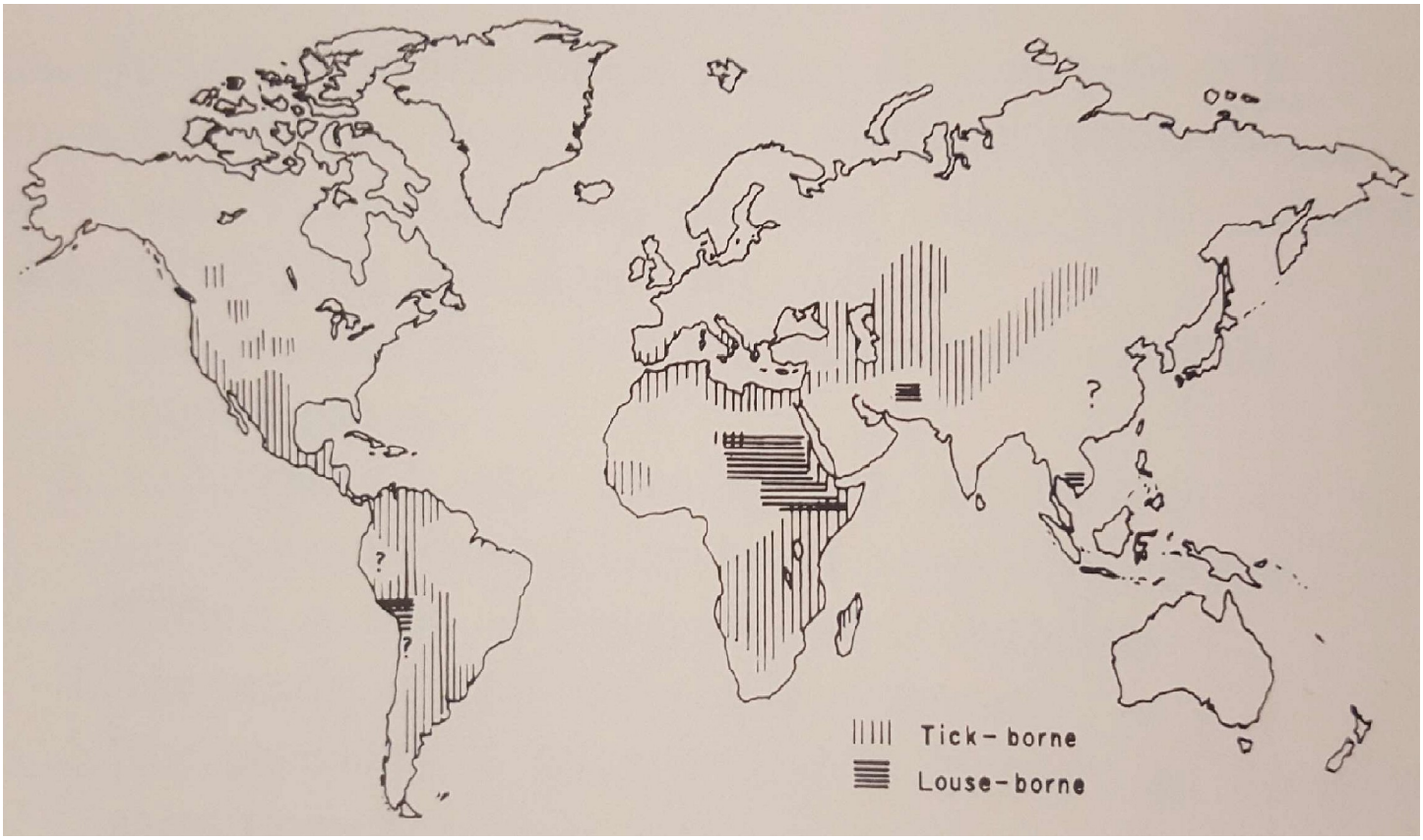


Fig 1. Assumed global distribution of TBRF and LBRF (1950–1969) [25]. LBRF, louse-borne relapsing fever; TBRF, tick-borne relapsing fever.

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performed an in-depth analysis of the rather unclear past and present situation of LBRF in Latin America and Asia.

Diagnostic aspects

Before the causative organism of LBRF was identified in 1873 [6], the diagnosis was exclusively based on signs and symptoms. However, since other febrile illnesses may present with similar signs and symptoms (e.g., "louse-borne typhus" caused by *Rickettsia prowazekii*, typhoid fever, or leptospirosis) as well as recurrent or periodic episodes of fever (e.g., tick-borne relapsing fever [TBRF], malaria, or the louse-borne "trench fever" caused by *Bartonella quintana*), it is probable that these diagnoses were often confused. After the discovery of the causative organism, microscopy of blood films became the diagnostic gold standard for LBRF. In thick and thin blood films (stained with Giemsa, May–Grünwald–Giemsa, Wright, Wright–Giemsa, Field's, or Diff-Quick stains or examined under dark field), *Borrelia* spirochetes are identifiable by their typical morphology (Fig 2). However, since *Borrelia recurrentis* is microscopically undifferentiatable from *Borrelia* spp. causing TBRF, true diagnostic confirmation of LBRF only became available with the introduction of polymerase chain reaction (PCR) and sequencing techniques in the 1980s. However, considering the disease's transmission, certain circumstances (e.g., outbreaks, epidemics, and occurrence in a vulnerable population) add conclusive epidemiological evidence and support the microscopical diagnosis as LBRF rather than TBRF. This does not apply to sporadic cases, where the way of transmission (i.e., ticks in an endemic

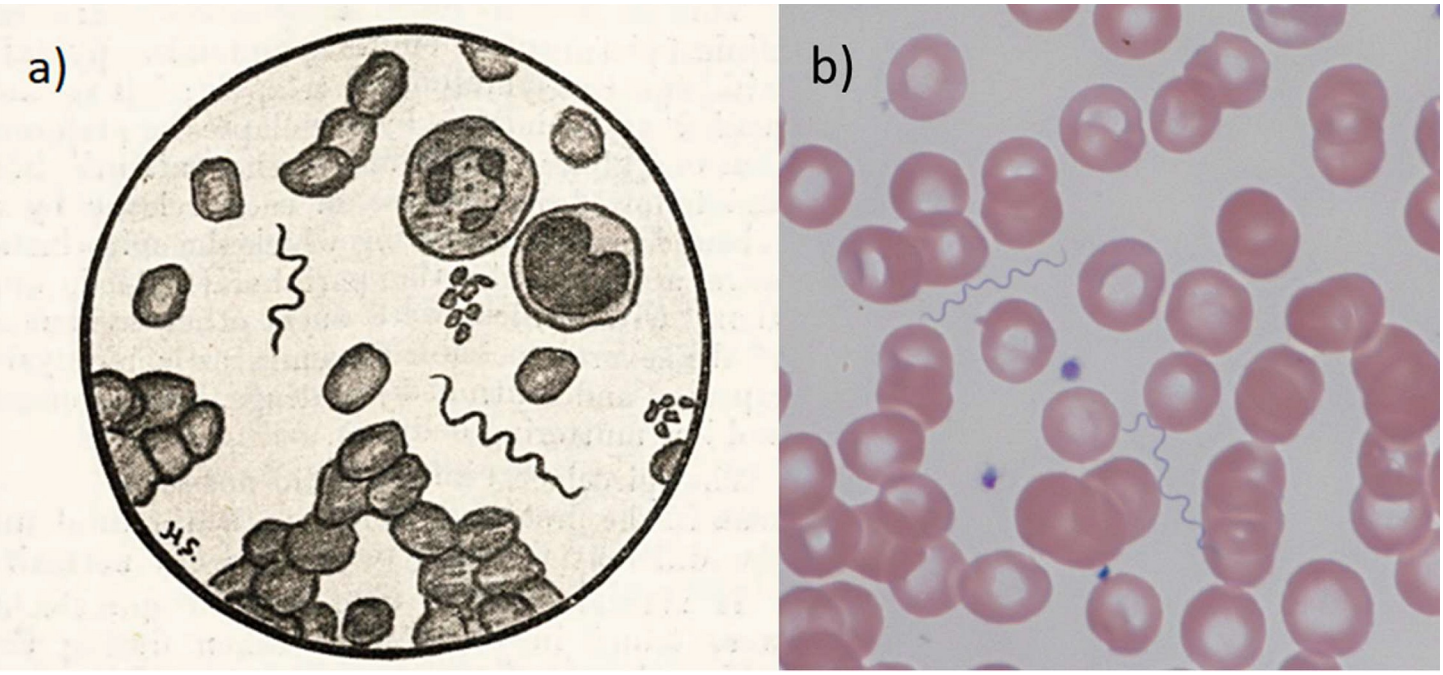


Fig 2. Microscopic detection of *B. recurrentis* in blood films. a) Drawing of *B. recurrentis* spirochetes found in a thin blood film obtained from a patient suffering from LBRF in 1909 in Jaipur, India [39]. b) Photography of *B. recurrentis* spirochetes found in a thin blood smear (May–Grünwald–Giemsa stain, magnification 1,000-fold) obtained from an Eritrean migrant suffering from LBRF in 2015 in Basel, Switzerland. LBRF, louse-borne relapsing fever; TBRF, tick-borne relapsing fever. *Image credit*: *Dr. Michael Osthoff*.

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region) rather supports TBRF. Nevertheless, microscopic examination has been the gold standard for diagnosing relapsing fever [37,38].

Serology has been used as alternative diagnostic method, but shows limited specificity due to cross-reactivity among *Borrelia* spp., including TBRF, as well as with other spirochetes (e.g., *Treponema pallidum*) [25,35,36]. Moreover, within endemic regions, the interpretation of serological assays is often complicated by high background reactivity and sero-scars (persistence of detectable antibodies after infection) in addition to the fact that the respective laboratory capacity and expertise are mostly unavailable. Furthermore, since seroconversion demands time, serology is not helpful to diagnose acute infection. Thus, serology was never developed into commercial available assays, and microscopy remains the sole and most widely available diagnostic tool to date [40]. With the introduction of PCR-based methods, the diagnostic sensitivity improved markedly, and species differentiation of relapsing fever *Borrelia* became available [41,42]. However, the availability is still largely restricted to affluent countries and research settings.

Table 1 lists the advantages and disadvantages of the laboratory diagnostic and research methods applied in LBRF.

In order to shed light on the evolution of LBRF diagnostics over the past 100 years and the accuracy of different test methods, we reviewed the available literature and summarized the available data.

Methods

A systematic review protocol established for this review is available in the Supporting information section (S1 Text). The electronic databases BIOSIS, CINAHL, Cochrane Library, Current

Table 1. Overview on laboratory methods applied in LBRF and their advantages, disadvantages, and use.

Method	Advantage	Disadvantage	Use
Microscopy	Fast; widely available	Variability (spirochete density, interobserver variability, and methodological differences); does not allow the differentiation between LBRF and TBRF *Borrelia*	Diagnostic gold standard
Serology	Allows retrospective evaluation	Not useful as acute diagnostic method due to delayed seroconversion; does not allow for species differentiation	Epidemiological studies
PCR	Species specific; high sensitivity; allows to differentiate LBRF from TBRF *Borrelia**	Currently no standardized protocol for discrimination between *Borrelia duttonii* and *B. recurrentis*	Largely restricted to research institutions
Culture	Isolation and growth of *B. recurrentis*	Time and resource demanding, overall challenging	Research only
Animal inoculation	Enhanced sensitivity in cases with negative microscopy; differentiation between LBRF and TBRF *Borreliae***	Time and resource demanding	Historical research method; also formerly used to "transport" *Borrelia*

LBRF, louse-borne relapsing fever; PCR, polymerase chain reaction; TBRF, tick-borne relapsing fever; WGS, whole genome sequencing.

* Note: With the increasing availability of pan-bacterial 16S rRNA PCR assays as well as WGS technology, the diagnostic repertoire has greatly improved in resource-rich settings in recent years.

** Note: Rodents are susceptible to TBRF *Borreliae*, but refractory to *B. recurrentis* infection.

https://doi.org/10.1371/journal.pntd.0008564.t001

Contents Connect, Elsevier, EMBASE Ovid, Ovid MEDLINE, PMC, PubMed, Scopus, and Web of Science were searched on October 4, 2017 using the search term ((Relapsing AND fever AND (Louse OR Lice OR (Pediculus AND humanus))) OR (Borrelia AND recurrentis) OR LBRF). A second and third search, using the same search term on the same databases, was conducted on August 7, 2018 and June 17, 2019, respectively. After checking for and removing duplicates (using EndNote software and manually [43]), publications were prescreened by checking titles and abstracts if they concerned patient(s) with the diagnosis of LBRF. Publications not reporting patient(s) with the diagnosis of LBRF were excluded. The remaining publications were then full-text assessed for fulfillment of the inclusion criteria: reporting conclusively diagnosed case(s) of LBRF and published after 1907 (the year when the disease's mode of transmission was discovered) and published in English, French, German, or Spanish. Publications not fulfilling the inclusion criteria were excluded. Publications that could neither be retrieved through their respective journals, nor by contacting libraries, or after contacting the authors, were classified as "not retrievable" and excluded. Additional relevant publications identified when reading the full-text articles or checking their reference lists were reviewed and included if they fulfilled the inclusion criteria ("snowball" search strategy).

A data extraction sheet for screening and selecting eligible publications was developed and is available in the Supporting information section (S2 Text). The following data were extracted from eligible publications using a standardized excel spreadsheet: patient characteristics (number of patients, age, gender, origin, occupation, social status, and way and duration of migration), diagnostic method (microscopy and molecular method), symptoms and signs (fever, chills, myalgia, headache, hepatomegaly, splenomegaly, signs of hemorrhage, and others), treatment (number of treated and untreated patients, drug, dosage, and duration and route of administration), and outcome (Jarisch–Herxheimer reaction (JHR), abortion/stillbirth, premature delivery, and mortality).

To minimize bias, the same reviewer conducted a second full data extraction more than 1 month after the first extraction. Discrepant results and unclear cases were resolved by consulting a second reviewer.

In order to consider the probability of a correct diagnosis of LBRF, all reviewed cases were graded according to the used diagnostic method and respectively classified (Table 2).

Table 2. Diagnostic grading system to judge the certainty of the correct diagnosis of LBRF.

Diagnostic method	Grade of diagnostic certainty	Case classification	Comment
PCR-based method	A	Confirmed diagnosis	Highest level of evidence for correct diagnosis
Microscopy	B	Microscopic diagnosis	Second highest level of evidence for correct diagnosis; microscopic identification of spirochetes during LBRF epidemics or in countries with current endemic foci leaves little doubt of the certainty of the diagnosis and may be regarded with an almost equal level of certainty as grade A
Paired serology	C	Indirect evidence	Intermediate level of evidence for correct diagnosis due to limited sensitivity and specificity of the method; paired serology, demonstrating seroconversion or increment of titer, is considered superior to single-titer serology
Single-titer serology	D	Indirect evidence	See comment under C above
Clinical diagnosis	E	Clinical diagnosis	Lowest level of evidence for correct diagnosis

LBRF, louse-borne relapsing fever; PCR, polymerase chain reaction.

* Note: Animal inoculation, historically used as supportive diagnostic method in LBRF research, was not considered a means of conclusive diagnosis, and thus not included in the evaluation.

https://doi.org/10.1371/journal.pntd.0008564.t002

To visualize the worldwide epidemiology of LBRF over the past century, data of all identified cases were entered into a geographic information system (GIS) application (https://www.qgis.org/en/site/) and graphically displayed using geodata from Natural Earth (https://www.naturalearthdata.com/about/terms-of-use/).

The review followed the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) statement (S1 Checklist).

Results

Our search strategy identified 4,943 publications of which 184 finally proved eligible for being included and analyzed (Fig 3, S1 Fig). A list of included and excluded publications is available in the Supporting information section (S3 Text).

From the 184 included publications, data of 18,613 LBRF cases were extracted (S1 Data). A total of 16,632 cases (90%) were microscopically diagnosed, 1,882 cases (10%) were clinically diagnosed, and 99 cases (0.5%) were confirmed by PCR (the majority after primarily being diagnosed by microscopy) (Table 3).

Epidemiology

The geographic localisation of all included LBRF cases is displayed in Fig 4 according to the following criteria: (i) time of occurrence; (ii) autochthonous versus imported; and (iii) underlying diagnostic certainty.

Publications reporting only indirect evidence for the presence of LBRF (but no clinical cases) are summarized in Table 4.

Reports on imported cases of LBRF are summarized in Tables 5 and 6.

Reports on the occurrence of LBRF in Latin America, Asia, and the Middle East are summarized in Tables 7–13.

In Fig 5, the number of published LBRF cases is visualized in relation to the number of publications.

Diagnostic aspects

Tables 14–17 summarize the milestones of LBRF diagnostics.

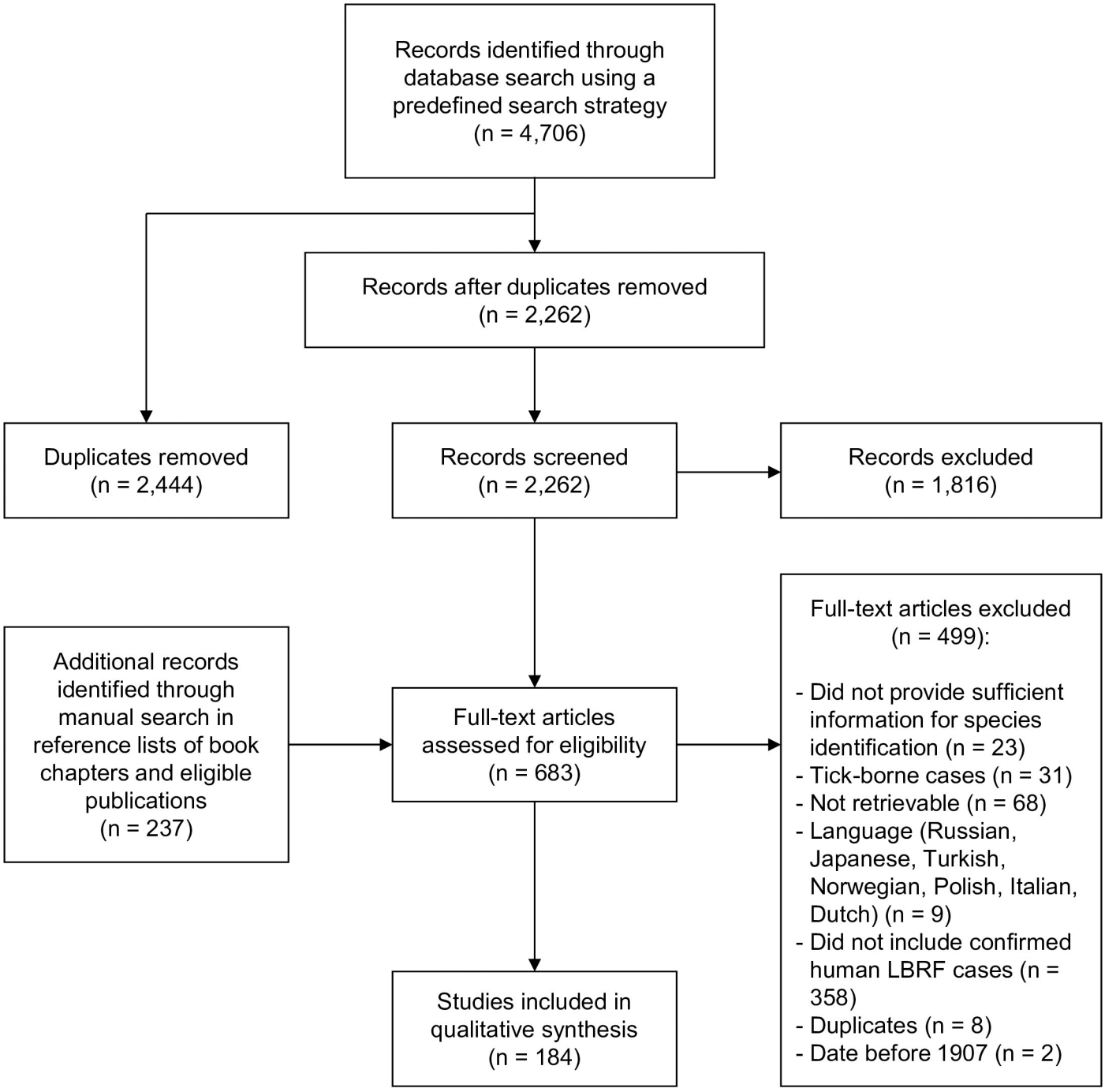


Fig 3. Flow diagram of search and selection of eligible publications. LBRF, louse-borne relapsing fever.

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Discussion

Epidemiology

Latin America. The last conclusive evidence of the occurrence of LBRF in Latin America dates back almost a century and is restricted to Peru. This notably contrasts the assumed occurence of the disease in South America in the mid-20th century published by Felsenfeld in

Table 3. Number of diagnosed LBRF cases over time according to the diagnostic method used.

Diagnostic method	Grade of diagnostic certainty	Case classification	Reported cases of LBRF diagnosed 1907–2019 *N* (%)											
			1907–1919	1920–1929	1930–1939	1940–1949	1950–1959	1960–1969	1970–1979	1980–1989	1990–1999	2000–2009	2010–2019	1907–2019
PCR-based method	A	Confirmed diagnosis	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	0	24[†] (0.7)	4 (0.2)	71[‡] (14)	99
Microscopy	B	Microscopic diagnosis	1,360 (60.1)	2,017 (93.3)	972 (97.8)	4,679 (88.7)	297 (100)	117 (100)	921 (100)	636 (100)	3,183 (97.7)	2,017 (92.4)	433 (85.1)	16,632
Paired serology	C	Indirect evidence	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	[33[§]]	n.a.	n.a.	n.a.
Single-titer serology	D	Indirect evidence	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
Clinical diagnosis	E	Clinical diagnosis	903 (39.9)	145 (6.7)	22 (2.2)	595 (11.3)	0	0	0	0	51 (1.6)	161 (7.4)	5 (1.0)	1,882
All methods	–	–	2,263	2,162	994	5,274	297	117	921	636	3,258	2,182	509	18,613

LBRF, louse-borne relapsing fever; n.a., not applicable (PCR: method not yet available [developed 1983]; serology: no commerical test developed; thus restricted to research institutions); PCR, polymerase chain reaction.

Note: Two publications did not report cases in absolute numbers and were thus not included in the numerical analysis [44,45].

† The first publication reporting the use of PCR to characterize and identify *B. recurrentis* was published in 1997 [28].

‡ Between 2010 and 2019, 72 PCR-confirmed cases were reported: 2 autochthonous cases from Ethiopia and 69 imported cases from Europe (68) and Israel (1).

§ In 2000, paired serology was used to retrospectively investigate 33 LBRF cases [34] which were microscopically diagnosed in a study conducted in 1977 [46].

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1971 (Fig 1) (Note: The method used to create the map is not known to us; the question marks within the map suggest some degree of uncertainty). We did not find any evidence supporting the persistent occurence of LBRF foci in Peru beyond the 1920s, nor did we find any reports on the occurrence of LBRF in other countries of Latin America.

Asia and the Middle East. Reports from Asia often coincide with times of colonialization as well as wars that western countries were involved in, when medical officers published their observations during their service. Subsequently, publications ceased shortly after the ends of conflicts and colonialization. The most recent examples were the Second World War and the Korean War. However, endemic foci may have persisted among a certain population at risk, as it currently does in Ethiopia, which simply were not detected or not published in western languages. It seems unlikely that the disease vanished as abruptly as publications ceased. A publication bias seems likely for these regions. However, our findings strikingly contrast the assumed occurence of LBRF in Asia in the mid-20th century published by Felsenfeld in 1971 (Fig 1).

China: The last published report dealing with microscopically diagnosed LBRF cases linked to China dates back to 1946, when relapsing fever was reported in Chinese soldiers just flown into Assam, India (Table 9). No reports on cases of LBRF exist from China thereafter.

Korea and Japan: Scarce data are available from Korea, where LBRF occurred and was confirmed until the end of the Korean War (Table 10). A retrospective description suggests a far wider spread than assumed in English literature and describes existing publications in Japanese or Korean language before the Korean War [90]. Thereafter, no information is available.

Southeast Asia: Data from Southeast Asia, where LBRF was reported in Vietnam and Cambodia, are very limited, and no information is available after 1912 and 1958, respectively (Table 11). Remaining residua past that can neither be confirmed nor excluded.

Indian subcontinent: Many accounts were published from the Indian subcontinent, where the disease often occurred in epidemics and endemic foci (Table 12). Investigations in the North-West Frontier province, nowadays located in Pakistan, led to the belief that both the

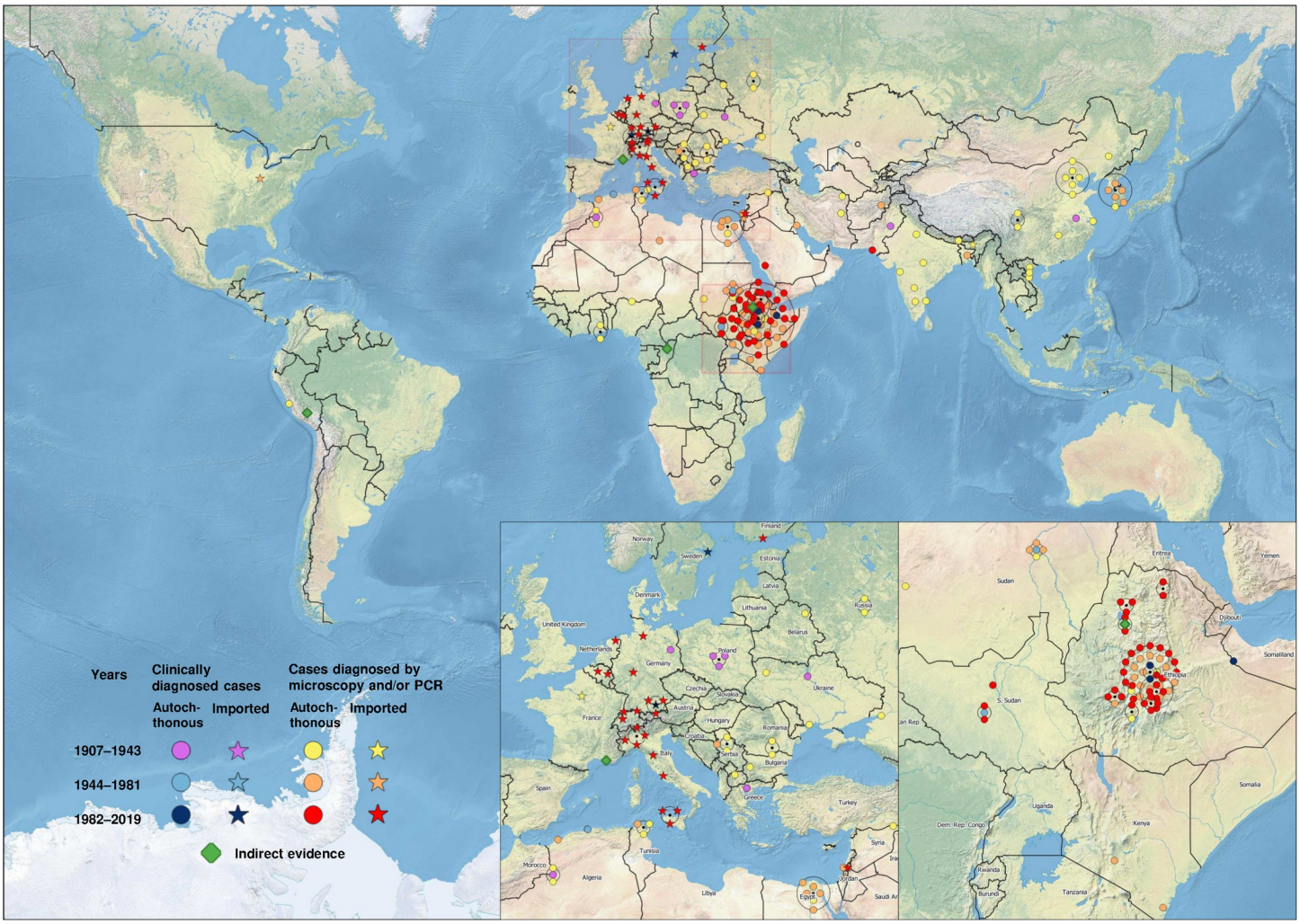


Fig 4. Geographic visualization of all identified LBRF cases published from 1907 to 2019. Each data point on the map corresponds to one of the analyzed 184 publications. If 2 or more studies reported cases from the same location, the dots representing these studies are connected by a circle with its center corresponding to this location. LBRF, louse-borne relapsing fever; PCR, polymerase chain reaction. (Map data: Natural Earth).

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louse-borne and the tick-borne variety are endemic in these areas, suggesting that careful watch should be kept for a sufficient differentiation of LBRF to TBRF [107]. The diagnosis of the reported sporadic cases was established after careful consideration of TBRF as a differential

Table 4. Publications reporting indirect evidence for the presence of LBRF.

Year	Country	Reported findings of the publication	Grd.	Ref.
1998	Peru	Detection of *B. recurrentis* specific antibodies in the blood of 2 out of 194 volunteers during a serosurvey conducted in rural Andean communities; based on single-titer testing; no clinical data reported	D	[30]
2000–2003	France	Detection of *B. recurrentis* specific antibodies in the blood of 15 out of 930 homeless people during a serosurvey conducted in Marseille; based on single-titer testing; no clinical data reported	D	[29]
2011	Ethiopia	Detection of *B. recurrentis* DNA in human head lice sampled; no clinical data reported	A	[47,48]
2015	Republic of Congo	Detection of *B. recurrentis* DNA in human head lice sampled; no clinical data reported	A	[47,48]

DNA, deoxyribonucleic acid; Grd., grade of diagnostic certainty according to Table 2; LBRF, louse-borne relapsing fever; Ref., reference.

https://doi.org/10.1371/journal.pntd.0008564.t004

Table 5. Imported cases of LBRF reported from America, Asia, the Middle East, and Africa.

Year	Number of cases	Country of exportation	Country of importation	Comment	Grd.	Ref.
1943–1944	134	China	India	Diagnosed in Chinese soldiers after arrival in a US Army Station Hospital in Assam	B	[49]
1945–1946	64	Morocco	Senegal	In the course of 9 months, 12 ships	E	[18]
1976	1	Ethiopia	USA, Ohio	Diagnosed in an immigrant shortly after arrival; detection in blood and lice	B	[50]
1985	2	Ethiopia	Israel	Two Ethiopian immigrants	B	[51]
2015	1	Ethiopia	Israel	Priest traveling with a group of pilgrims	A	[52]

Grd., grade of diagnostic certainty according to Table 2; LBRF, louse-borne relapsing fever; Ref., reference.

https://doi.org/10.1371/journal.pntd.0008564.t005

Table 6. Imported cases of LBRF reported from Europe.

Year	Number of cases	Country of exportation	Country of importation	Comment	Grd.	Ref.
1948	1	Greece	France	Greek migrant; Migration route: unknown	B	[53]
2015	1	Eritrea	Switzerland	Eritrean refugee; Migration route: Sudan, Libya, and Italy	A	[54]
2015	15	Somalia (n = 12); Eritrea (n = 2); Ethiopia (n = 1)	Germany	Somali, Eritrean, and Ethiopian refugees; Migration route: Sudan, Libya, and Italy	14A, 1B	[32]
2015	2	Eritrea	the Netherlands	Eritrean refugee; Migration route: Ethiopia, Sudan, Libya, and Italy	A	[55]
2016	1	Unkown	Sweden	Migration route: unknown	E	[56]
2015	1	Somalia	Italy	Somali refugee; Migration route: unknown	A	[57]
2015	3	Somalia	Italy	Somali refugees; Migration route: Libya	A	[58]
2015	1	Somalia	Italy	Somali refugee; Migration route: Kenya, South Sudan, Sudan, and Libya	A	[59]
2015	5	Somalia	Italy	Somali refugees; Migration route: Kenya, Uganda, Sudan, and Libya	A	[60]
2015	1	Somalia	Italy	Somali refugee; Migration route: Sudan and Libya	A	[61]
2015	4	Somalia (n = 3), Eritrea (n = 1)	Switzerland	Somali and Eritrean refugees; Migration route: Sudan, Libya, and Italy	A	[16]
2016	1	Eritrea	Switzerland	Eritrean refugee; Migration route: Sudan, Libya, and Italy	B	[62]
2015–2016	25	Somalia (n = 23), Eritrea (n = 2)	Germany	Somali and Eritrean refugee; Migration route: Sudan, Libya, Yemen, and Italy	A	[13]
2016	1	Somalia	Germany	Somali refugee; Migration route: unknown	A	[63]
2015	1	Somalia	Belgium	Somali refugee; Migration route: Italy	A	[17]
2015	1	Somalia	Germany	Somali refugee; Migration route: Ethiopia, Sudan, Libya, and Italy	A	[64]
2015	2	Somalia	Belgium	Somali refugees; Migration route: Ethiopia, Sudan, and Libya	B	[65]
2014–2015	2	African region	Switzerland	Exact date and migration route: unknown	E	[66]
2017	1	East Africa	Italy	Date and migration route: unknown	A	[67]
2016	1	Mali	Italy	Malian refugee; Migration route: Algeria and Libya	A	[15]
2015	1	Somalia	Germany	Somali refugee; Migration route: Ethiopia, Sudan, and Libya	A	[68]
2015	2	Somalia	Finnland	Somali refugees; Migration routes 1: Yemen, Egypt, Greece, and Italy; 2: Uganda, Libya, Italy, and Germany	A	[14]
2016	2	Somalia	Italy	Somali refugees; Migration route: Libya	A	[69]
2016	1	Somalia	Italy	Somali refugees; Migration route: unknown	A	[70]
2014–2015	2	Somalia	Germany	Somali refugees; Date and migration route: unknown	E	[71]

Grd., grade of diagnostic certainty according to Table 2; LBRF, louse-borne relapsing fever; Ref., reference.

https://doi.org/10.1371/journal.pntd.0008564.t006

Table 7. Reports on LBRF from Latin America.

Year	Comment	Grd.	Ref.
16th century	The first appearances of LBRF may date back to the times when Spanish conquistadors arrived in South America	NA	[72]
1917	**Peru:** Retrospective description of the first microscopically diagnosed case of LBRF in Peru	NA	[72–74]
1918–1919	**Peru:** First microscopically diagnosed LBRF cases published in a study in Peru	B	[73]
around 1920	**Peru:** Relapsing fever is reported from various parts of the country, mostly from central and southern regions. The regions Ayacucho, Huancavelica, Junín, Cajamarca, Ancash, Lima, Arequipa, Cuzco, Apurímac, and Puno were reported to be affected	NA	[72,73]
1946	**Peru:** A putative case of LBRF was published in 1946, but excluded from this review because of insufficient information regarding the differentiation between TBRF and LBRF, being simply named "relapsing fever"	NA	[75]
1999	**Peru:** Detection of *B. recurrentis* specific antibodies in the blood of 2 out of 194 volunteers during a serosurvey conducted in rural Andean communities; based on single-titer testing; no clinical data reported	D	[30]

Grd., grade of diagnostic certainty according to Table 2; LBRF, louse-borne relapsing fever; NA, not applicable; Ref., reference; TBRF, tick-borne relapsing fever.

https://doi.org/10.1371/journal.pntd.0008564.t007

diagnosis. The author notes further that in most cases, blood was taken after symptoms resolved. Therefore, most microscopic examinations were negative. As there was no further data on the positive results, all cases were considered as clinically diagnosed. A worker informed the authors in a personal communication that LBRF had been known in the area for many years [107]. The last published report from 1990 microscopically diagnosed LBRF in 2 patients in 1984 to 1985 [111]. Despite the authors titling this discovery *B. recurrentis*, there is neither sufficient clinical evidence nor conclusive case histories to retrospectively comprehend how the differentiation was achieved. The authors noted that the disease is common in the Karachi region; however, their work lacks discussion as to whether these cases belong to the tick-borne or louse-borne species [111]. The study was included due to positive blood smears and the authors publishing the data as *B. recurrentis*. However, considering the circumstances, the evidence should be regarded with caution. Further studies are needed from this area to either confirm or rebut the presence of LBRF in these areas.

Afghanistan and Middle East: Accounts on LBRF from former Persia and Mesopotamia are closely related to the World Wars and the medical officers who published their reports, with no further information published thereafter (Table 13). Migration of refugees, prisoners of war, and the movement of Russian, Turkish, and Indian troops were often reported to be the main cause for the infection of western troops with the disease [113,114,116]. Many of the occurrences mentioned were traced back to troop movements, close contact with the native population or the recruiting of refugees for labor corps [113,114,116–118]. A report from Iran in 1976 found borreliae in a febrile patient, suggesting a new species in the area, although noting a similarity to the malady known from Ethiopia. Despite the suspicions, no further

Table 8. Reports on LBRF from Northern America.

Year	Comment	Grd.	Ref.
1844–1874	Retrospective description of several epidemics of LBRF that may have occurred between 1844 and 1874	NA	[50]

Grd., grade of diagnostic certainty according to Table 2; LBRF, louse-borne relapsing fever; NA, not applicable; Ref., reference.

https://doi.org/10.1371/journal.pntd.0008564.t008

research on the means of transmission has been conducted. For this reason, the species of borreliae found in Iran remains inconclusive [139]. Caution should be taken regarding sporadic cases, as the authors themselves and others noted the presence of TBRF in these areas [119,139].

Imported cases. Living in cramped and poor hygienic conditions provides favorable conditions for the transmission of LBRF [19,20,45,134,140–145]. As this is the case for many immigrants, refugees, and seasonal workers, human migration is a critical component in the development of LBRF epidemics. In the last century, when the presence of LBRF was almost worldwide, only sporadic cases of imported LBRF were reported. In recent years, the proportional discrepancy between the number of reports from endemic regions and reports on imported cases to non-endemic countries is striking: Between 2010 and 2019, 25 publications reported 78 imported cases of LBRF in non-endemic countries, whereas from endemic regions, 7 publications reported 431 autochthonous cases (Fig 5). This increase in reports on imported cases of LBRF is primarily attributable to the increased migration flow from Africa to Europe observed in 2015 and 2016. Most of these cases were diagnosed using PCR-based methods.

Persistence of LBRF and factors perpetuating the disease. The vector: The transmitting vector *P. humanus* [146] is a specialized human ectoparasite that flourishes in hygienically poor and overcrowded conditions [11,146]. Besides *B. recurrentis*, it also transmits *R.*

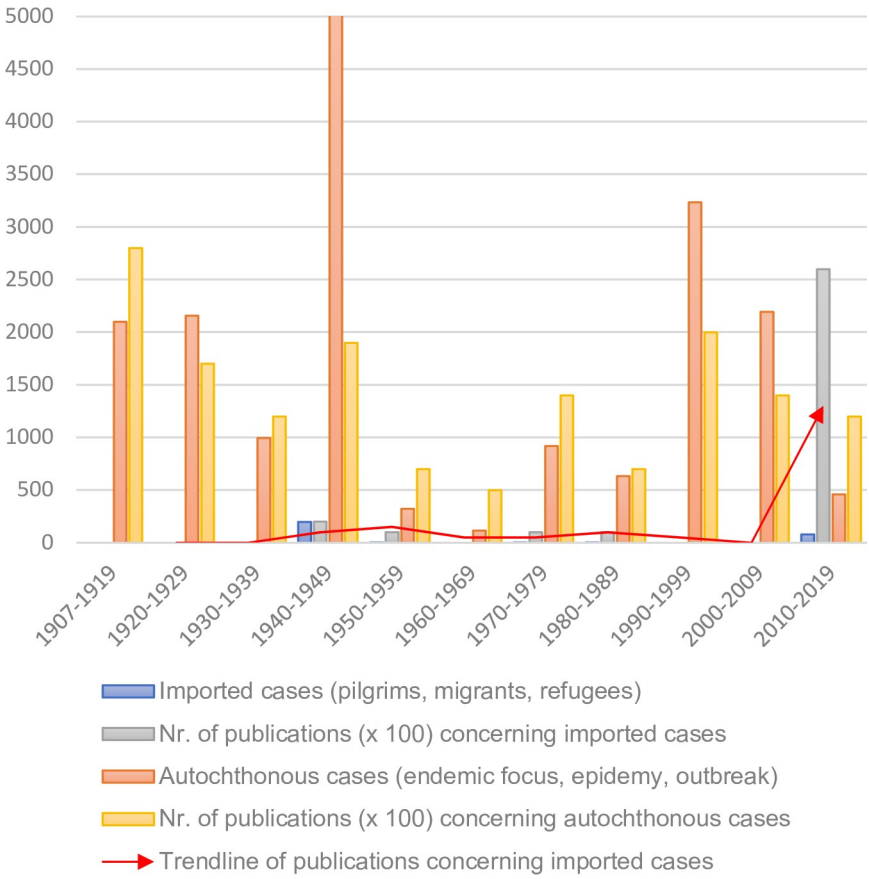


Fig 5. Number of published LBRF cases in relation to the number of publications. For better visualization, the number of publications is multiplied by 100. The red arrow depicts the trendline of publications reporting imported cases of LBRF. LBRF, louse-borne relapsing fever.

https://doi.org/10.1371/journal.pntd.0008564.g005

Table 9. Reports on LBRF from China.

Year	Comment	Grd.	Ref.
?	LBRF may have been around for a very long time and may have been the cause of 2 outbreaks reported from Beijing in 1864 and Hong Kong in 1865	NA	[76–80]
1904	Retrospective description of LBRF in Pakhoi in the South of China	NA	[81,82]
1905–1906	Retrospective description of LBRF in Shanghai, Tien-Tsin, Hankou, and Hong Kong	NA	[73,82]
1909	Retrospective description of LBRF in the southern region of Yunnan	NA	[72,73,82]
1911–1912	Report of an LBRF epidemic with clinically diagnosed cases in Hwaiyuan and an endemic focus in Chongqing	E; NA	[82,83]
1911–1919	Retrospective description of annual outbreaks of LBRF in Sichuan; published case series of microscopically confirmed cases from Sichuan in 1919	NA; B	[82,84]
1913	Retrospective description of LBRF among prisoners in Shanghai	NA	[82]
1913–1917	Retrospective description of LBRF in Manchuria until 1917; published notes on microscopically confirmed cases from an outbreak in a mine in 1913	B	[82,85]
1918–1938	Report of sporadic cases and occasional small outbreaks of LBRF in Hunan, including 41 microscopically diagnosed cases	B	[79]
1919	Retrospective description of LBRF cases among soldiers in Fujian	NA	[77,82]
1920	Descriptive note that relapsing fever had been found at any location in China where laboratories had been built	NA	[82]
1931	Reports of microscopically diagnosed LBRF cases in Beijing	B	[77]
1932	Retrospective description of endemic LBRF in all provinces along the Yangtse River and sporadic epidemics. Further notes dating back to 1924 and 1925 indicate the presence of LBRF in Tibetan regions from where the disease spread both along the eastern trading route toward Tachienlu (today Kangding) and the southern trading route along the Mekong river	NA	[76]
1932	Report on a minor epidemic with microscopically diagnosed cases of LBRF in Shanghai (including the note of the disease's constant presence in the hospital records over the past 25 years)	B	[76]
1936–1939	Several reports on microscopically diagnosed LBRF cases in Beijing	B	[80,86–89]
1943–1944	Report of microscopically diagnosed LBRF cases in Chinese soldiers just flown into Assam, India. The infections were considered to have been acquired in China	B	[49]

Grd., grade of diagnostic certainty according to Table 2; LBRF, louse-borne relapsing fever; NA, not applicable; Ref., reference.

https://doi.org/10.1371/journal.pntd.0008564.t009

Table 10. Reports on LBRF from Korea and Japan.

Year	Comment	Grd.	Ref.
?–1913	Notes of publications in Japanese about LBRF cases in Japan. Description of the presence of LBRF in Tokyo during 1913	NA	[85]
?–1913	LBRF is believed to have been clinically diagnosed before 1913 in Korea	NA	[90]
1913–1943	Report of the first microscopical diagnosis in 1913; notes of repeated incidence in local hospital admission records; description of an epidemic among railway laborers and of accounts in other languages such as Korean or Japanese	B; NA	[90]
1950–1955	Several reports on microscopically diagnosed LBRF cases during the Korean War, especially among Chinese and Korean prisoners of war or United Nations personnel. The presence of LBRF in the native population is noted. Movement of Chinese troops is suggested to have imported and perpetuated the disease in some regions	B	[91–95]

Grd., grade of diagnostic certainty according to Table 2; LBRF, louse-borne relapsing fever; NA, not applicable; Ref., reference.

https://doi.org/10.1371/journal.pntd.0008564.t010

Table 11. Reports on LBRF from Southeast Asia.

Year	Comment	Grd.	Ref.
?–1906	LBRF may have been around for a long time, but has been confused with malaria in Vietnam	NA	[96]
1907–1912	Report of microscopically diagnosed LBRF cases in Hanoi. The disease has been reported yearly in hospital statistics since 1907; in 1912, case numbers began to rise	B	[97]
1908	Restrospective description of 4 Chinese LBRF patients from Yunnan treated in Hanoi	NA	[82]
1907–1909	Retrospective description of cases in the province Thanh-Hoa	NA	[96]
1912	Reports of microscopically diagnosed LBRF cases in the provinces of Thanh-Hoa and Nghê-An in 1912. Retrospective descriptions of clinically suspected cases in earlier years, endemic foci in these and more southern reagions, such as Ha-Tinh	B	[96,98]
1950–1958	Retrospective descriptions of cases in Cambodia between 1950 and 1958	NA	[25]

Grd., grade of diagnostic certainty according to Table 2; LBRF, louse-borne relapsing fever; NA, not applicable; Ref., reference.

https://doi.org/10.1371/journal.pntd.0008564.t011

prowazekii and *B. quintana*, the causative agents of epidemic typhus and trench fever, respectively [146]. Experiments demonstrated that *B. recurrentis* is not transmitted by ticks [147]. The body louse is currently the only proven vector, and humans are the only known reservoir [11]. Body lice live and lay eggs in clothing and only approach the human body for an obligate blood meal [40]. A factor that perpetuates transmission, especially during epidemics, is that lice are temperature sensitive and tend to leave patients clothes during febrile episodes [22]. Interestingly, body lice were reported on secondhand clothing found on a street market in Italy in 2018, making it the first report of human body lice since 1945 in Italy [148]. This finding challenges the paradigm that body lice die quickly once off the host.

Transmission: Unlike most vector borne infections, *B. recurrentis* is not transmitted to the human host during the blood sucking act of the vector, as the digestive tract and the salivary glands of the body louse are not affected by the infection [11,86,149,150]. Since the louse's hemolymph was found to harbor *B. recurrentis*, human infection is traditionally considered to result from damaging or crushing the louse, thus liberating the insect's hemolymph. The co-liberated bacteria were then considered to enter the human host through microlesions of the skin, which are either caused by the bite itself or by scratching induced by the itchiness of the bites [11]. This transcutaneous route of infection is supported by animal [80] as well as human experiments [86,151]. In 1938, Chung and colleagues identified *B. recurrentis* in the feces of lice and suggested this as an additional source of infection, with the caveat that in their experiments, the excreted spirochetes were dead [86]. The issue remained dormant until 2005, when Houhamdi and Raoult reported the detection of living *B. recurrentis* in excreted feces of an infected louse, which revived the discussion [152]. However, whether the fecal excretion of *B. recurrentis* is relevant for the pathogens transmission still remains to be clarified.

Unnoticed reservoir? Asymptomatic cases: Asymptomatic cases of LBRF in Sudan were observed by Atkey in 1930 [153]. These cases appeared toward the end of an epidemic, and spirochetes were readily found in their blood. The author further observed a general milder course of the disease toward the end of the epidemic [153]. Another author reported latent and atypical LBRF infections [154]. During an outbreak survey conducted in Khartoum in 1969, 22 microscopically positive but asymptomatic cases of LBRF were identified among 979 immigrant laborers from southern Sudan [134]. The fact that such cases have been reported may imply a high number of overlooked cases. Further research is needed to confirm the

Table 12. Reports on LBRF from the Indian subcontinent.

Year	Comment	Grd.	Ref.
18th century	LBRF may have existed on the Indian subcontinent since the mid-18th century	NA	[81]
1836–1877	Retrospective description of severe epidemics in the United Provinces in 1836, 1837, and 1862; retrospective description of the first recognized outbreak in 1852 in Usufzai Valley, in present day Pakistan; retrospective description of an epidemic in Patna in 1856 and Bombay in 1877; retrospective description of an epidemic in the Réunion due to infected coolies shipped from said epidemic in Bombay	NA	[81,99,100]
1869–1911	Retrospective description of endemic foci in northern India and the Himalaya region with several minor outbreaks and occasional epidemics, such as in Punjab and the United Provinces (in 1869, 1878, 1891, 1896, 1899, 1906, and 1911)	NA	[81,99]
1905	Report of microscopically diagnosed LBRF cases in an epidemic in Peshawar Valley, in present day Pakistan	B	[101]
1906	Report of an epidemic in Sirur with notes of microscopically diagnosed cases	B	[102]
1907	Report of microscopically diagnosed LBRF cases and first description of the mode of transmission and discovery of *B. recurrentis* in lice	B	[7]
1908–1911	Report of microscopically diagnosed LBRF cases in Bulandshahr. It further suggests the yearly occurrence, but also a certain neglectance of the disease, being often misdiagnosed for malaria in the area	B	[103]
1911	Report of microscopically diagnosed cases in Bangalore	B	[39]
1912	Report of microscopically diagnosed cases in Darjeeling District	B	[104]
1917–1920	Description of an LBRF epidemic affecting Punjab, the United Provinces, the Central Provinces; report of microscopically diagnosed cases during this epidemic in the Seoni District (Central Provinces). Description of the presence of LBRF before the epidemic in these areas	NA; B	[99,105]
1923	Report of microscopically diagnosed LBRF cases in Raichur	B	[106]
1923–1924	Retrospective description of an epidemic in Madras	NA	[107]
1923–1926	Account on diseases in India suggesting LBRF to be far more widely spread at the time than formerly suspected, with affected areas scattered in both central and southern areas with a few spots in eastern areas	NA	[108]
1924	Description of microscopically diagnosed cases in Nilgiri Hills, Madras Presidency, and further research with sera of patients	B	[107,109]
1925–1929	Description of sporadic cases and the occurrence of small outbreaks in Punjab between 1925 and 1929, indicating endemic residua in the North after the epidemic. Following the description of sporadic LBRF cases in the North-West Frontier, in present day Pakistan, the report further suggests both the louse-borne and the tick-borne variety to be endemic in these areas	E	[107]
1948	Report of microscopically diagnosed LBRF cases in North-East Bengal among military personnel	B	[110]
1984–1985	Report of 2 microscopically diagnosed LBRF cases in the Karachi region. Finding was called "*B. recurrentis*" by authors; however, the publication lacks data on differentiation	B	[111]

Grd., grade of diagnostic certainty according to Table 2; LBRF, louse-borne relapsing fever; NA, not applicable; Ref., reference.

https://doi.org/10.1371/journal.pntd.0008564.t012

existence of asymptomatic cases and to assess the rate of asymptomatic infections since they may contribute to the persistence of the disease in certain areas.

Residual brain infection: Residual brain infection (RBI) describes the tendency of spirochetes to persist in the brain after they have cleared from the blood and was first described by Buschke and Kroo with TBRF in an animal model [38]. Data for *B. recurrentis* are scarce. One study demonstrated infection of squirrels using cerebrospinal fluids of LBRF patients [87]. Other studies demonstrated *B. recurrentis* in the cerebrospinal fluid of patients [155] and

Table 13. Reports on LBRF from the Middle East and Afghanistan.

Year	Comment	Grd.	Ref.
?–1950	Report of microscopically diagnosed LBRF cases in Kabul. The disease was said to be frequently encountered in Afghanistan	B	[112]
?–First World War	LBRF may have been around for a long time. Descriptions of LBRF being endemic among the native population	NA	[113–115]
First World War	Description that Baghdad and the northern Persian areas were most affected, as well as Birjand, Busra, Meshed, and several other locations	NA	[113,114,116]
1918	Report of microscopically diagnosed LBRF cases in Meshed, Birjand, and a railway construction camp near Nisibin (today Iran). Many of the occurrences were traced back to troop movements, close contact with the native population, or recruiting of refugees for labor corps	B	[114,117,118]
1945–1946	Report of microscopically diagnosed LBRF cases during an epidemic in the aftermath of the Second World War in Abadan. Authors discussed former putative LBRF epidemic during the Second Word War mislabelled as "typhus" and further highlight the presence of TBRF in the area	B	[119]

Grd., grade of diagnostic certainty according to Table 2; LBRF, louse-borne relapsing fever; NA, not applicable; Ref., reference; TBRF, tick-borne relapsing fever.

https://doi.org/10.1371/journal.pntd.0008564.t013

infection of the brain of primates [121]. The involvement of the central nervous system in the acute phase of LBRF infection was suggested in up to 30% of cases [36]. However, a review article suggested that most of neurologic symptoms in LBRF infection may be due to hemorrhages in the central nervous system, rather than direct involvement of the spirochetes [38]. In animal experiments with *Borrelia turicatae*, RBI was found in 19% of immunocompentent mice [156]. Another study using a murine model to investigate RBI using *B. turicatae*, *Borrelia crocidurae*, *Borrelia hermsii*, and *B. duttonii* found limited brain persistence in *B. crocidurae* and longest persistence for *B. duttonii*. Additionally, reactivation of the infection was demonstrated in the case of immunosuppression. The authors suggested the brain as reservoir [157]. Further studies are needed to investigate RBI.

Historic point of view: Many hypotheses were discussed regarding the question of persistence and maintenance of the disease between large epidemics. As an example, in North Africa, despite systematic research, no LBRF cases were identified between the 2 large epidemics of 1908 to 1920 and 1943 to 1945 [2]. Considering that LBRF has no currently known host

Table 14. Milestones in microscopy.

Year	Comment	Ref.
1873	Discovery of the causative organism	[6,120]
1907	Discovery of the organism in the vector	[7]
1947	A publication reported 65% of suspected relapsing fever cases to yield a positive result on microscopic examination	[121]
1969	A retrospective review of 2,825 cases of relapsing fever suggested a sensitivity of approximately 70% on initial blood smears to find spirochetes	[36]
1983	Enhancement of sensitivity was reported when using either fluorescence microscopy on acridine orange-stained blood smears	[122]
1996	A review suggested animal inoculation or culture to enhance sensitivity and specificity	[27]
2001	Enhancement of sensitivity was reported using QBC fluorescent technique. (QBC: detection of spirochetes down to 10 organisms/mm3, Wright-stained blood film: did not detect organisms <82 organisms/mm3; positive readings found by means of blood film fell significantly after dilutions below 3,263 organisms/mm3, in contrast to the QBC, the accuracy of which fell only at dilutions of <82 organisms/mm3)	[123]
2009	A publication suggested a centrifugation-based method to concentrate spirochetes, followed by Giemsa staining, to be equally sensitive to PCR-based methods. According to the study results, detection at a bacterial concentration of 1 bacteria/ml was achieved	[124]

PCR, polymerase chain reaction; QBC, quantitative buffy coat; Ref., reference.

https://doi.org/10.1371/journal.pntd.0008564.t014

Table 15. Milestones in serology.

Year	Comment	Ref.
?–2019	Antigenic variation and cross-reactivity within the species, and with other closely related organisms, were 2 major issues in the development of reliable serological diagnostic methods	[25,34,38,125–128]
1996 and 2000	Publications suggest the use of *glpQ* gene, which was found to be absent in Lyme disease borreliae, and very different in amino acid sequences in *Treponema pallidum*, hence enabling the serological distinction from relapsing fever cases	[34,127]
2000	A report demonstrated seroconversion 1 to 2 weeks after clinical presentation, hence recommending the use of paired acute-phase and convalescent-phase sera. Persistence of detectable antibodies to recombinant GlpQ in a serum sample taken 27 years after infection with LBRF. Immunoblotting with recombinant GlpQ was found to be more sensitive than ELISA with purified His-tagged GlpQ	[34]

ELISA, enzyme-linked immunosorbent assay; GlpQ (protein)/glpQ (gene), glycerophosphodiester phosphodiesterase; LBRF, louse-borne relapsing fever; Ref., reference.

https://doi.org/10.1371/journal.pntd.0008564.t015

other than humans, that there is currently no known reservoir, that lice do not transmit *B. recurrentis* to their progeny, and that transmission requires injury or crushing of the louse, the persistence of the disease was a mystery. Two hypotheses were formulated around 1960 concerning the origins of epidemics: (i) the epidemics of LBRF are only a temporary phenomenon due to a mutation of a tick-borne species by passage through the human into the louse; and (ii) the existence of reservoirs in an endemic focus from where epidemics originate [2]. Regarding the first hypothesis, it was experimentally achieved to infect lice with tick-borne species [2]. In one study, lice were infected with *B. duttonii* and successfully transmitted back to primates [158]. However, no mutations or changes in pathogenic character were observed, and no evidence supporting this hypothesis was found in vivo. Regarding the second hypothesis, Ethiopia was recognized to be an endemic focus by Sparrow [133]. Since then, Ethiopia has remained an endemic focus. It seems likely that the persistence of endemic foci were the origins of the

Table 16. Milestones in PCR.

Year	Comment	Ref.
1996	Phylogenetic analysis, using PCR-based methods targeting 16S ribosomal DNA sequences, reported a close relationship between the different relapsing fever borreliae, and researchers have suggested this strategy for diagnosis	[129]
2003	Report of successful discrimination using a real-time PCR assay targeting the flagellin gene in *B. recurrentis*, which differs only by a single nucleotide from the sequence in *B. duttonii*. Successful recognition has been achieved at annealing/extension temperatures of 64.5˚C, 65˚C, and 66˚C. Sensitivity of 3 copies of the target sequence was reported	[41]
2008–2012	Phylogenetic analysis has led to the concept that *B. recurrentis* is a degraded subset of *B. duttonii*. Using a MLSA approach, gene sequence identities greater than 99% were reported	[130–132]
2012	Using MST method, 3% sequence divergence was observed when using the MST7 spacer to discriminate between *B. duttonii* and *B. recurrentis*	[131]
2013	Development of an MR-TPCR assay reporting a 100% sensitivity and specificity for both *B. duttonii/recurrentis* and *Borrelia hispanica*, as well as a 99% sensitivity and specificity for *Borrelia crocidurae*. 16S rRNA gene probe for the detection of any relapsing fever borreliae, combined with species-specific primers (recN gene detecting *B. duttonii/B. recurrentis*). Accuracy of detecting 100 copies was reported. Successful discrimination between *B. duttonii* and *B. recurrentis* was not achieved	[42]

DNA, deoxyribonucleic acid; MR-TPCR, multiplex real-time PCR; MSLA, multilocus sequence analysis; MST, multispacer sequence typing; PCR, polymerase chain reaction; Ref., reference; rRNA, ribosomal ribonucleic acid.

https://doi.org/10.1371/journal.pntd.0008564.t016

Table 17. Milestones in animal inoculation and culture.

Year	Comment	Ref.
1954	The inoculation of guinea pigs was suggested for use in differential diagnosis of TBRF and LBRF: Adult rodents are susceptible to TBRF borreliae, but refractory to *B. recurrentis* infection	[120]
1958 and 1969	Susceptibility tests in adult rodents were reported to be the most reliable method for the differentiation between tick-borne and louse-borne borreliae	[133,134]
1965 and 1968	Reviews on LBRF describing that monkeys are susceptible to infection with *B. recurrentis*, while adult mice and rats have limited susceptibility. Young mice and rats were found to be susceptible	[125,135]
1969 and 1971	Reviews retrospectively describing that maintenance and arguably even limited growth of *B. recurrentis* was achieved in vitro	[25,36]
1971	Monograph describing animal inoculation with *B. recurrentis* on various animals with differing results regarding their susceptibility until 1971, pointing out lack of details in many reports which limits the comparability	[25]
1984	Retrospective description of the successful multiplication of *Borrelia hermsii* in 1971 on a media, which led to the creation of the BSK II medium in 1984 (originally used to cultivate *Borrelia burgdorferi*)	[136]
1994	Report of the first growing isolate of *B. recurrentis* in BSK II medium	[137]
2009	Report showing improved results using immunodeficient mouse strains	[138]

BSK, Barbour Stoenner Kelly; LBRF, louse-borne relapsing fever; Ref., reference; TBRF, tick-borne relapsing fever.

https://doi.org/10.1371/journal.pntd.0008564.t017

former epidemics, rather than the first mentioned, or any other hypothesis. Asymptomatic infections and RBIs most probably serve as factors that further perpetuate the persistence of LBRF in an endemic focus.

New endemic foci? Most of the recently imported cases originated from the Horn of Africa. In one refugee from Mali, contact with people migrating from this area was reported. The location of a current endemic focus has been suspected in Libya, which reportedly serves as a focal point for smugglers to bring refugees across the Mediterranean Sea [15,32,159]. Even though there is no published evidence to confirm the suspicions, a temporary unnoticed endemic focus in certain refugee camps or places where migrants have gathered is likely. The duration of a migrant's journey from East Africa to Europe largely exceeds the reported incubation periods of LBRF. It is probable that refugees go through the first attack and the following relapses during the first month of their journey. The fact that these patients have showed first symptoms upon arrival in Europe suggests an endemic focus around the Mediterranean Sea. This could have been a temporary focus, as it vanished after 2016 without further reports from imported cases thereafter. However, the epidemiological investigation in this review has shown that there is currently no other known endemic focus than Ethiopia.

Migrants and the homeless population, overlooked small outbreaks in Europe? Since the Second World War, there has been no reported outbreak of LBRF in Europe. In 2005, one study indicated that there may have been a small outbreak among the homeless population in Marseille based on the detection of immunoglobulin G (IgG) antibodies to *B. recurrentis* [29]. In Marseille, about 60% of the homeless population notably consists of migrants [29,160]. It is possible that an imported, unnoticed case could have caused a small epidemic among the high-risk population of homeless people. According to several studies, these people are commonly found infested with body lice, and in some instances further infected with other louse-borne pathogens, such as *B. quintana* [161–164]. A case report from Saudi-Arabia described *B. recurrentis* in a homeless man [165]. A report from Italy suggested that 2 migrants acquired the disease in a housing facility of newly arrived refugees. The authors noted that the disease

was only diagnosed because of the microscopic blood smear investigations for malaria conducted due to the patients' recent migration history. They further stated that febrile patients without travel history may receive empirical antibiotic treatment, which may result in the resolution of symptoms without further investigation [60]. Given the susceptibility of borreliae to common empirical antibiotic treatment and the unspecific symptoms of the disease, such minor outbreaks may be easily overlooked. Moreover, vulnerable population groups, such as the homeless, may have limited access to medical care. Under these premises, the introduction of a single case into such surroundings may be sufficient to create a small, unnoticed outbreak.

Thus, in regard to LBRF, close attention should be paid to patients from vulnerable population groups, such as migrants or the homeless that display febrile symptoms. Considering asymptomatic cases, publication bias, and possible temporary foci somewhere along migration routes, another reemergence of the disease should neither be neglected nor its epidemic potential be underestimated.

New vector? Interestingly, two studies found DNA of *B. recurrentis* in head lice obtained from humans in the Republic of Congo and Ethiopia [47,48]. They raise the question whether head lice can transmit human louse-borne pathogens. The evidence suggests that head lice, contrary to former belief, may act as a vector. However, further research is necessary to investigate the role of head lice in the transmission of louse-borne pathogens.

Diagnostic aspects

Microscopy. Microscopy remains the gold standard for diagnosing LBRF since the discovery of the organism in 1867. The sensitivity, however, is directly affected both by the number of spirochetes in the blood and whether the blood sample was taken during a febrile or afebrile period of the infection. Blood should be obtained during a febrile period [19,36–38,124], yet it is possible, though very hard, to find spirochetes during afebrile periods [124]. Although data on sensitivity are scarce, one study reported spirochetes in 38% of patients whose blood had been taken during an afebrile period [155], and another study reported positive microscopy in 5% during afebrile periods [27]. The number of positive results may be increased through the examination of repeated blood smears [36]. Furthermore, the results are often dependant on the fixative and staining method used [25]. Additionally, the level of experience the observer has is another issue that can influence sensitivity [166]. Five cases were identified that reported negative microscopy but positive PCR. In one case, blood was taken the day after initiation of adequate empirical antibiotic therapy [57], and in the other four, reasons remain unknown [69,167]. One study reported positive blood smears, but only after reexamination by an experienced microbiologist [65]. The sensitivity of 70% reported in 1969 was obtained through analysis of both tick-borne and LBRF cases [36], hence may not be fully representative for *B. recurrentis* specifically. Research into the validation of the enhancement methods is needed and has already been suggested [136].

Animal inoculation, serology, and PCR. Animal vaccination has been used to aid diagnosis, for research purposes and to some extent as a tool to differentiate between TBRF and LBRF. However, animal inoculation has never been routinely used for diagnostic reasons alone. Lack of standardized data and protocol is further limiting this method [25]. Historically, serological methods were extensively investigated, but the development was hampered by antigenetic variability and cross-reactivity [25,27,34,38,125–128]. To date, serology is not routinely used for diagnosis and is not recommended. PCR-based methods lack availability in poor countries, however, are unquestionably the best and arguably only means for a certain diagnosis of LBRF. Still, protocols for differentiation of *B. recurrentis* and *B. duttonii* need to be

established. Due to the proximity of these relapsing fever borreliae, development of specific diagnostic tools and accurate discrimination between the species are challenging [131].

Considering that roughly 90% of all published LBRF cases have been diagnosed by microscopy and that microscopy continues to be the most widely used and available diagnostic method, there is a need to determine its sensitivity and to evaluate how to increase sensitivity by serial investigations and/or enhancement methods. Negative microscopy results should be regarded critically, since the last available sensitivity data from 1969 suggested a sensitivity of 70% on a single blood smear. Positive microscopy should be regarded critically, especially in areas where TBRF is endemic. The data show that microbiological methods were mainly used in Europe. In Ethiopia, the country most affected by the disease, microscopy remains the main diagnostic tool.

Key Learning Points

- Currently, East Africa remains the only endemic focus of louse-borne relapsing fever (LBRF).
- Human migration has repeatedly imported the disease into non-endemic countries.
- Although polymerase chain reaction (PCR)-based methods are the only means of species identification, microscopy remains the gold standard in diagnosing LBRF.

Top Five Papers

1. Warrell DA. Louse-borne relapsing fever (Borrelia recurrentis infection). Epidemiol Infect. 2019;147:e106. doi: 10.1017/S0950268819000116
2. Bryceson ADM, Parry EHO, Perine PL, Warrell DA, Vukotich D, Leithead CS. A clinical and laboratory study of 62 cases in ethiopia and a reconsideration of the literature. QJM. 1970;39(1):129–70. doi: 10.1093/oxfordjournals.qjmed.a067198
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Supporting information

S1 Checklist. PRISMA Checklist. Twenty-seven-item checklist for systematic reviews. PRISMA, Preferred Reporting Items for Systematic Reviews and Meta-Analyses.
(DOC)

S1 Text. Review protocol. Established to conduct this systematic review.
(DOCX)

S2 Text. Data extraction sheet. Used for screening and selecting eligible publications.
(DOCX)

S3 Text. References. Reference list of included and excluded publications.
(DOCX)

S1 Fig. PRISMA flow diagram.
(PDF)

S1 Data. Data extracted from included studies. Excel spreadsheet containing, in separate sheets, the underlying numerical data.
(XLSX)

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PLOS NEGLECTED TROPICAL DISEASES

REVIEW

Louse-borne relapsing fever—A systematic review and analysis of the literature: Part 2—Mortality, Jarisch–Herxheimer reaction, impact on pregnancy

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Abstract

Louse-borne relapsing fever (LBRF) is a classical epidemic disease, which in the past was associated with war, famine, poverty, forced migration, and crowding under poor hygienic conditions around the world. The disease's causative pathogen, the spirochete bacterium *Borrelia recurrentis*, is confined to humans and transmitted by a single vector, the human body louse *Pediculus humanus corporis*. Since the disease was at its peak before the days of modern medicine, many of its aspects have never been formally studied and to date remain incompletely understood. In order to shed light on some of these aspects, we have systematically reviewed the accessible literature on LBRF since the recognition of its mode of transmission in 1907, and summarized the existing data on mortality, Jarisch–Herxheimer reaction (JHR), and impact on pregnancy.

Publications were identified by using a predefined search strategy of electronic databases and a subsequent review of the reference lists of the obtained publications. All publications reporting patients with a confirmed diagnosis of LBRF published in English, French, German, and Spanish since 1907 were included. Data extraction followed a predefined protocol and included a grading system to judge the certainty of the diagnosis of reported cases.

The high mortality rates often found in literature are confined to extreme scenarios. The case fatality rate (CFR) of untreated cases is on average significantly lower than frequently assumed. In recent years, a rise in the overall CFRs is documented, for which reasons remain unknown.

Lacking standardized criteria, a clear diagnostic threshold defining antibiotic treatment-induced JHR does not exist. This explains the wide range of occurrence rates found in literature. Pre-antibiotic era data suggest the existence of a JHR-like reaction also in cases treated with arsenicals and even in untreated cases.

LBRF-related adverse outcomes are observed in 3 out of 4 pregnancies.

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Citation: Kahlig P, Neumayr A, Paris DH (2021) Louse-borne relapsing fever—A systematic review and analysis of the literature: Part 2—Mortality, Jarisch–Herxheimer reaction, impact on pregnancy. PLoS Negl Trop Dis 15(3): e0008656. https://doi.org/10.1371/journal.pntd.0008656

Editor: Aparna Krishnavajhala, Baylor College of Medicine, UNITED STATES

Published: March 11, 2021

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Funding: The authors received no specific funding for this work.

Competing interests: The authors have declared that no competing interests exist.

Author summary

Louse-borne relapsing fever (LBRF) is an ancient epidemic disease, with descriptions dating back to Hippocrates' times. Linked to war, famine, poverty, forced migration, and crowding under poor hygienic conditions, the disease has accompanied humankind throughout history and, until 100 years ago, the disease was well known among physicians in Europe and North America. Since then, the disease has fallen into oblivion, and reports of the disease are largely confined to Ethiopia. However, the disease has recently resurfaced when it was recognized as a cause of fever in East African migrants (originating from Somalia, Eritrea, and Ethiopia) arriving to Europe. Since the disease was at its peak before the days of modern medicine, many aspects of the disease have never been formally studied and, to date, remain incompletely understood. In order to shed light on some of these aspects, we have reviewed and analysed the accessible literature on LBRF since the recognition of its mode of transmission in 1907, and compiled this two-part review focusing on epidemiology and diagnostic aspects (part 1) and mortality, JHR, and impact on pregnancy (part 2). We deliberately did not include an analysis on the antimicrobial treatment of LBRF, since Guerrier and Doherty have published a comprehensive meta-analysis on this aspect in 2011.

Introduction

Relapsing fevers are potentially fatal ectoparasite-borne diseases which are caused by spirochetes of the genus *Borrelia* and characterized by recurring episodes of fever [1].

Relapsing fevers are classified according to their transmitting vector which include tick-borne relapsing fever (TBRF) and louse-borne relapsing fever (LBRF). While several *Borrelia* spp. have been identified to cause TBRF, *Borrelia recurrentis* is the only species known to cause LBRF [2]. The incubation period of LBRF ranges from 4 to 18 (average 7) days. The attack starts abruptly with a fever that increases to nearly 40˚C in a few days, accompanied by rigors. Early symptoms include headache, dizziness, generalized aches and pains (affecting especially the lower back, knees, and elbows), anorexia, nausea, vomiting, and diarrhea. Upper abdominal pain, cough, and epistaxis develop later [3]. A petechial or ecchymotic rash, particularly involving the trunk, is seen in 2% to 8% of patients, and 7% to >70% of patients develop jaundice [4]. Subconjunctival hemorrhages and epistaxis are common (25%), hemoptysis, gastrointestinal bleeding, and retinal hemorrhages less so [3]. In severe cases, neurological involvement, myocarditis, acute respiratory distress syndrome, hepatic failure, spleen rupture, disseminated intravascular coagulation leading to intracranial, massive gastrointestinal, pulmonary or peripartum hemorrhage may occur [3]. Case fatalities between 30% and 80% [4–12] have been reported in untreated patients during major epidemics, but with antibiotic treatment mortality is reduced to 2% to 5% [13]. Untreated attacks resolve by crisis after 4 to 10 (average 5) days. This is followed by an afebrile remission of 5 to 9 days and succeeded by up to 5 relapses of diminishing severity, during which there may be epistaxis, but no petechial rashes [3]. The crisis which abruptly terminates untreated attacks is a consequence of specific bactericidal antibodies lysing spirochetes in the blood. The relapse phenomenon can be attributed to the antigenic variation of the bacterium's outer membrane lipoproteins (vmp) [14]. Once antibodies have been generated against a specific vmp variant, a new vmp variant is expressed by the *Borrelia*. After the removal of the majority population through antibodies, the minority variant population expands until antibodies are also generated against the new vmp variant. Also linked to vmp is the phenomenon of Jarisch–Herxheimer reaction (JHR) [15].

Named after the researchers who first described the reaction in patients with syphilis [16–18], the JHR has been described to occur after the elimination of spirochetes in patients with syphilis, leptospirosis, Lyme disease, and relapsing fever [18]. Around 1970, several authors had studied this reaction in LBRF patients after treatment with antibiotics [4,18–21]. The observations resembled the 4 phases described for fever induced by endotoxin [22]. JHR occurs about 45 minutes to 2 hours after administration of antibiotic treatment and is characterized by restlessness followed by a chill phase of acute and intense rigors lasting 10 to 30 minutes. During this chill phase, body temperature, pulse, and respiratory rates rise steeply, and associated delirium (sometimes leading to dangerous behavior) as well as potentially fatal hyperpyrexia may occur. The flush phase following the chill phase is characterized by a systemic vasodilatation with a fall in blood pressure (possibly leading to collapse and intractable hypotension). This is accompanied by profuse sweating and a slowly declining body temperature, threatening the patient for several hours. The prodromal phase, the chill phase, the flush phase, and the phase of defervescence have been described in detail by several authors [4,20]. Both the spontaneous crisis that terminates untreated attacks and the JHR induced by antibiotic treatment show pathophysiological features of a classic endotoxin reaction mediated by proinflammatory cytokines (tumor necrosis factor α [TNF-α], interleukin 6 (IL-6), IL-8) [18]. A recent review has addressed the current understanding of the pathogenesis of JHR, which seems to be of multifactorial genesis and is not fully understood yet [18]. A JHR occurs in up to 80% to 90% of treated patients [13], and symptoms usually resolve in a few hours. Even though JHR is rarely fatal, it does enhance the risk for complications and a fatal outcome (liver and renal function impairment, acute respiratory distress syndrome, myocardial injury, hypotensive shock, seizures, strokes, induction of uterine contractions in pregnancy) [13,18]. Besides supportive treatment, administration of antibodies against TNF-α have been proven effective in the management of the JHR [23–25].

A variety of antibiotic drugs, including tetracyclines, erythromycin, chloramphenicol, and penicillin, are effective as single-dose treatment (given orally or parenterally) in LBRF [13]. A meta-analysis of antibiotic treatment of LBRF published by Guerrier and Doherty in 2011 found no significant difference between tetracycline and penicillin with regard to mortality rate. Tetracycline use was found to be associated with faster resolution of fever and a lower risk of relapse compared to penicillin treatment. However, tetracycline use appears to be associated with a higher risk for JHR compared to penicillin treatment [26].

In a study on TBRF in pregnant women conducted in Rwanda, the risk of pregnancy loss was 33% and that of perinatal mortality 15% [27]. In another study on TBRF in pregnant women in Tanzania, the risk of pregnancy loss was 30% and that of perinatal mortality 15% [28]. In contrast to TBRF, data on the impact of LBRF on pregnancy is scarce. A publication on LBRF published in 1970 states: "*Abortion or miscarriage is the usual fate of pregnancy. That we had one live premature birth and three uninterrupted pregnancies is probably due to a standard of care not normally available in an epidemic. That congenital infection and abortion are the rule is clear from El Ramly's post-mortem studies (1946)*" [4].

With this systematic review, we aim to summarize the available data and address the following main review questions: What is the mortality of untreated and treated LBRF? What are the reported frequencies of the JHR in LBRF? What is the impact of LBRF on pregnancy?

Methods

A systematic review protocol established for this review is available in the Supporting information section (S1 Text). The electronic databases Biosis, CINAHL, Cochrane Library, Current Contents Connect, Elsevier, EMBASE ovid, Ovid MEDLINE, PMC, PUBMED, SCOPUS, and

Web of Science were searched on 04/Oct/2017 using the search term: ((Relapsing AND fever AND (Louse OR Lice OR (Pediculus AND humanus))) OR (Borrelia AND recurrentis) OR LBRF). A second and third search, using the same search term on the same databases, was conducted on 07/Aug/2018 and 17/Jun/2019, respectively. After checking for and removing duplicates (using Endnote software and manually [29]), publications were prescreened by checking titles and abstracts if they concerned patient(s) with the diagnosis of LBRF. Publications not reporting patient(s) with the diagnosis of LBRF were excluded. The remaining publications were then assessed in full text for eligibility according to the inclusion criteria: reporting conclusively diagnosed case(s) of LBRF and published after 1907 (the year of the discovery of the disease's transmission) and published in English, French, German, or Spanish. Publications not fulfilling these inclusion criteria were excluded. Publications that could neither be retrieved through their respective journals, nor by contacting libraries, or after contacting the authors, were classified as "not retrievable" and excluded. Additional relevant publications identified when reading the full-text articles or checking their reference lists were reviewed and included if they fulfilled the inclusion criteria ("snowball" search strategy). Finally, a subsearch was conducted on (i) outcome, (ii) JHR, and (iii) impact on pregnancies. Inclusion criteria for (i) outcome were: available conclusive data on number of patients treated/untreated plus treatment plus mortality. Inclusion criteria for (ii) JHR were: available conclusive data on number of patients treated/untreated plus number of patients with a JHR/JHR rate plus reaction after treatment described as JHR. In order to investigate a potential influence of the study objectives/endpoints on the reported incidence rates of JHR, we conducted a subgroup analysis. Thus, the 42 included studies were divided into 2 groups, one primarily focusing on JHR and the other primarily focusing on other aspects of LBRF. Inclusion criteria for (iii) impact on pregnancies were: available conclusive data on the number of cases plus pregnancy outcome.

A data extraction sheet for screening and selecting eligible publications was developed and is available in the Supporting information section (S2 Text). The following data were extracted from eligible publications using a standardized excel spreadsheet: patient characteristics (number of patients, age, gender, origin, occupation, social status, and way and duration of migration), diagnostic method (microscopy and molecular method), symptoms and signs (fever, chills, myalgia, headache, hepatomegaly, splenomegaly, signs of hemorrhage, and others), treatment (number of treated and untreated patients, drug, dosage, and duration and route of administration), and outcome (JHR, abortion/stillbirth, premature delivery, and mortality).

To minimize bias, the same reviewer conducted a second full data extraction $\geq$1 month after the first extraction. Discrepant results and unclear cases were resolved by consulting a second reviewer. In order to consider the probability of a correct diagnosis of LBRF, all reviewed cases were graded according to the used diagnostic method and respectively classified (Table 1).

The review followed the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) statement (S1 Checklist).

Results

Our search strategy identified 4,943 publications of which 184 proved eligible for being included and analysed (Fig 1; S1 Fig; S1 Data). Lists of included and excluded publications are available in the Supporting information section (S3–S6 Text).

Mortality

After full-text assessment of the eligible 184 publications, 102 proved eligible after applying subsearch inclusion criteria for "outcome."

Table 1. Diagnostic grading system to judge the certainty of the correct diagnosis of LBRF.

Diagnostic method	Grade of diagnostic certainty	Case classification	Comment
PCR-based method	A	Confirmed diagnosis	Highest level of evidence for correct diagnosis
Microscopy	B	Microscopy diagnosis	Second highest level of evidence for correct diagnosis; microscopic identification of spirochetes during LBRF epidemics or in countries with current endemic foci leaves little doubt of the certainty of the diagnosis and may be regarded with an almost equal level of certainty as grade A
Paired serology	C	Indirect evidence	Intermediate level of evidence for correct diagnosis due to limited sensitivity and specificity of the method; paired serology, demonstrating seroconversion or increment of titer, is considered superior to single titer serology
Single titer serology	D	Indirect evidence	See comment under C above
Clinical diagnosis	E	Clinical diagnosis	Lowest level of evidence for correct diagnosis

PCR, polymerase chain reaction.

Note: Animal inoculation, historically used as supportive diagnostic method in LBRF research, was not considered a means of conclusive diagnosis and was thus not included in the evaluation.

https://doi.org/10.1371/journal.pntd.0008656.t001

The identified treated cases (n = 5,969) were divided into 2 groups according to the treatments used: treated with arsenicals (n = 2,338) and treated with antibiotics (n = 3,631). The group treated with antibiotics was not divided further, since a recent systematic review has already compared the antibiotic treatments of LBRF [26]. Identified studies with data on untreated cases are listed in Table 2.

An overview on all identified LBRF cases with the according case fatality rates (CFRs) is shown (Table 3).

Fig 2 depicts the CFR of LBRF cases from the reviewed studies according to treatment modality over time.

Between 2010 and 2019, 127 patients were treated with an overall CFR of 6.3%. Of these, 53 were treated in the frame of a study conducted in Ethiopia which reported a CFR of 13.21%. The other patients were treated in Europe (n = 72), Saudi Arabia (n = 1), and Israel (n = 1). Among these, 1 fatal case occurred in Europe, which corresponds to a CFR of 1.35%. An overview on CFR in relation to treatment and geographical region is shown in Fig 3. Reported signs, symptoms, and factors associated with mortality and poor prognosis are summarized in Table 4.

Jarisch–Herxheimer reaction (JHR)

After full-text assessment, 184 publications were eligible. After applying subsearch inclusion criteria for "JHR," 42 publications were included and are summarized in Table 5.

We identified 3 studies [24,59,86] in which already previously published patient data were republished [4,23,50]. Data from these 3 studies were compiled (data of identical patients/cases).

Overall, JHR was observed in 1,452 of the 2,618 reported cases, corresponding to an incidence rate of 55.8%. Of note, the incidence rate of JHR rate in studies focusing on JHR (n_{cases} = 603) versus studies not focusing on JHR (n_{cases} = 2015) was 71.78% and 50.62%, respectively.

Pregnancies

After full-text assessment of the eligible 184 publications, 14 proved eligible after applying subsearch inclusion criteria for "pregnancies." The characteristics of these 14 included studies are summarized in Table 6.

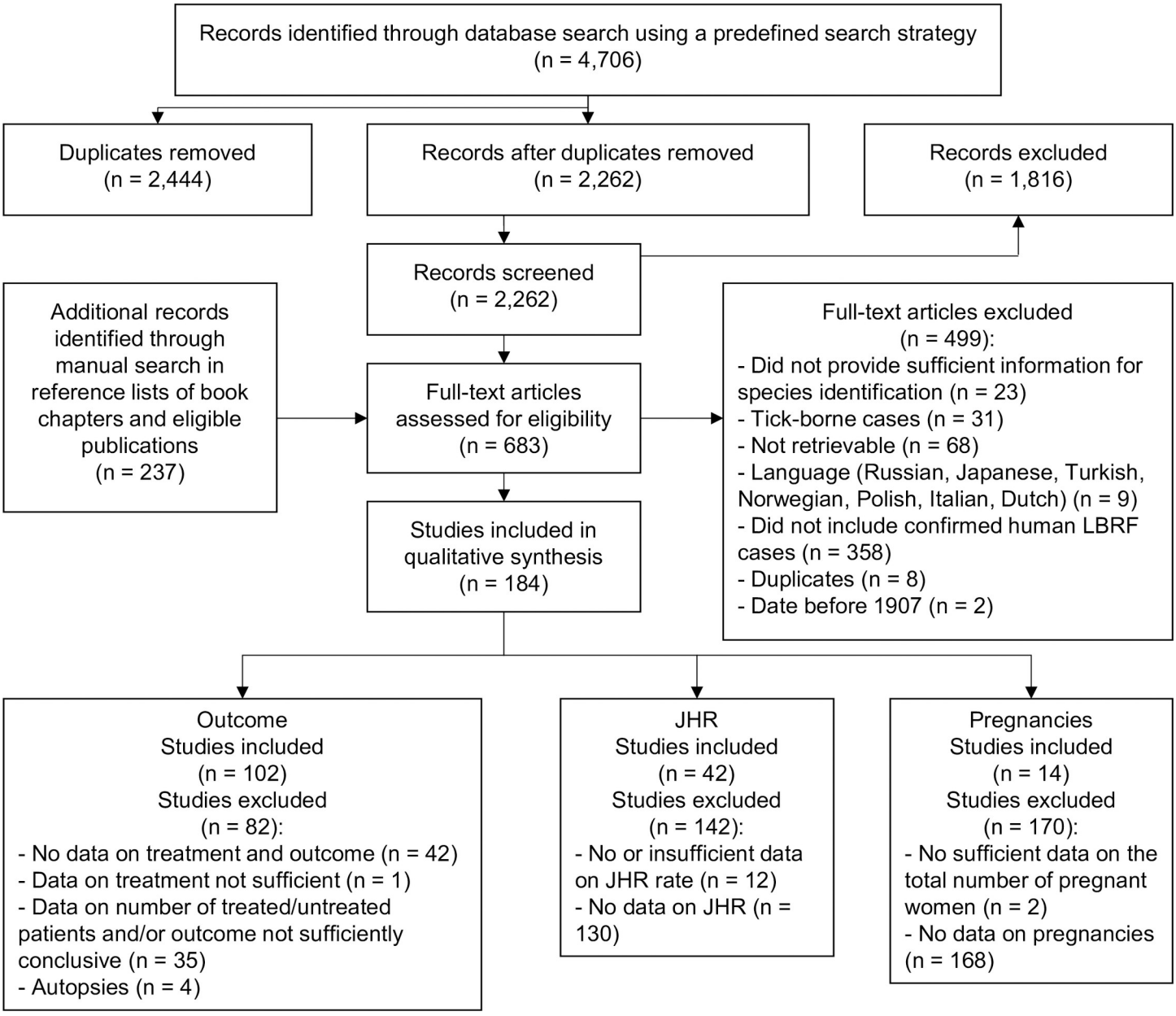


Fig 1. Flow diagram of search and selection of eligible publications.

https://doi.org/10.1371/journal.pntd.0008656.g001

In total, 206 pregnancies and their outcome were reported in the 14 included studies (Table 6): The overall incidence of an adverse pregnancy outcome was 70.9% (n = 146). When considering only cases reported by studies graded at least "B" (n = 122 pregnancies), adverse pregnancy outcomes were reported in 76.2% of the cases (n = 93) (Fig 4).

Discussion

Mortality

Observational factors potentially related to high case fatality. High mortality rates in untreated cases (30% to 80%) [4–12] were mostly traced back to reports of epidemics based

Table 2. Details of all identified studies that included untreated cases, in chronological order and according to diagnostic grading.

First Author	Year of pub.	Country	Grade of diagnostic certainty	Number of patients	Number of fatal cases	Case fatality rate in %	Ref.
Duchamp	1917	Serbia	B	68	0	0	[30]
Porot	1917	Algeria	E†	2	0	0	[31]
Margolis	1919	Poland	E‡	267	0	0	[32]
Del Prado	1920	Peru	B	2	0	0	[33]
Fry	1920	Iran	B	11	1	9.1	[34]
Jouveau-Dubreuil	1920	China	B	89	12	13.5	[35]
Sinton	1921	Iran	B	21	0	0	[36]
Sergent	1922	Algeria	E‡	145	0	0	[37]
McCulloch	1925	Nigeria	B	68	32	47.1	[38]
McCulloch§	1925	Nigeria	B	16	2	12.5	[38]
Chu	1931	China	B	19	0	0	[39]
Chung	1936	China	B	1	0	0	[40]
Chung	1939	China	B	56	8	14.3	[41]
Wolman	1944	Ethiopia	B	103	5	4.9	[42]
Wolman	1945	Egypt	B	80	1	1.3	[43]
Ingraham	1946	Egypt	B	53	0	0	[44]
Perine	1983	Ethiopia	B	6	0	0	[45]

Pub, publication; Ref, reference.

† Report of treated cases, including 2 untreated cases which were diagnosed retrospectively based on clinical presentation.

‡ The use of microscopy is mentioned in the descriptions of a few cases. However, whether microscopy was systematically used or only in some cases is unclear. Thus, the studies were conservatively graded "clinically diagnosed."

§ There is a striking difference in CFR among the untreated group, especially in McCulloch's study [38], which contains a cohort of prisoners responsible for the high CFR. A further group was added to evaluate the impact when the cohort of prisoners is excluded.

https://doi.org/10.1371/journal.pntd.0008656.t002

either on observations by the authors or on renditions of observational reports by colleagues. In both cases, these are rough estimations without any diagnostic evidence [5,8,90]. In other instances, high rates trace back to coinfections with other diseases [38,90–93]. Additionally, some authors attribute the striking differences in reported mortality rates to the general poor health and nutritional condition of the affected population, the lack of medical and sanitary measures, and sociopolitical factors like war [10–12,35,49,92]. Exemplarily, observations from

Table 3. Numbers of diagnosed cases and respective CFR.

Cases diagnosed by microscopy and/or PCR			Overall†	
	Number of cases	Case fatality rate	Number of cases	Case fatality rate
Untreated	*n* = 577	10.2%	*n* = 991	6%
Untreated‡	*n* = 525	5.5%	*n* = 939	3.1%
Treated	*n* = 5,893	4%	*n* = 5,969	4%
Arsenicals	*n* = 2,262	5.2%	*n* = 2,338	5%
Antibiotics	*n* = 3,631	3.3%	*n* = 3,631	3.3%

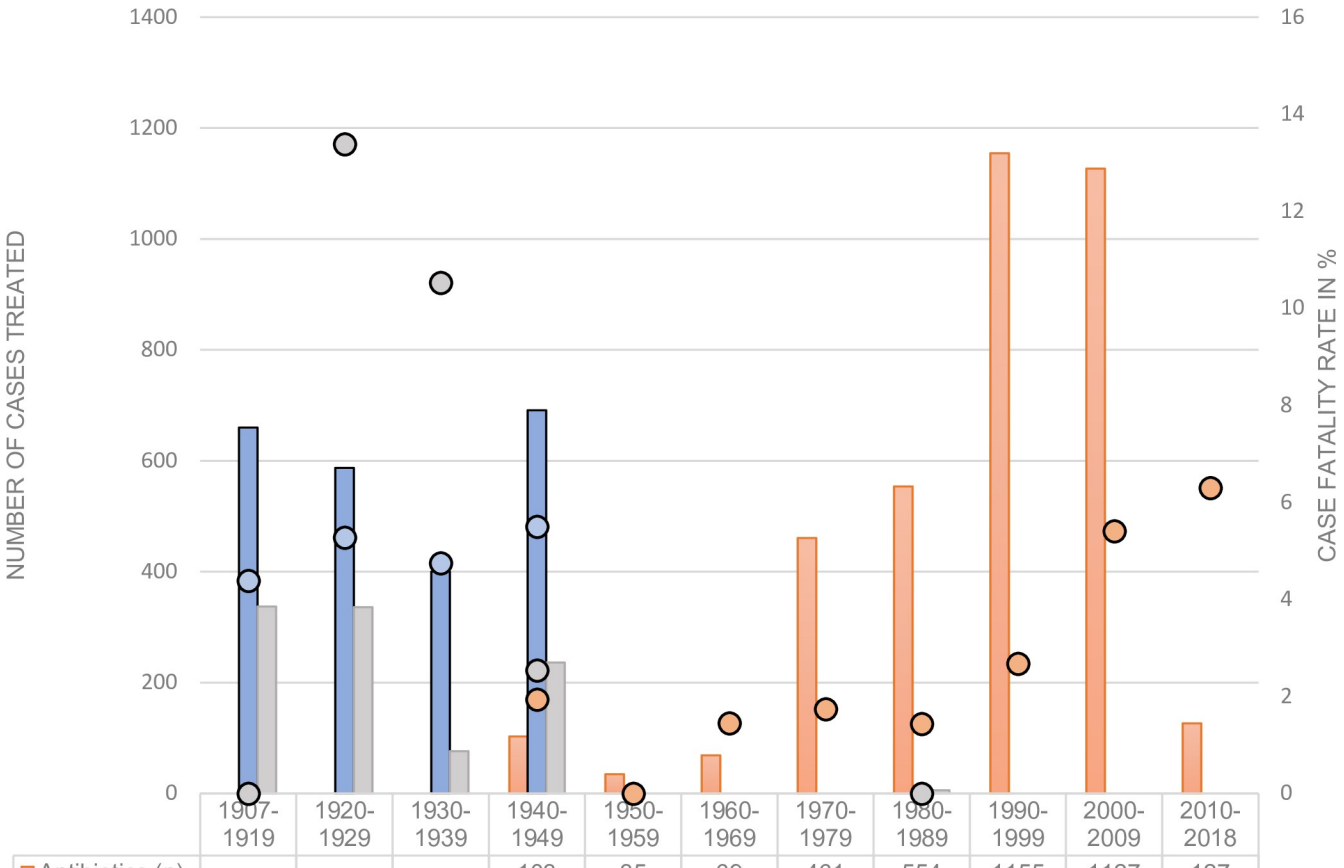
† Includes cases that were categorised as clinically diagnosed according to diagnostic grading E.

‡ Cohort of prisoners is excluded.

PCR, polymerase chain reaction.

Evaluated antibiotics: Ampicillin, Amikacin, Cefotaxime, Ceftriaxone, Cefuroxime, Chloramphenicol, Chlortetracycline, Clarithromycin, Doxycycline, Erythromycin, Meropenem, Metronidazole, Penicillin, Tetracycline-HCl; evaluated arsenicals: Acetylarsan, Arrhenal, Arsalyt, Arsphenamine (Arsenobenzol, Salvarsan), Arsenobillon, Galyl, Ludyl, Mepharsen, Neoarsphenamine (Neosalvarsan, Novarsenobenzol), Neoiacol, Olarsol

https://doi.org/10.1371/journal.pntd.0008656.t003



	1907-1919	1920-1929	1930-1939	1940-1949	1950-1959	1960-1969	1970-1979	1980-1989	1990-1999	2000-2009	2010-2018
Antibiotics (n)				103	35	69	461	554	1155	1127	127
Arsenicals (n)	660	587	400	691							
Untreated (n)	337	336	76	236				6			
Antibiotics (%)				1.94	0	1.45	1.74	1.44	2.68	5.41	6.3
Arsenicals (%)	4.39	5.28	4.75	5.5							
Untreated (%)	0	13.39	10.53	2.54				0			

Fig 2. CFR of LBRF cases from the reviewed studies according to treatment modality over time. The dots correspond to the CFR of the respective group. The columns represent the absolute number of cases. The table gives detailed information on the data for each segment of time. Note: Since the data set is compiled from published literature reporting on cases, outbreaks and clinical studies from different regions, different populations as well as different seasons, no trend in annual incidence of LBRF can be inferred.

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China noted the difference between an outbreak among "robust" soldiers and among poor villagers, where mortality had been near 0% and 50%, respectively [35]. CFRs and the course of the disease depend on the general condition and health of the population [94,95]. Additionally, a study found the disease to be more common and have an atypical course during famine years, compared to other years [94]. However, some authors noted that the disease is relatively benign, given appropriate treatment within a short time after the manifestation of symptoms [49], as Gaud and Morgan's quote of Charles Nicolle shows: "No one need to die of relapsing fever provided that it is diagnosed in time" [49].

Untreated cases and analysis of CFR. Anecdotal CFR of up to 80% are described in literature [4–12]; however, evidence-based data only show CFR up to 47.1% (Table 2). One study stands out with a CFR of 47.1%, which is the only data reaching the reported high CFRs (Table 2). The study investigated relapsing fever in Nigeria, including 52 of the untreated cases

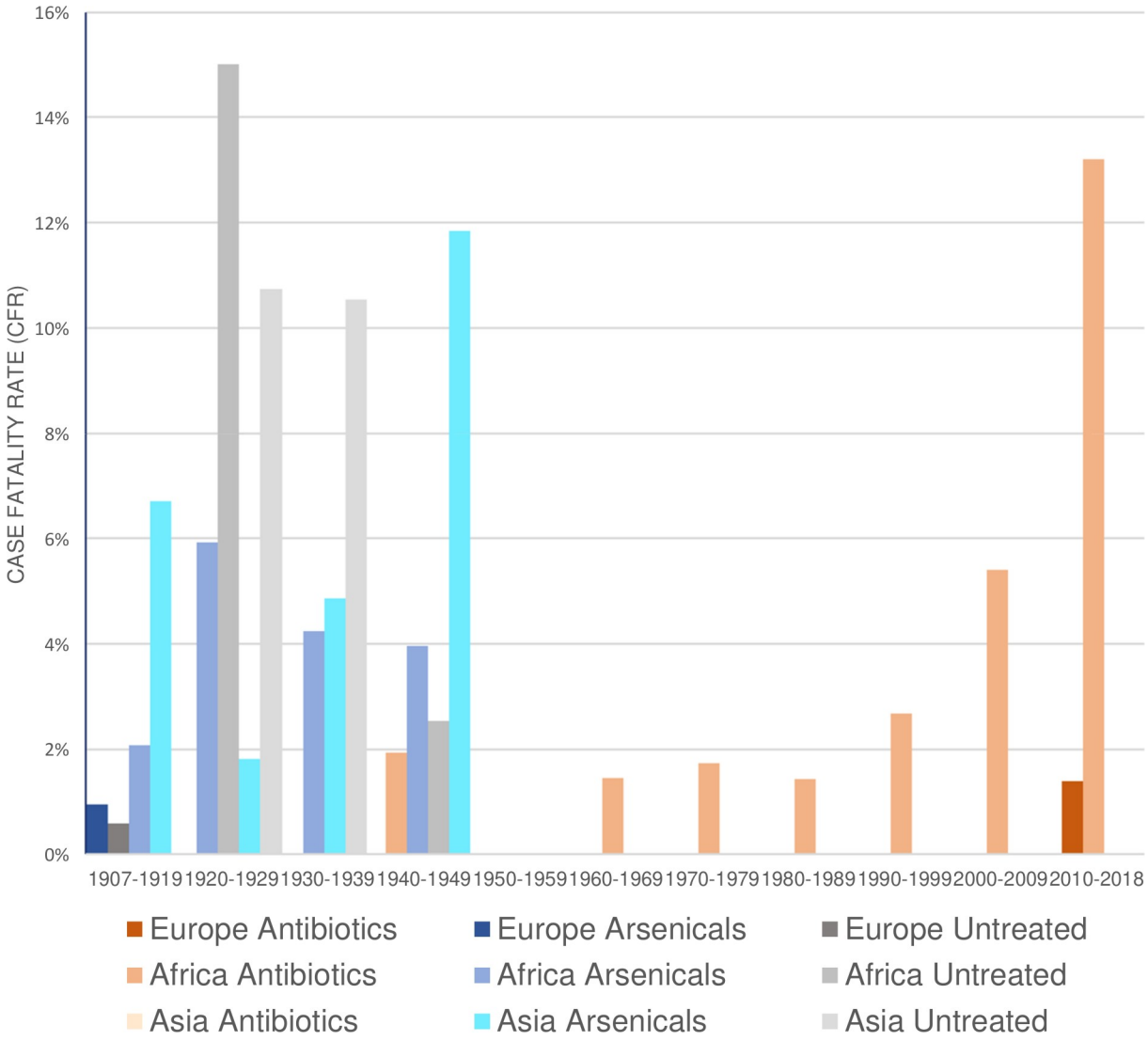


Fig 3. CFR in relation to treatment modality, time, and geographical region.

https://doi.org/10.1371/journal.pntd.0008656.g003

which originated from a prison in Kano, among whom 30 died resulting in a mortality rate of 57.69%. The reasons for the strikingly high CFR remain unknown. The only information available is that the author conducted weekly inspections only on the inmates. Coinfections or poor nutritional, medical, and health conditions cannot be excluded. Only 2 patients died out of the remaining 16 reported untreated patients, accounting for a mortality rate of 12.5% [38]. The prison series may be regarded as an extreme scenario. Interestingly, CFR drops roughly by half, when the extreme scenario is excluded (Table 3). The overall CFR for untreated cases was only slightly higher than the overall CFR for patients treated with arsenicals. When excluding the prison series, the overall CFR in untreated patients is 3.1%, which is lower than the overall CFR for patients treated with antibiotics. This demonstrates the impact of an insufficiently investigated extraordinary scenario on reported CFRs. Still, the overall results strikingly contrast with the rates of 40% to 80% that have been reported earlier, and published available data suggest that untreated LBRF may not be as deadly as reported earlier. Since the mortality rate is influenced by many factors, the numbers given may not reflect every situation. However,

Table 4. Risk factors, symptoms, and signs for mortality and poor prognosis reported in literature.

First Author	Year of pub.	Country	Symptoms/Signs/Factors associated with high mortality and poor prognosis	Ref.
Jouveau-Dubreuil	1920	China	Poor health condition prior to infection	[35]
Russell	1931	Ghana	Jaundice	[46]
Chang	1938	China	Jaundice, enlargement of liver	[47]
Chung	1939	China	Jaundice, pneumonia	[41]
El Ramley	1946	Egypt	Critical health condition due to late admission, jaundice, pneumonic signs	[48]
Gaud	1948	North Africa	Poor health condition prior to infection, epistaxis, hemoptysis, intestinal hemorrhages	[49]
Salih	1977	Ethiopia	Epistaxis, jaundice	[50]
Borgnolo	1993	Ethiopia	Abdominal guarding, jaundice, disturbed consciousness, high spirochetemia	[51]
Seboxa	1995	Ethiopia	JHR	[7]
Mitiku	2002	Ethiopia	JHR	[52]
Ramos	2004	Ethiopia	Vomiting, admission more than 4 days after the onset of symptoms	[53]
Nordmann	2018	Ethiopia	Critical health condition due to late admission	[54]

JHR, Jarisch–Herxheimer reaction; Pub, publication; Ref, reference.

https://doi.org/10.1371/journal.pntd.0008656.t004

they reflect the overall rates and tendencies from all confirmed LBRF cases that were identified through published literature research.

Benefit of therapy. Since antibiotic treatment of LBRF may trigger a strong and possibly even lethal JHR, the benefit of specific therapy has been questioned in certain situations. However, there is also no evidence to support a beneficial outcome when treatment is withheld. As suggested by Wolff and agreed to by Bryceson and colleagues, patients in a critical condition should primarily be treated symptomatically. Only after the attack should specific treatment be given [4,95]. Today, this approach of "resuscitate first—but then treat" appears to be the most widely accepted consensus regarding the management of LBRF patients.

Rise of CFR in treated cases. Figs 2 and 3 demonstrate that antibiotic treatment reduced the CFR of LBFR compared to untreated cases and treatment using arsenicals in mid-20th century. However, the figures also show that over the past few decades, the CFR reported in the literature seems to have increased in patients treated with antibiotics. Fig 3 demonstrates both the geographic distribution and the increased number of studies publishing cases from the African continent, primarily East Africa. The patients treated in Europe were immigrants and refugees that were taken care of in European hospital centers. The patients in Ethiopia were mostly men from poor socioeconomic status, who could not afford costs for diagnosis and treatment. The poverty and the lack of awareness of LBRF prevented or delayed admission to medical care [54]. Two factors may have been responsible for this striking difference in CFR: on the one hand, the difference in the general condition and nutritional status of the 2 populations; on the other hand, the availability of medical care to the people affected. Unknown factors such as additional coinfections in the population from Arsi (Ethiopia) may have contributed. The rise over the past few decades may indicate a decline in health status of the affected population in Ethiopia. Further studies investigating the reasons for these high CFR in Ethiopia are needed.

The Jarisch–Herxheimer reaction

"Resolvement by crisis" in untreated cases. In LBRF, a JHR-like reaction, often called crisis, has been observed after treatment since the beginning of the last century [34,41,47,88,95–98]. Even untreated cases have been reported to exhibit a spontaneous resolvement of fever through crisis, which has been recognized by researchers to resemble the features and

Table 5. Details of all the included studies that published data on the JHR, in chronological order and according to diagnostic grading.

First Author	Year of pub.	Country	Diagnostic Grade	Number of patients	JHR *n* (%)	Study focus	Ref.
Schofield	1968	Ethiopia	B	10	10 (100)	JHR	[19]
Bryceson	1970	Ethiopia	B	62	62 (100)	JHR	[4]
Warrell	1970	Ethiopia	B	19	19 (100)	JHR	[20]
Bryceson	1972	Ethiopia	B	9	9 (100)	JHR	[55]
Knaack	1972	Ethiopia	B	25	14 (56)	Treatment	[56]
Perine	1974	Ethiopia	B	26	26 (100)	Treatment	[57]
Galloway	1977	Ethiopia	B	15	15 (100)	JHR	[58]
Salih	1977	Sudan	B	160	23 (14.4)	Treatment	[59]
Butler	1979	Ethiopia	B	90	78 (86.7)	JHR, endotoxins	[60]
Butler	1980	Ethiopia	B	11	9 (81.8)	JHR, phagocytosis	[61]
Perine	1983	Ethiopia	B	377	377 (100)	Treatment, 6 untreated cases	[45]
Teklu	1983	Ethiopia	B	33	33 (100)	JHR	[62]
Warrell	1983	Ethiopia	B	12	12 (100)	JHR	[63]
Zein	1987	Ethiopia	B	132	28 (21.2)	JHR	[64]
Brown	1988	Somalia	B	37	0 (0)	Symptoms, Epidemiology	[65]
Daniel	1992	Ethiopia	B	80	26 (32.5)	Children	[66]
Gebrehiwot	1992	Ethiopia	B	120	48 (40)	Treatment	[67]
Mekasha	1992	Ethiopia	B	63	10 (15.9)	Children	[68]
Negussie	1992	Ethiopia	B	17	14 (82.4)	JHR	[69]
Borgnolo	1993	Ethiopia	B	103	63 (61.2)	Children	[51]
Borgnolo	1993	Ethiopia	B	389	168 (43.2)	Symptoms, epidemiology	[6]
Knox	1994	Ethiopia	E	51	26 (51)	JHR, anti-TNF	[25]
Cuevas	1995	Ethiopia	B	25	14 (56)	JHR, cytokines	[70]
De Jong	1995	Southern Sudan	B	22	4 (18.2)	Epidemiology	[71]
Seboxa	1995	Ethiopia	B	184	54 (29.4)	Treatment	[7]
Fekade	1996	Ethiopia	B	49	36 (73.5)	JHR, anti-TNF	[23]
Remick	1996	Ethiopia	B	19	19 (100)	JHR	[72]
Cooper	2000	Ethiopia	B	49	48 (98)	JHR	[73]
Mitiku	2002	Ethiopia	B	262	83 (31.7)	Epidemiology	[52]
Tewdros	2002	Ethiopia	B	106	80 (75.5)	Symptoms, epidemiology	[74]
Alfaifi	2014	Saudi Arabia	B	1	1 (100)	Case report	[75]
Hoch	2015	Germany	14A,1B	15	10 (66.7)	Epidemiology	[76]
Wilting	2015	Netherlands	A	2	2 (100)	Case series	[77]
Ciervo	2016	Italy	A	3	1 (33.3)	Case series	[78]
Costescu	2016	Belgium	B	2	1 (50)	Case series	[79]
Lucchini	2016	Italy	A	5	2 (40)	Case series	[80]
Osthoff	2016	Switzerland	A	4	1 (25)	Case series	[81]
Seilmaier	2016	Germany	A	25	22 (88)	Case series	[82]
Von Both	2016	Germany	A	1	1 (100)	Case report	[83]
Zammarchi	2016	Italy	A	1	1 (100)	Case report	[84]
Hytonen	2017	Finland	A	2	2 (100)	Case series	[85]
Nordmann	2018	Ethiopia	B	54	4 (7.4)	Symptoms, epidemiology	[54]

JHR, Jarisch–Herxheimer reaction; Pub, publication; Ref, reference; TNF, tumor necrosis factor.

https://doi.org/10.1371/journal.pntd.0008656.t005

severity of the JHR in certain cases [50,63,99]. A study comparing the effects of different antibiotic treatments followed 6 untreated cases that had been admitted during this spontaneous crisis. These observations showed that the JHR was indistinguishable from the spontaneous

Table 6. Details of all the studies including data on LBRF infection during pregnancy, in chronological order and according to grade of diagnostic certainty.

First Author	Year of pub.	Country	Grade of diagnostic certainty	Number of pregnancies	Adverse pregnancy outcomes *n* (%)	Ref.
Jukes	1912	India	B	1	1 (100)	[87]
Sergent	1922	Algeria	E†	30	4 (13.3)	[37]
McCulloch	1925	Nigeria	B	2	2 (100)	[38]
Robertson	1932	China	B	5	3 (60)	[88]
Chung	1939	China	B	2	1 (50)	[41]
Benhamou	1945	Algeria	E‡	54	49 (90.7)	[89]
El Ramley	1946	Egypt	B	79	68 (86.1)	[48]
Ingraham	1946	Egypt	B	2	2 (100)	[44]
Garnham	1947	Kenya	B	3	3 (100)	[12]
Bryceson	1970	Ethiopia	B	6	3 (50)	[4]
Brown	1988	Somalia	B	4	2 (50)	[65]
Gebrehiwot	1992	Ethiopia	B	2	1 (50)	[67]
Borgnolo	1993	Ethiopia	B	15	6 (40)	[6]
De Jong	1995	Southern Sudan	B	1	1 (100)	[71]

LBRF, louse-borne relapsing fever; Pub, publication; Ref, reference.

† The use of microscopy is mentioned in the descriptions of a few cases. However, whether microscopy was systematically used or only in some cases is unclear. Thus, the study was conservatively graded "clinically diagnosed."

‡ No information on the use of microscopy. Thus, the study was graded "clinically diagnosed."

https://doi.org/10.1371/journal.pntd.0008656.t006

crisis [45]. Unfortunately, there are no published studies which have investigated the process in detail during the natural course of the disease. It would be of great academic interest to compare the clinical and laboratory changes during the spontaneous reaction and the reaction after antibiotic treatment.

Detection bias. Lack of standardized defining criteria: The observed incidence rate of JHR reported in studies range from 0% to 100% (Table 5). A reason for the discrepancies may be that there was neither a uniform protocol or guidelines for the classification of severity, nor a standardized definition or threshold from which point on the observations may be regarded as a JHR. Only 3 studies noted a predefined grading system [7,23,24,55]. A study stated that comparison is hampered due to this issue and noted further that some studies include critically

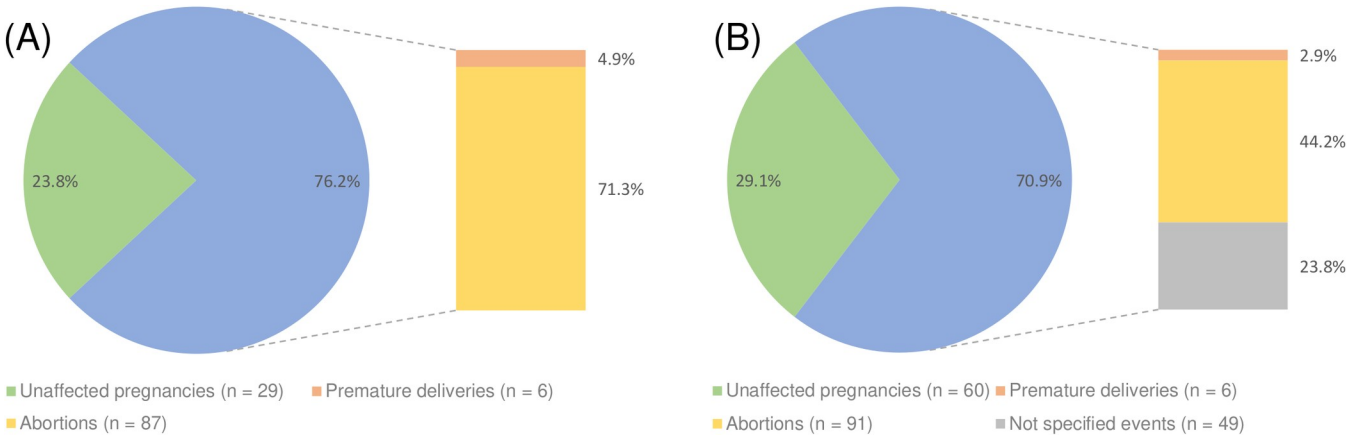


Fig 4. Adverse pregnancy outcomes in LBRF cases. (A) Microscopically diagnosed cases. (B) Microscopically and clinically diagnosed cases. One study reported adverse pregnancy outcomes without specifying them (gray color) [89].

https://doi.org/10.1371/journal.pntd.0008656.g004

ill patients, while others exclude this group of patients [7]. Another study reflected on its own results of a JHR rate of only 7.4% and states that these low results may have been caused by the lack of defining criteria for the occurrence of a JHR [54].

Influence of monitoring: Our data suggest that studies investigating JHR report higher incidence rates than studies which primarily focus on other aspects (Table 5). A review suggested that JHRs are often not recognized, thus underreported and easily overlooked [18]. Most studies investigating the pathophysiology after treatment included close monitoring of patients, often observing most of the parameters that are likely to change during JHR, hence increasing the likelihood of the reaction being detected. Mild reactions, on the other hand, may be easily overlooked without close monitoring, considering that they already peak 4 hours after treatment [18]. The issue may be demonstrated in the 37 cases studied by Brown and colleagues, who reported no JHR in the series. However, there was no information available about a threshold or definition used to identify a potential JHR. Intriguingly, rigors were observed in 30% of patients. The aims of the study were to survey a refugee camp and identify clinical criteria for diagnosing LBRF in the absence of laboratory facilities. As the resources in the given situation were not suitable for proper monitoring, the question may retrospectively be raised as to whether a certain number of JHRs could have been missed [65]. Another issue may be the degree of experience of the involved personnel and scientific researchers with diagnosing JHR. Perine, an author focusing on JHR [20,55,63,100], published JHR rates of 100% when studying the treatment of LBRF both in 1974 and 1983 (Table 5). Salih reported a JHR rate of 14.4% when researching treatment of LBRF in 1977. It seems likely that experience in detection influenced the striking difference. The pathophysiology of the reaction after treatment suggests a chance of missing the diagnosis when vital signs are monitored only 3 to 4 times a day, which may be assumed standard procedure in hospitalized stable patients.

Developing a grading system: There is a need to develop and widely apply a clear definition as to what should be regarded as a JHR. Until then, published data can hardly be compared to one another, since most studies do not report the process of monitoring and the threshold from which they consider a reaction as a JHR. Seboxa and Rahlenbeck defined a JHR as an increase in body temperature (>1˚C), tachycardia, and a drop in systolic and diastolic blood pressure greater than 10 mmHg during 4 hours after treatment [7]. If further factors were to be added, the JHR might be objectified more precisely. An initial rise in body temperature, a rise in heart rate, rigors, an initial rise in blood pressure, tachypnoea, a late drop in arterial blood pressure, and a decline of body temperature could be easily measured and added to a protocol. If some of these changes were to be detected after treatment, a JHR might be diagnosed.

Redefining the reaction after treatment: The data suggest that defining the JHR as a reaction after antibiotic treatment should be reconsidered. It seems likely that there is always a reaction, which is altered by administering antimicrobial drugs, and thus more likely to be detected. Detection is favored by the circumstance of patients being monitored to various degrees in a medical facility.

Pregnancies

Lack of data. To the best of our knowledge, this is the first review regarding the question of the influence of the *B. recurrentis* infection on pregnancy outcomes. Finding an explanation for these high rates is difficult due to scarce data. Most cases are simply noted in the reports without any further information [37,38,44,65,67,71].

Descriptions of cases and the question of congenital infection. However, there are a few cases of LBRF described in detail. In one case, a mother was admitted on the third day of fever

with the pregnancy intact. On the fifth day of fever, labor began just as her body temperature began to fall, and she gave birth to a stillborn child. The second case was of a woman admitted in labor on the seventh day of fever, when birth was given to a premature infant. This infant developed a fever with spirochetemia on the sixth day and died on the seventh day after birth. The postmortem examination found jaundiced conjunctivae, a collapsed left lung, and a congested right lung. Spirochetes were found in sections of the spleen, kidney, liver, lung and, as noted, "suprarenal." Additional direct films were taken from the umbilical cord, lungs, liver, kidneys, and spleen, which all revealed positive results. Regarding the route of transmission, the author found the evidence to be inconclusive. When examining the placenta, the maternal surface contained a large number of spirochetes in the maternal blood sinuses. However, no positive results were yielded from the fetal vili. Nevertheless, all possible sources of infection other than intra-utero were found to be unlikely, suggesting an intrauterine infection as a probable source of infection in this case [88]. A similar scenario was described by Chung and Chang, in which a premature infant died within a week after birth. This was supposed to be a case of congenital infection. The mother was admitted with relapsing fever during parturition [41]. Another study gave a detailed account on the adverse pregnancy outcomes. The report concludes that both age of the mother and age of gestation had no influence on adverse events. Adverse events were observed at any gestation week during pregnancy. Further, the study reported spirochetes in a stillborn infant's blood, obtained from its heart, and intrauterine infection was suspected in 2 new born infants [48]. Bryceson and colleagues stated that abortion and miscarriage seem to be usual outcomes of pregnancies where the mother is infected with *B*. *recurrentis*. The authors further referenced a colleague, El Ramley A., who found congenital infection and abortion to be a standard in his studies [4].

Association between gestational progress and adverse outcome?. In one publication, all adverse events took place before the 20th week of gestation, and no stillbirths had been observed past that point [6]. By contrast, several studies reported adverse pregnancy outcomes past that time: 2 stillbirths in the ninth month of pregnancy [65], a premature delivery in the seventh month of gestation [41], a premature delivery at 7.5 months of gestation [44], a stillbirth in the fifth month, and in the case of a further 2 women, a stillbirth and a premature delivery in the seventh month of gestation [88]. The evidence suggests that adverse pregnancy outcomes may occur at any time, as another study has observed [89].

Need for further studies. Results of the analysis demonstrated an enormous rate of adverse pregnancy outcomes for pregnant women infected with *B*. *recurrentis* (Fig 4). While most aspects of LBRF were extensively researched, the effect on pregnant women and the unborn child has not been investigated yet. Factors that influence the risk of adverse pregnancy outcome during LBRF are currently unknown or not retrievable in published literature. The findings suggest a need for studies specifically looking at LBRF during pregnancy. Regarding adverse pregnancy outcomes during TBRF infection, rates between 30% and 44% were reported [27,28,101–103]. Larsson and colleagues recently demonstrated the effects of *Borrelia duttonii* infection in a mouse model, where spirochetes frequently caused congenital infection [102]. A treatable, neglected disease with a probability of up to 75% of adverse pregnancy outcomes should receive close attention, especially when dealing with women in, or from, endemic countries. Parallels can be drawn to similar neglected diseases, such as scrub typhus and murine typhus, in which a poor neonatal outcome (stillborn, premature, and/or small for gestation age) has been reported in 43.6% and 33.3%, respectively [104]. There is a need for further studies and increased awareness, especially among women in endemic countries.

Key learning points

- Reported CFRs in Africa have been rising throughout the past few decades.
- There is no standardized protocol on diagnosing JHR and reported JHR rates depend on the focus of the study, monitoring of patients, and awareness.
- Three out of 4 pregnancies are negatively affected by LBRF.

Top five papers

- Warrell DA. Louse-borne relapsing fever (Borrelia recurrentis infection). Epidemiol Infect. 2019;147:e106. doi: 10.1017/S0950268819000116
- Bryceson ADM, Parry EHO, Perine PL, Warrell DA, Vukotich D, Leithead CS. A clinical and laboratory study of 62 cases in ethiopia and a reconsideration of the literature1. QJM. 1970;39(1):129–70. doi: 10.1093/oxfordjournals.qjmed.a067198
- Felsenfeld O. Borrelia; Strains, Vectors, Human and Animal Borreliosis. St. Louis: Warren H. Green; 1971.
- Butler T. The Jarisch-Herxheimer Reaction After Antibiotic Treatment of Spirochetal Infections: A Review of Recent Cases and Our Understanding of Pathogenesis. Am J Trop Med Hyg. 2017;96(1):46–52. PubMed PMID: CCC:000397822900010.
- Guerrier G, Doherty T. Comparison of antibiotic regimens for treating louse-borne relapsing fever: A meta-analysis. Trans R Soc Trop Med Hyg. 2011;105(9):483–90. doi: http://dx.doi.org/10.1016/j.trstmh.2011.04.004. PubMed PMID: 51548792.

Supporting information

S1 Checklist. PRISMA checklist. 27-item checklist for systematic reviews.
(DOC)

S1 Text. Review Protocol. Established to conduct this systematic review.
(DOCX)

S2 Text. Data Extraction Sheet. Used for screening and selecting eligible publications.
(DOCX)

S3 Text. References before sub-search. Included and excluded references for qualitative synthesis, before a subsearch was conducted on (i) outcome, (ii) JHR, and (iii) impact on pregnancies.
(DOCX)

S4 Text. References–mortality. Included and excluded references in mortality section.
(DOCX)

S5 Text. References–JHR. Included and excluded references in JHR section.
(DOCX)

S6 Text. References–pregnancies. Included and excluded references in pregnancies section.
(DOCX)

S1 Fig. PRISMA flow diagram.
(PDF)

S1 Data. Data extracted from included studies. Excel spreadsheet containing, in separate sheets, the underlying numerical data.
(XLSX)

Author Contributions

Conceptualization: Pascal Kahlig, Andreas Neumayr, Daniel H. Paris.

Data curation: Pascal Kahlig.

Formal analysis: Pascal Kahlig.

Investigation: Pascal Kahlig.

Methodology: Pascal Kahlig, Andreas Neumayr, Daniel H. Paris.

Supervision: Andreas Neumayr, Daniel H. Paris.

Visualization: Pascal Kahlig.

Writing – original draft: Pascal Kahlig.

Writing – review & editing: Andreas Neumayr.

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